Self-Targeting in U.S. Transfer Programs*

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Abstract

This paper examines a classic rationale for voluntary take-up over automatic enrollment in transfer programs: Take-up may be advantageously "self-targeted" on characteristics that are infeasible to use as eligibility criteria. Using a correlation test, we find self-targeting in eight U.S. transfers: on average, recipients have lower consumption and lifetime incomes than similar eligible nonrecipients. Self-targeting focuses redistribution toward the lifetime poor and also increases transfers' within-lifetime insurance value. With a new sufficient-statistics result, we value the social benefits of self-targeting at approximately nine cents per transfer dollar, offsetting the social costs of take-up in some programs.

Keywords: transfer programs, take-up, eligibility, self-targeting
JEL Codes: H23, H53, I38

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1 Introduction

The take-up of U.S. means-tested transfer programs is voluntary, and in some programs, as many as one in three eligible households does not receive benefits (Currie, 2006; Ko and Moffitt, 2024). Instead of these voluntary transfers, the government could help all eligible people automatically, adjusting benefit levels to hold spending fixed. Going automatic would save the costs that recipients incur to claim benefits, but it would also give up a potential advantage of voluntary transfers: self-targeting. Self-targeting occurs when selective take-up among the eligible implicitly reveals dimensions of need which the government cannot itself use in eligibility rules. How much self-targeting is there in U.S. transfer programs, and can it justify why they are not automatic?

Answers to these questions would inform contentious debates over the role of voluntary take-up in the U.S. social safety net. Many critics of "administrative burdens" promote reforms to raise take-up, including automatic enrollment, over alternatives that would raise benefit levels or expand eligibility (e.g., Herd and Moynihan, 2019). Going automatic would make the U.S. safety net more like those in other developed countries.[1] It would also build on U.S. policy experiments in the Covid-19 pandemic, when two transfers—Medicaid and school meals—became partly automatic in the context of a temporary expansion of the safety net.[2] The issue of voluntary versus automatic redistribution bears further on other longstanding policy issues, such as the complexity of eligibility rules and fundamental reforms like a negative income tax or a basic income.

The classic theoretical rationale for voluntary transfers is that selection into take-up may be "advantageous." That is, a household's choice to take up a voluntary transfer in the face of costs or "ordeals" may reveal that it has a higher level of unobservable need (Nichols and Zeckhauser, 1982; Besley and Coate, 1992). However, there are also prominent arguments against voluntary transfers. Ordeals may instead perversely screen out the neediest households, who may face greater take-up costs or behavioral frictions (Currie and Gahvari, 2008; Mullainathan and Shafir, 2013). Even if selection is advantageous in direction, it may be too small in magnitude to offset take-up costs (Herd et al., 2023). Finally, ordeals may reduce the insurance value of transfers if they deter take-up among households hit by adverse shocks (Hoynes and Luttmer, 2011). Theory alone thus cannot say whether any benefits of self-targeting outweigh ordeal costs, and economists lack a way to connect empirical estimates of self-targeting to theory.

This paper studies self-targeting in U.S. transfer programs. First, we measure the extent of self-targeting in eight transfers that constitute together most of the U.S. safety net. Second, we estimate

[1] This is a key conclusion in international comparisons of welfare states (e.g., Esping-Andersen, 1990; Bartels and Neumann, 2021). However, these countries also see incomplete take-up in voluntary programs (Eurofound, 2015).

[2] Before the end of Medicaid auto-enrollment, U.S. Department of Health and Human Services (2022) forecasted that 8.2 million people would lose benefits, of which 6.8 million (83 percent) were expected to be from non-take-up among the eligible. Some policy reforms have already occurred in the context of school meals: Ten states have extended the pandemic-era expansion as of Spring 2024 (see `https://frac.org/healthy-school-meals-for-all`).

the expected annual payments and the insurance values of existing voluntary transfers at each point in the lifetime-income distribution, and we contrast these estimates with ones for counterfactual automatic transfers. Third, we derive a new sufficient-statistics formula for the welfare impact of moving from voluntary to automatic transfers, and we implement this formula using values for the sufficient statistics from our data and external estimates.

The first part of our analysis documents self-targeting through a correlation test. Using data from the Panel Study of Income Dynamics (PSID), we compare the consumption levels of transfer recipients to those of eligible nonrecipients with similar current annual incomes. We also examine self-targeting with respect to lifetime income, which allows us to distinguish between-lifetime redistribution and within-lifetime insurance value. These selection measures follow in a tradition in public finance dating to Vickrey (1947): We take as a premise that consumption is a superior measure of living standards to current income, since households can save or dissave to smooth transient income shocks, at least in part. However, practical or political constraints are often thought to rule out nonlinear taxes on consumption or lifetime income. If transfer take-up is self-targeted on either measure when current income is held fixed, then ordeal costs relax a key constraint on redistribution and social insurance.

We find self-targeting on both consumption and lifetime income for all eight transfers we study, often of substantial magnitude. In the Supplemental Nutrition Assistance Program (SNAP), for instance, the average consumption rank of recipients is 15 percentiles lower than eligible nonrecipients with similar incomes, a difference of about \$8,000 per person per year or 50 percent of their average per-capita consumption. These gaps are somewhat smaller for other transfers. Around half of this self-targeting on consumption is between-lifetime and half is within-lifetime, implying that take-up helps to identify the permanently poor as well as households facing temporary drops in consumption.

Our correlation test complements recent research that finds mixed evidence on ordeals' targeting properties.[3] However, there are three key differences. First, our test measures selection for the average transfer recipient, rather than the marginal one. As we show, this is the welfare-relevant parameter for a question that follows naturally from prior research: If ordeals are of mixed effectiveness on the margin, why not abolish voluntary take-up entirely? Second, our test distinguishes whether take-up reveals unobserved need rather than observed characteristics, a key distinction when eligibility rules or taxes can be adjusted in addition to ordeals. Prior work has instead examined heterogeneity in take-up responses according to characteristics that are common eligibility criteria, like age or current income. Third, our test is easy to implement via a cross-

[3]Recent papers studying the targeting properties of ordeals and information include Bhargava and Manoli (2015), Ganong and Liebman (2018), Deshpande and Li (2019), Finkelstein and Notowidigdo (2019), Gray (2019), Lieber and Lockwood (2019), Domurat et al. (2021), Arbogast et al. (2022), Ericson et al. (2023), Giannella et al. (2023), Shepard and Wagner (2024), Unrath (2024), and Wu and Meyer (2024).

sectional descriptive regression (Chiappori and Salanie, 2000; Jacobsen et al., 2020) and does not require exogenous variation in the ordeal of interest. Altogether, these differences yield a different economic conclusion: We establish that take-up costs induce desirable and robust self-targeting on average, whereas prior work suggests the marginal impacts are more ambiguous.

The second part of our analysis contrasts voluntary and automatic transfers in terms of their expected value and insurance value over the distribution of lifetime income. Such a comparison requires us to choose a specific "voluntary-to-automatic" reform that we use throughout the rest of the paper. Our reform implements a flat cut to a voluntary transfer and, at each level of current income, pays out the savings as automatic transfers to all eligible people with that income, including current nonrecipients. By imposing budget balance, our reform contrasts two versions of the same program and thus highlights self-targeting, rather than comparing the transfer to other uses of funds, as in Hendren and Sprung-Keyser (2020).[4] Among flat cuts, the reform is also uniquely distributionally-neutral with respect to current income, so that we distinguish the welfare impacts of self-targeting from incidental changes in progressivity. The reform can equally be viewed as a marginal shift from a fully-voluntary transfer $\$b$ to a \$1 automatic transfer with a voluntary "top-up" of $\$(b-1)$, thus holding fixed other aspects of the transfer.[5] Of course, automating real-world transfers would likely raise myriad other issues, so we use this hypothetical reform as a thought exercise to isolate a single fundamental trade-off in transfer design.

We find that, relative to our automatic counterfactual, existing transfers provide larger expected payments to households at the bottom of the lifetime-income distribution. They also provide more insurance against variation in consumption over the lifetime, holding fixed lifetime income. Overall, existing transfers provide about 30 percent more in expectation and in insurance value to the lifetime poor. These additional transfers are highly progressive, tapering off by the twentieth percentile of the lifetime-income distribution. Take-up costs thus endow transfers with a implicit ability to tax consumption and lifetime income progressively, a fact that income-based distributional analysis would obscure.[6] The reason take-up costs raise insurance value is that eligible households take up in low-consumption states relative to their lifetime, a correlation that cannot emerge in automatic transfers. By consequence, our results imply that self-targeting does not trade off between two key goals of safety nets, between-lifetime redistribution and within-lifetime insurance.

[4]Kleven and Kopczuk (2011), who also hold the transfer budget fixed, describe the same allocation decision as the inner level of the government's optimization problem, where the outer level is to set the budget.

[5]Holding fixed other aspects is important, given rationales for voluntary transfers related to their in-kind nature (Cunha et al., 2019; Gadenne and Singhal, 2023), and treatment effects (e.g., Bailey et al., forthcoming).

[6]In this way, our work connects to studies of lifetime tax and transfer incidence (Fullerton and Lim Rogers, 1993; Fullerton and Metcalf, 2002; Liebman, 2002; Bhattacharya and Lakdawalla, 2006; McClellan and Skinner, 2006; Hoynes and Luttmer, 2011; Blundell et al., 2015; Bengtsson et al., 2016; Roantree and Shaw, 2018; Fullerton and Rao, 2019; Levell et al., 2021; Auerbach et al., forthcoming). We build on this literature by focusing on voluntary programs, in which we show the importance of self-targeting to lifetime incidence. Another related paper is Blank and Ruggles (1996), which documents heterogeneous income dynamics that are correlated with take-up.

The third part of our analysis asks whether, at the margin, a transfer should be automatic, given the extent of self-targeting when it is voluntary.[7] We answer this question within a theoretical model of redistribution by income taxes and voluntary transfers. For this model, we prove that the social benefits of self-targeting from our "voluntary-to-automatic" reform are summarized by the regression coefficient from our correlation test. We also show that, for the same reform, the social costs of ordeals are summarized by a take-up elasticity with respect to the benefit level. However, the reform is generically not incentive-neutral in labor supply, and we derive an additional term for the welfare impact of the labor-supply response.

Calibrating the formula, we draw three conclusions about the social costs and benefits of self-targeting. First, self-targeting yields quantitatively important social benefits: In our baseline calibration, we estimate benefits of approximately nine cents per transfer dollar, taking a dollar-weighted average across programs. About two-thirds of these benefits arise from stronger insurance against within-lifetime consumption fluctuations, and the remaining third reflects an increase in progressivity with respect to lifetime income.

Second, overall across transfers, the social benefits of self-targeting likely exceed the social costs of ordeals. Our theoretical framework obtains upper-bound estimates of ordeal costs through the envelope theorem, as when take-up choices are made optimally, the welfare-relevant ordeal cost equals the fiscal cost of marginal recipients. These upper bounds are large: In annual per-recipient terms, we find ordeal costs could be as high as $240 for SNAP or $500 for Medicaid. Models in which incomplete take-up is, at least in part, a result of non-optimizing behavior imply smaller ordeal costs, as the marginal costs paid by non-optimizers must be less than their marginal fiscal costs. Behavioral frictions that reduce take-up would then further weaken the case for automatic redistribution. Our results thus establish self-targeting as a credible argument for existing U.S. transfers and cast doubt on the merits of going automatic.

The third conclusion from our welfare analysis is that programs vary greatly in the welfare effects of going automatic, stemming from differences in the magnitudes of self-targeting. In SNAP and housing assistance, self-targeting is strong and valued at about twelve cents per transfer dollar to society. This amount vastly exceeds upper-bound estimates of ordeal costs, so making these transfers automatic is unlikely to be socially desirable. By contrast, self-targeting is of little social value in several transfers. The programs nevertheless inflict ordeal costs, and so it may be valuable on net to go automatic in some cases.

Survey data like the PSID are subject to important concerns about measurement error in transfer receipt, income, and consumption (Meyer et al., 2009, 2015). We take this challenge seriously, as U.S. administrative data lack consumption and transfer receipt linked across programs.

[7] We focus on welfarist rationales for transfers. Non-welfarist normative frameworks, such as specific egalitarianism, can also justify transfers—as can externalities, paternalism, or market imperfections.

First, on misreporting of receipt, we adopt corrections from a recent literature that estimates how misreporting probabilities vary with observable characteristics (Davern et al., 2019; Mittag, 2019; Meyer et al., 2020). These corrections actually strengthen our results. Second, on consumption misreporting, self-targeting holds for types of consumption thought to be well-measured and for durable goods ownership (Meyer and Sullivan, 2023). Third, on lifetime income, we extend methods in Haider and Solon (2006) to address potential bias from incomplete income histories. Our analysis contains several implicit replications: first, the complementary analysis of consumption and lifetime income (Caspersen and Metcalf, 1994) and second, the consistency across programs.

We also implement several tests to address measurement error in transfer eligibility. Eligibility imputation is a difficult and pervasive challenge in analyses of U.S. transfers, whether using surveys or administrative data, as both lack eligibility information about nonrecipients. First, our results are robust to reclassifying simulated-ineligible recipients as eligible (Duclos, 1995). Second, we show that our results hold among the poorest subsets of the population that are almost certainly eligible and are most welfare-relevant. Third, results persist after further controlling for any characteristic observed in any eligibility rule across our eight transfers. These sensitivity analyses suggest it is unlikely that measurement issues in survey data explain our findings.

Our paper relates most closely to several other analyses of the welfare consequences of selective take-up in social programs. Two papers, Lieber and Lockwood (2019) and Shepard and Wagner (2024), develop sufficient-statistics formulas similar to to ours, which they respectively use to study ordeal-based targeting in Medicaid home care and markets for subsidized health insurance. Compared to these papers, we study the full breadth of the U.S. safety net and substantially generalize the sufficient-statistics result.[8] Also related is Deshpande and Lockwood (2022), which shows that disability benefits provide substantial implicit insurance of non-health risk, as take-up identifies need even among the marginally health-eligible. Finkelstein and Notowidigdo (2019) present a welfare analysis of changes in ordeal costs, and we discuss later the key differences between their reform and ours, in which benefit levels change and not ordeals. Finally, in a quite different income-maintenance framework, Kleven and Kopczuk (2011) also argue that take-up costs may be justified by their targeting benefits.

2 Data and Measurement

Our main source of data is the Panel Study of Income Dynamics (PSID) in its eleven biennial survey waves from 1997 to 2019. In each PSID wave, we observe heads of household and spouses ages 18 to 65. Here we first review key aspects of the data, leaving further details to Appendix

[8]We prove it extends to a rich dynamic environment with ability heterogeneity and nonlinear taxation, and we newly incorporate ordeal costs and labor supply responses into the welfare analysis.

B. We then explain three imputation procedures that augment the PSID data: for cash-equivalent values of in-kind transfers, transfer eligibility, and lifetime income.

Our goal is to measure selection into transfers on consumption and lifetime income. The PSID data has several crucial features for this purpose, including its long panel dimension to estimate lifetime income, its consumption data, and its information on the receipt of all major U.S. transfer programs. Its major limitations are the reporting issues that we discuss in depth in Section 3.

2.1 Income, Consumption, and Transfer Receipt

Current Income. We define household income as total annual income of the head and spouse before taxes and transfers, excluding other household members. Income includes labor, business, and capital income. Following the National Academy of Sciences (Citro and Michael, 1995), we adjust for household size using the equivalence scale $e_h = (N_{h,\text{adult}} + 0.7N_{h,\text{child}})^{-0.7}$, where $N_{h,\text{adult}}$ and $N_{h,\text{child}}$ respectively denote the numbers of adults and of children in household h.[9] We compute income ranks within year, pooling across birth-year cohorts.[10]

Current Consumption. The PSID has extensive coverage of consumption expenditures since 1999. Expenditure categories include food, housing, health, transportation, education, child care, and several smaller topics. We adjust the data in two ways to better reflect consumption rather than expenditure, following Meyer and Sullivan (2023). These adjustments aim to convert durable-goods ownership into consumption flows. First, for homeowners, we replace mortgage and property tax payments with equivalent rents based on reported home values. Second, for vehicle owners, we replace loan payments with estimates of lease-cost equivalents. Household consumption is then equivalized as above. Consumption ranks are also computed within year.

Transfer Receipt. We observe self-reported household-level receipt for ten transfers. These are the Supplemental Assistance Nutrition Program (SNAP); Medicaid; Section 8; public housing; Temporary Assistance for Needy Families (TANF); Supplemental Security Income (SSI); Women, Infants, and Children (WIC); the Low Income Home Energy Assistance Program (LIHEAP); and the National School Lunch Program and School Breakfast Program. We combine public housing and Section 8 into one program to which we refer as "housing assistance," and the lunch and breakfast programs into "school meals." Table 1 provides summary statistics on the transfers.

[9] Appendix A includes results not adjusted for household size and composition. These are typically quite similar to those with equivalized households. By implication, other equivalence scales are also likely to yield similar results.

[10] We do not rank households within both year and cohort, as this would remove life-cycle effects of rising incomes and falling transfer receipt rates with age. Reranking within year and cohort would thus reduce estimated transfer progressivity in current income relative to lifetime income, so it would only strengthen our findings.

Table 1: Means-Tested Transfer Programs in the U.S.

	SNAP	Medicaid	Housing Assistance	TANF	SSI	WIC	LIHEAP	School Meals	Any Transfer	U.S. Population
Budgetary Cost in 2019 (billions)	60.4	613.5	41.7	30.9	55.8	5.3	3.7	18.7	n.a.	n.a.
Receipt Rate	10.7	16.3	5.3	1.0	6.8	4.7	4.1	11.7	28.5	n.a.
Take-Up Rate, Simulated Eligibles	42.2	54.6	10.1	10.4	59.8	41.4	17.0	46.2	47.3	n.a.
Mean Annual Benefit, Recipients	3,390	5,455	5,574	14,199	2,768	574	531	628	n.a.	n.a.
Characteristics of Households or Heads of Recipient Households										
Mean Age, Head	41.6	41.8	40.6	35.5	45.2	33.9	44.6	39.8	42.0	43.1
% Married	20.0	33.2	10.8	15.7	27.6	41.3	26.7	41.7	34.5	48.7
Mean Household Size	3.1	3.4	2.5	3.5	2.6	4.3	3.1	4.2	3.2	2.6
% Children at Home	51.9	59.9	40.5	91.0	28.9	92.1	49.1	94.3	51.7	32.1
% Nonwhite or Hispanic	64.6	62.8	73.1	74.5	62.0	70.5	59.1	69.0	62.9	41.4
% H.S. Graduate	71.1	74.0	74.6	60.3	73.9	71.6	71.5	72.2	76.7	89.0
Mean Household Income	16,270	27,995	17,830	11,691	20,510	32,091	17,221	34,860	31,741	78,503
% Employed	46.3	54.9	50.5	40.1	38.9	70.0	45.8	71.3	59.9	79.2
Mean Rank, Equivalized Households										
Current Income	16.7	22.3	19.8	12.5	18.6	24.1	17.1	25.5	25.2	50.0
Consumption	17.0	22.8	16.9	11.9	27.8	19.3	19.7	22.4	26.4	50.0
Lifetime Income	24.4	30.6	24.9	22.6	27.0	34.4	26.1	34.3	33.3	50.0

Notes: This table reports summary statistics on the eight means-tested transfer programs we study. See Appendix B for sources on budgetary costs. All other data is from the PSID (waves 1997–2019). Average lifetime ranks are computed as means weighted by life-years of transfer receipt. Monetary values are expressed in 2020 constant dollars.

2.2 Imputation of Other Variables

Cash Equivalents of In-Kind Transfers. We measure the dollar value of transfers by combining information from the PSID and the Supplemental Poverty Measure module of the U.S. Current Population Survey (CPS). For SNAP, TANF, SSI, UI, and LIHEAP, the PSID records the nominal value of transfers over various time periods, which we rebase as the per-capita annualized amount in 2020 constant dollars. The PSID does not include cash-equivalent values for in-kind transfers, namely Medicaid, Section 8, public housing, and WIC. We impute these amounts with the average values by household size and year reported in the CPS for all but WIC, where we use the national average benefit. The CPS generally values in-kind transfers dollar-for-dollar with expenditure.[11]

Lifetime Income. We construct a lifetime concept of household income from incomplete income histories. To begin, we estimate a Poisson regression model with individual fixed effects, interacted with age-specific coefficients as recommended by Haider and Solon (2006). Letting i index individuals, t index calendar years, and a index age in years, the model takes the following form:

$$\mathrm{E}[y_{it} \mid X_{it}] = \exp(\alpha_i \lambda_a + X_{it}' \beta_a), \tag{1}$$

where α_i is an individual fixed effect, X_{it} is a matrix of time-varying demographic characteristics, and λ_a and β_a are vectors of age-specific coefficients. The outcome y_{it} is individual income.

We then perform several adjustments, explained in Appendix B, before using the regression results to impute lifetime income. These adjustments shrink the fixed effects to account for sampling variation and impute demographic characteristics to balance the panel. We calculate lifetime average income from ages 18 to 65, and then we account for spousal income in a way that permits changes in household composition over time. In particular, let $j(i,t)$ indicate i's spouse in year t. Our concept of lifetime household income follows each individual through the sequence of households they experience as adults, without discounting for time. That is, the lifetime household income of individual i is

$$\overline{y}_i^h = \sum_t e(\widehat{y}_{it}^h) = \sum_t e(\widehat{y}_{it} + \widehat{y}_{j(i,t),t}) \tag{2}$$

where t is again summed over the years in which i is between ages 18 and 65, $\widehat{y}$ is a predicted income, and $e(\cdot)$ is the equivalence-scale function. If we restrict our sample to stable households (as in, e.g., Fullerton and Lim Rogers, 1993), our definition of lifetime income would coincide with the standard concept. We compute lifetime-income ranks within birth-year cohorts.

[11] Except for Medicaid, for which the Census produces household-level "fungible values" and individual-level "market values." We use fungible values, so as to remain at the household level. The PSID variables for Medicaid also include state medical-assistance programs.

Simulated Eligibility. Studying transfer take-up among the eligible requires measures of transfer eligibility, so that one can distinguish the ineligible from eligible nonrecipients. We simulate eligibility by compiling information on program rules, mainly from primary-source documents and research databases of such rules, similar to the Urban Institute's TRIM program (Zedlewski and Giannarelli, 2015). See Appendix B for details on these eligibility simulations.

Eligibility simulations cannot perfectly capture true eligibility, as information used in actual eligibility determinations differs from survey-data variables. In Appendix B, we show that our simulated-eligibility measure is strongly predictive of transfer receipt, though misclassification is apparent. Considerable fractions of recipients are simulated to be ineligible, and take-up rates are lower than official estimates. Both are routine issues in microsimulations of eligibility (Duclos, 1995).[12] Mismeasured eligibility predisposes us to understate the importance of eligibility rules relative to self-targeting among the eligible, and so we consider this threat carefully.

2.3 Preliminary Evidence of Self-Targeting

Before turning to the main analysis, we show evidence of self-targeting in nearly the raw data. We focus on SNAP as an illustrative example. Table 2 reports rates of receipt, simulated eligibility, and take-up among the eligible, doing so jointly by quintile of income and consumption.

Panel A shows that, holding income fixed, low-consumption households are more likely to receive SNAP than high-consumption households. For instance, 34 percent of households in the first quintile of income receive SNAP on average. But 51 percent of households in both the first quintile of income and the first quintile of consumption receive SNAP. Thus, holding income fixed, take-up is concentrated among people who have low consumption. This finding differs from the well-known fact that transfer recipients have lower incomes and are generally needier (e.g., Tiehen et al., 2017), in that we show gaps between recipients and non-recipients persist within income. Indeed, these gaps will prove robust to controls that span the government's information set.

[12]Appendix B provides two supplementary analyses that probe why our estimated take-up rates are lower than figures published by government agencies. First, focusing on SNAP, we decompose the gap into three components: differences in receipt, differences in simulated eligibility, and a "misalignment" component. This third component reflects a little-discussed flaw in official figures that biases them upward: As they simply divide total enrollment by total simulated eligibility, there is no adjustment for recipients in the administrative data who would be simulated ineligible in survey data. We find the underreporting of receipt (see Section 3.3) and the misalignment issue are both quantitatively important to the gap in take-up rates. Second, we review estimates of take-up rates for other programs. Our estimates appear in line with other survey-based estimates but are below administrative-data estimates, consistent with transfer underreporting and misalignment.

Table 2: SNAP Receipt, Eligibility, and Take-Up Rates by Income and Consumption Quintile

Panel A: Receipt Rate

		Income Quintile					
		1	2	3	4	5	Avg.
	1	51.2	22.3	7.8	4.8	4.9	35.3
	2	23.7	9.6	2.8	1.0	0.5	8.5
Consumption Quintile	3	12.2	5.9	2.2	0.5	0.3	3.3
	4	6.2	3.4	1.4	0.2	0.2	1.3
	5	5.5	3.0	1.7	0.3	0.1	0.9
	Avg.	33.6	12.2	2.8	0.5	0.2	

Panel B: Simulated Eligibility Rate

		Income Quintile					
		1	2	3	4	5	Avg.
	1	70.6	19.9	2.0	2.9	2.9	43.9
	2	43.7	12.1	1.7	0.9	0.4	12.6
Consumption Quintile	3	33.3	10.7	2.2	1.3	0.4	6.8
	4	27.8	10.9	1.7	0.8	0.7	4.4
	5	29.0	12.3	2.5	1.5	0.7	4.2
	Avg.	53.9	14.4	1.9	1.1	0.6	

Panel C: Take-Up Rate Among Simulated Eligibles

		Income Quintile					
		1	2	3	4	5	Avg.
	1	58.5	40.9	.	.	.	55.5
	2	34.7	23.1	.	.	.	29.6
Consumption Quintile	3	14.2	13.2	.	.	.	17.9
	4	21.8	14.1	.	.	.	11.8
	5	16.0	8.7	.	.	.	11.5
	Avg.	48.3	28.5	.	.	.	

Notes: This table reports the shares of households that receive SNAP (Panel A), are simulated to be eligible for SNAP (Panel B), and take up SNAP conditional on being simulated to eligible (Panel C). Households are split by quintiles of equivalized household consumption and income. Due to low rates of simulated eligibility, we do not report take-up rates for the top three income quintiles. See Appendix Table A1 for a tabulation by income and lifetime income.

There are two reasons why, in principle, receipt might be sensitive to consumption as well as income. First, transfers may have eligibility criteria that correlate with consumption, even conditional on income, such as asset tests or categorical eligibility for some groups (e.g., people with disabilities). Making many high-consumption households ineligible would directly reduce receipt in that population. Second, such patterns may arise due to "self-targeting," or rates of transfer take-up among the eligible that depend on their consumption.

Panels B and C show the pattern we highlight in Panel A results from both self-targeting and eligibility rules. Take-up among the simulated eligible drops sharply in consumption given income, as much as does the rate of simulated eligibility (in percentage points).[13] In the rest of the paper, we establish self-targeting more carefully as an empirical fact, contrast the distributional impacts of voluntary and automatic transfers, and assess the welfare consequences of a marginal shift from voluntary toward automatic transfers.

3 Estimates of Self-Targeting in Transfers

This section quantifies advantageous self-targeting in transfers. First, we introduce and implement a correlation test to detect self-targeting. Second, we provide four facts about self-targeting using this test. Third, we assess the sensitivity of our results to measurement issues.

3.1 Empirical Framework

Definition. We say there is *advantageous self-targeting* on an outcome C_i (e.g., consumption) if transfer recipients are negatively selected on the outcome relative to eligible nonrecipients with the same current income Y_i. That is:

$$\mathrm{E}[C_i \mid D_i = E_i = 1, Y_i] \;\leq\; \mathrm{E}[C_i \mid D_i = 0, E_i = 1, Y_i], \tag{3}$$

where, for households indexed by i, D_i and E_i respectively indicate receipt and eligibility.

Equation 3 is motivated by the correlation test of Chiappori and Salanie (2000). Consider, in particular, the following joint semiparametric model of the outcome and transfer receipt:

$$C_i = \beta D_i + f(Y_i) + X_i\delta + \nu_i$$

$$D_i = \begin{cases} 1[g(Y_i) + X_i\gamma + \varepsilon_i \geq 0] & \text{if } E_i = 1 \\ 0 & \text{if } E_i = 0. \end{cases}$$

[13]While take-up rates are sensitive in *levels* to the general expansiveness or conservativeness of any eligibility simulation, measurement issues can less easily explain the vast *differences* in take-up by consumption given income.

where X_i contains eligibility-rule observables, so that eligibility $E_i = E(X_i)$, and $f(\cdot)$ and $g(\cdot)$ are flexible functions of current income.

In this model, transfer receipt D_i may be correlated with the outcome C_i for two reasons, a causal relation (given by β) and a non-causal association ($\text{corr}(\nu_i, \varepsilon_i) \neq 0$). The latter is self-targeting: Households which are unobservably more or less likely to take up the transfer when eligible may also be also positively or negatively selected on the unobservable component of the outcome. If $\beta \geq 0$ (transfer receipt has a weakly positive causal effect on the outcome), then Equation 3 implies advantageous self-targeting, that is, the errors ν_i and ε_i are negatively correlated.

To test for self-targeting, we always control for current income and restrict the sample to eligible people. In robustness checks, we also control for eligibility-rule observable characteristics. These specifications are motivated by what the government might adjust to redistribute in lieu of self-targeting. Current income is a sufficient control if the government adjusts the income tax, whereas controlling for eligibility-rule observables would be appropriate if these rules are also adjusted.

Furthermore, a motive for transfers beyond redistribution is to insure households against shocks. This is socially valuable to the extent that households cannot self-insure or there are insurance-market imperfections. We will use our correlation test to study if households use voluntary transfers as insurance—that is, if their take-up responds to variation in their consumption over time. Accounting for insurance, households may value the average transfer dollar in excess of one dollar, if transfers tend to come in states of the world with high marginal utilities of income.

Connection to Prior Research. Section 1 reviews research on the targeting properties of changes in ordeal costs, which may enter the take-up error term ε_i. This literature uses policy variation in specific ordeals to measure the characteristics of eligible people whose take-up choice responds to the policy change—reform "compliers" for whom $E_i = 1$ and $X_i\gamma + \varepsilon_i \approx 0$. Complier characteristics are relevant for the welfare consequences of changes in ordeals.

By contrast, our correlation test includes non-marginal households, who would always or never take up across levels of ordeal costs. It produces the welfare-relevant measure of targeting to study a budgetary shift from voluntary to automatic transfers. Phrased precisely, the question it addresses is: "Given an existing transfer program, should the government spend a marginal dollar on an automatic transfer to all eligible people, or on greater benefits to those who have already taken up a transfer?" Intuitively, that policy choice is between the voluntary transfer's always-takers and its eligible never-takers.

Studying both ordeal changes and benefit shifts offers valuable insights. Practical and political considerations may sometimes facilitate or preclude one but not the other. Furthermore, the literature finds mixed results on targeting under the first approach (e.g., Deshpande and Li 2019 versus Finkelstein and Notowidigdo 2019), seemingly due to heterogeneity in marginal households across settings and ordeals. The resultant uncertainty gives reason to examine other policy hypotheticals, like our "voluntary-to-automatic" reform, which might (and indeed does) offer a more-robust empirical conclusion. Another virtue of the correlation test is that it is easy to estimate with observational data, since it is fundamentally descriptive. Identifying valid complier groups for ordeals would require exogenous variation, such as through a randomized trial.

3.2 Four Facts About Self-Targeting

To implement our correlation test, we estimate specifications of the following form:

$$\bar{R}_{it} = \beta D_{it} + f(R_{it}) + u_{it}, \tag{4}$$

where $\bar{R}_{it}$ is the consumption rank or lifetime-income rank for household i in year t, R_{it} is i's current-income rank, $f(R_{it})$ is a flexible function of this rank, and D_{it} indicates i's receipt status for a given transfer program. The regression is estimated on the simulated-eligible ($E_{it} = 1$) unless otherwise noted. The coefficient β summarizes the extent of advantageous selection into a transfer. We parameterize $f(R_{it})$ using cubic splines with knots at the 10th, 25th, and 50th percentiles of the current-income distribution.

Figure 1 presents the results, which we organize into four facts.

Fact 1: Self-targeting is advantageous with respect to consumption. Overall, recipients of a given transfer rank up to 20 percentiles lower in the consumption distribution than eligible nonrecipients with similar current incomes (see Panel A). In level terms, these results are consistent with differences of approximately 20 to 60 percent in these outcomes, or around \$2,500 to \$10,000 per person per year in consumption (see Appendix Table A2).

Another implication of Panel A is that self-targeting varies in magnitude, but not in sign, across programs. For instance, the receipt of SNAP and housing assistance is highly informative about consumption given income, whereas SSI receipt is less informative.[14] Some "non-differences" are also interesting: for example, it is not the case that cash-like transfers (e.g., SNAP) are systematically more self-targeted than the transfers most unlike cash (e.g., housing or Medicaid).

Fact 2: Self-targeting among the eligible often appears more important than eligibility rules. Panels A and B further show differences in consumption between recipients and *all* similar-income nonrecipients, including those simulated to be ineligible. Conceptually, that comparison pools selection by eligibility rules with self-targeting by the eligible. The results are broadly similar to our main results, which restrict the sample to the simulated-eligible. Taking these results together, self-targeting often appears to be the primary force, and eligibility rules somewhat secondary,

[14]Stringent eligibility rules in SSI mean few households are simulated to be eligible, which may perhaps leave little room for self-targeting among this population.

in inducing selection into transfers. However, this interpretation requires eligibility to be well-measured, a topic to which we return below.

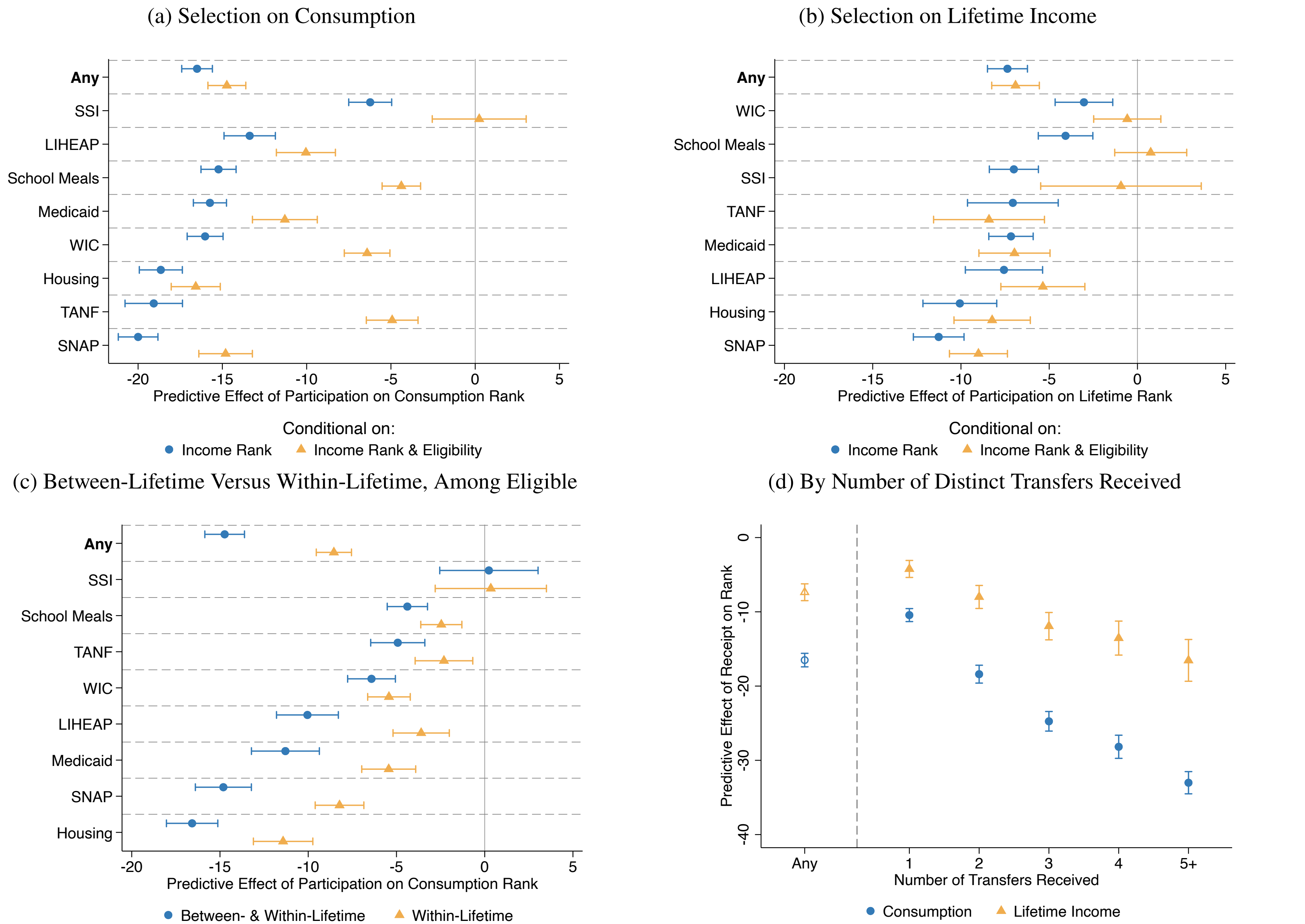
Fact 3: Unobserved shocks within life, not just low lifetime incomes, induce transfer take-up. Panels B and C study the extent to which self-targeting on consumption is on lifetime income (and thus between lifetimes) or is on consumption given lifetime income (thus within lifetimes). In Panel B, we find roughly half as much self-targeting on lifetime income as we found on consumption in Panel A. That is, some of what take-up reveals is who among the eligible is lifetime-poor and thus has little past income or future income to smooth during a period of transfer eligibility.

Panel C reaches the same conclusion in a different way. Here we augment the self-targeting regression (Equation 4) with person fixed effects and restrict to the simulated-eligible. For comparison, the blue series in Panel C repeats results from Panel A (originally in yellow) which do not include fixed effects and thus pool between- and within-person self-targeting. We continue to find self-targeting on consumption: Within-lifetime consumption drops correlate with changes in take-up, holding current income fixed. Importantly, these results continue to condition on current income, so they go beyond well-documented empirical patterns of take-up around observable shocks (e.g., Wu and Zhang, 2024). These empirical patterns lead to our conclusion in Section 4 that voluntary transfers have substantial insurance value.

Fact 4: There is economically interesting heterogeneity in self-targeting. Panel D of Figure 1 estimates the extent of self-targeting into transfers, distinguishing by the number of distinct transfers received. Eligible households that receive multiple transfers are more self-targeted on consumption and lifetime income than those receiving only one transfer. Such a pattern could arise if take-up costs were positively correlated across programs within households and negatively correlated with need, so that multiple receipt indicates deep inframarginality to transfers. This explanation would reconcile our finding (strong self-targeting on average) with the findings of prior research (mixed evidence of selection on the margin).

Next, we decompose our self-targeting results by consumption, lifetime income, and current income. Appendix Figure A5 shows our self-targeting results are driven by households with very low consumption and lifetime income. Comparing households with similar incomes, households who take-up SNAP and Medicaid are respectively 20 and 12 percentage points more likely than eligible nonrecipients to be in the bottom ventile of the consumption distribution. We also split the sample by current income: Appendix Figure A7 shows a roughly similar extent on self-targeting on consumption and lifetime income above and below the tenth percentile of current-income rank. Taken together, these results show that self-targeting is present even among the neediest households, who are most relevant to our welfare calculations.

Figure 1: Self-Targeting in Transfer Programs



Notes: This figure displays estimates of the predictive effect of transfer receipt on consumption rank (Panel A) or lifetime-income rank (Panel B), conditional on current-income rank (coefficient γ from Equation 4). For the yellow diamonds, we estimate the regression only on people whom we simulate to be eligible. Panel C augments the specification with person-level fixed effects. The “any” row of Panels A–C is an indicator for receipt of at least one of the eight transfers. Panel D adapts Equation 4 by replacing the transfer indicator with indicators for the number of transfers received in that year. In all panels, 95-percent confidence intervals reflect clustered standard errors by household.

Finally, we examine whether self-targeting may be less informative of need for members of disadvantaged demographic groups. Such heterogeneity is motivated by the concern that ordeals screen out the most vulnerable groups (Mullainathan and Shafir, 2013), often reflected in the existence of targeted policy accommodations (such as social workers and native-language forms). We divide the population by education and race/ethnicity, and we also break out populations of particular interest, such as non-native English speakers and car non-owners. Appendix Figure A8 shows some suggestive heterogeneity in self-targeting, focusing on SNAP and Medicaid. In SNAP, for example, self-targeting appears to be weaker for Hispanics and non-native English speakers in SNAP, although these patterns are not so evident in Medicaid.

3.3 Sensitivity to Mismeasurement

Survey data are imperfect. Here we consider the potential for bias in our results due to measurement error in simulated eligibility, self-reported transfer receipt, current income, consumption, and lifetime income. Overall, measurement issues seem unlikely to explain our results, given their magnitude and robustness.

We are especially careful with the survey data because our analysis is mostly infeasible in U.S. administrative data. First, such datasets lack appropriate measures of consumption. Second, they rarely link information across transfer programs. Third, while administrative data would improve measurement for some inputs to the analysis (e.g., income and transfer receipt), it would be harder to impute eligibility. Administrative data largely does not record eligibility information for nonrecipients, lacks the detailed covariates of survey data useful for imputing it, and may not capture some eligible nonrecipients who may appear in surveys (e.g., income-tax nonfilers).

Eligibility. Viewing eligibility as a control variable that is measured with noise, the threat to our analysis is that we may overstate the strength of self-targeting.[15] In several robustness checks, we assess this threat. We show that adjustments to our measures of simulated eligibility can shrink our estimates of self-targeting, but they leave standing our conclusion that advantageous self-targeting is a key force in most transfers.

First, simple adjustments to the eligibility simulations, such as imposing income limits or liquid-asset tests, have modest impacts on our estimates of self-targeting (Appendix Figure A9). These results suggest that further improvements to the eligibility simulations are unlikely to change our estimates materially. Furthermore, we reclassify all simulated-ineligible recipients of a given program as eligible, we still find that self-targeting concentrates incidence among the consumption-

[15] We see this as the likely direction of the bias because, conditional on income rank, simulated eligibility and receipt are positively correlated, and simulated eligibility and consumption rank are negatively correlated. Our analysis does not require us to accurately estimate overall take-up rates, only that we capture differences in take-up by consumption.

poor and lifetime-poor (Appendix Figure A10). This reclassification test is quite demanding, as by construction it "overfits" eligibility rules to explain receipt.

Second, when we restrict the sample to demographic groups with a high probability of eligibility, we continue to find self-targeting (Appendix Figure A11). Among such groups, patterns of transfer receipt are less likely to be confounded by eligibility mismeasurement and are, by consequence, more directly informative about self-targeting. This result echoes our earlier finding that self-targeting is present among the worst-off households, which also can be viewed as a robustness check against measurement error.

Third, we test measurement error by examining how the self-targeting coefficient moves when we add additional controls to proxy for unobserved eligibility rules. Following Oster (2019), coefficient stability suggests mismeasurement in eligibility rules is unlikely to reverse our conclusions. We first augment Equation 4 with controls for any variable that enters into the eligibility simulation for any transfer we study.[16] This attenuates but does not eliminate selection into receipt for most transfers (Appendix Figure A12). We then also include controls for variables that do not enter any eligibility simulation but predict both consumption or lifetime income and transfer receipt: race, education, and marital status. Adding in these further controls does little to move our estimates. This test builds confidence in our results, as it seems unlikely that any unobserved variable would be much more important than these "unused observables."

Transfer Receipt. Using linked survey and administrative data, Mittag (2019) and Davern et al. (2019) estimate statistical models of household survey reporting behavior for SNAP and Medicaid receipt respectively. Their models, intended for use as misreporting corrections, predict the probability of true transfer receipt given survey-reported receipt and demographic characteristics. These models allow researchers to replace assumptions of constant misreporting rates with misreporting probabilities that are functions of demographic observables.

Their corrections consistently increase our estimates of advantageous selection (Appendix Table A5). There are two reasons why. First, under constant misreporting rates, our estimates are attenuated. Consider misreporting probabilities $p_0 = \Pr(\widetilde{D}_i = 0 | D_i = 1)$ and $p_1 = \Pr(\widetilde{D}_i = 1 | D_i = 0)$, where D_i indicates true receipt and $\widetilde{D}_i$ indicates reported receipt. Comparing the feasible regression of $y_i = \widetilde{\beta}\widetilde{D}_i + u_i$ to the infeasible regression $y_i = \beta D_i + u_i$, one can show that $\beta = \widetilde{\beta}/(1 - p_0 - p_1)$.[17] Second, the parameter estimates in Mittag (2019) and Davern et al. (2019) both imply that underreporting of transfer receipt is somewhat more common among households

[16] These are the household's state of residence by year, household size and composition, income, earnings, ages of household members, disability status, unemployment duration and reason, and basic measures of wealth (value of any automobiles and liquid assets).

[17] Meyer et al. (2009) finds rates of under-reporting rates in the PSID in the range of 7 to 27 percent across programs we study. This suggests a presumption that our main estimates in Figures 2 and 3 are understated, even for transfer programs where heterogeneous-misreporting corrections have not yet been estimated.

with low consumption and lifetime income, holding income constant. Thus, their adjustments amplify the increase in the selection that we would find under constant misreporting rates.

We also compute the rates of transfer underreporting at the top of the consumption distribution that would be necessary to yield zero selection among the eligible (Appendix B). Overturning our conclusions requires a degree of underreporting that we view as implausible, such as "false-negative" rates of 50 percent in the top quarter of the consumption distribution. Though misreporting of transfer receipt is an important phenomenon, it is unlikely to explain our results.

Income. Income is often said to be poorly measured at the bottom of the distribution, and while there is also mismeasurement in consumption, it appears less severe than for income (Meyer and Sullivan, 2003; Brewer et al., 2017).

One story of intentional misreporting likely pushes in the opposite direction of our results. In this story, transfer recipients have incentives to underreport income so as to maintain eligibility, and they may do so in any quasi-official setting, including in surveys. Such incentives could apply less strongly to consumption, and to nonrecipients. All else equal, transfer recipients would thus appear *positively* selected on consumption given income. This is opposite to what we find.

We further explore concerns about income misreporting by predicting household income using other labor variables, such as weekly hours and occupation.[18] This approach provides an external check against issues in reporting, assuming accurate reporting of these variables. Appendix Figure A13 show that controlling for predicted income, in addition to reported income, has a modest impact on our results. Even this modest attenuation may reflect that occupation and education account for much of the year-to-year persistence in income.

Consumption. We show in Appendix Table A6 that similar selection patterns appear when only looking at categories of consumption deemed "well-measured" in Meyer and Sullivan (2023). For instance, on average, Medicaid recipients consume 38 percent less in housing, 32 percent less in vehicles, and 32 percent less in food at home than similar-income eligibles who are not on Medicaid. Similar patterns also manifest in PSID data on durable-goods ownership, such as whether the household owns a home, car, computer, and the number of rooms or presence of air conditioning in the home (Meyer and Sullivan, 2012), as we show in Appendix Table A7. SNAP recipients are 12 percentage points less likely to own a home, 8 percentage points less likely to own a car, and 9 percentage points less likely to own a computer than similar-income eligibles not on SNAP. The consistency of selection across measures, some of which seem truly unlikely to suffer from meaningful mismeasurement, bolsters our findings.

[18]Using the March Supplements to the Current Population Survey that match our PSID data years, we estimate Poisson regression models of individual income using occupation, industry, weeks per year, weekly hours, self-employment, in addition to basic demographic information. We then apply these predicted incomes to our PSID data.

Lifetime Income. Inferring lifetime income from "snapshots" is known to be difficult (Haider and Solon, 2006). We employ a new, transparent check against this issue. Lifetime-income ranks for households with many years-in-sample are mostly data, whereas those with few years-in-sample are mostly imputation. If the imputation process has imparted a systematic bias in our results, then the estimated extent of selection into transfers be related to a household's number of years-in-sample. We re-estimate the predictive effects of transfer receipt on lifetime rank as in Equation 4, retaining only individuals with progressively more years-in-sample.

Appendix Figure A14 shows that selection on lifetime income is essentially constant across households when split by their years-in-sample. Moreover, this "drift" effect sometimes appears for consumption, which has no across-year imputation step. In these cases, the phenomenon likely reflects considerations other than a bias in lifetime-income estimation, such as sample attrition.[19]

3.4 Is It Self-Targeting or a Behavioral Response?

We have interpreted our correlation-test results as showing that take-up acts as a "tag" of the persistent component of earnings ability. This interpretation seems to exclude possible behavioral responses to transfers, though we have not yet shown evidence to that end. For instance, some households may reduce their consumption or lifetime income so as to receive transfers.

Appendix Figure A15 presents evidence in favor of our "tagging" or selection interpretation. We show that, among *current* eligible nonrecipients of a given transfer, *future* recipients have on average a lower current-consumption rank than similar *future* eligible nonrecipients.[20]

Our argument is that, at some future horizon, the claim that our correlation-test results reflect behavioral responses is implausible a priori. As the findings are robust even when one performs this comparison in the distant future, they cannot be explained by households that might strategically reduce their consumption just before taking up. Indeed, the results are not much weaker for distant-future take-up than contemporaneous take-up. The figure also addresses the symmetric concern that transfers raise lifetime income, such as by improving productivity.

4 The Value of Voluntary and Automatic Transfers

This section contrasts existing voluntary transfers with counterfactual automatic versions of the same programs. We do so through a distributional analysis of the lifetime gains from each transfer in both its automatic and voluntary forms, considering both its expected value and insurance value.

[19] Attrition from the PSID is, of course, also a source of concern. Yet is not obvious that households that stay in the sample longer are more representative than those who attrit more quickly.

[20] The regression of interest is $\bar{R}_{it} = \alpha_{ct} + \beta D_{i,t+k} + f(R_{it}) + u_{it}$ within the subsample such that $D_{it} = 0$, for $k > 0$.

4.1 Approach

We focus on the distribution of lifetime income for two reasons. First, distributional analyses with respect to current income obscure that many low-income households do not have low consumption (Poterba, 1989, 1991). Second, a lifetime perspective provides a natural way to capture the ex-ante insurance value of transfers, in particular as insurance against within-lifetime shocks to consumption. Appendix C contains distributional analyses by current income and consumption.

Automatic Counterfactual. For each transfer, we will consider an automatic version that pays to all eligible households at a given income level exactly the dollar amount that was received in the voluntary transfer by the average eligible household at that income, without conditioning on take-up.[21] Among eligible households, the automatic transfer amount at an income rank y is

$$s_A(y, E_{it} = 1) = \mathrm{E}[s_{it} \mid y_{it} = y, E_{it} = 1],$$

where s_{it} is the observed transfer amount to household i in year t. We set $s_A(y, E = 0) = 0$ for all y, so that eligibility rules are enforced. For our income measure, we group people by their rank in the equivalized household income distribution, as in Section 3.

Among reforms that eliminate the voluntary transfer, this one is uniquely budget-neutral and distributionally neutral with respect to income. Intuitively, we study the lifetime incidence of a reform which shuffles the transfers received among eligible households with the same current income. For example, suppose eligible households at a given income rank receive \$1,000 if they take up a voluntary transfer, and half take up. In the counterfactual, the government would give \$500 to all eligible households with that income.

Expected Value. We measure the annual per-capita dollar amount that the average household at lifetime-income rank $\bar{R}_i = r$ receives under each existing transfer, $\mathrm{E}[s_{it} \mid \bar{R}_i = r]$. The amount under our automatic counterfactual is $\mathrm{E}[s_A(y_{it}, E_{it}) \mid \bar{R}_i = r]$. Integrating over the lifetime distribution, the population averages of these two quantities must be equal, given budget balance.

By conditioning on simulated eligibility, the automatic counterfactual $s_A(y_{it}, E_{it})$ accounts for rules which select on correlates of consumption and lifetime income. The difference in the expected payments between $\mathrm{E}[s_{it} \mid \bar{R}_i = r]$ and $\mathrm{E}[s_A(y_{it}, E_{it}) \mid \bar{R}_i = r]$ thus arises from self-targeting among the eligible at each lifetime-income rank. To simulate the counterfactual, we apply the empirical transition matrices from current ranks to lifetime ranks conditional on eligibility.

Insurance Value. To measure insurance value, we mark up or down the dollar amount of transfers

[21] The reform here is non-marginal, and so it is suited to positive rather than normative analysis. In Section 5, we focus on a marginal version of the reform.

to a household i in year t according to a marginal-utility weight ω_{it}:

$$\omega_{it} = c_{it}^{-\gamma} \,/\, \mathrm{E}[c_{it}^{-\gamma} \mid \bar{R}_i = r],$$

where $\bar{R}_i$ is i's lifetime income. The term $c_{it}^{-\gamma}$ is the marginal utility of income for a household with constant relative risk aversion over consumption, and the denominator recasts the utility impact into money-metric terms. By conditioning on lifetime rank in the denominator, we estimate average willingness to pay for insurance among households at that rank, not among all households.

The weights ω_{it} will be greater than or less than one when a household's consumption is respectively below or above the average of households with the same lifetime income. A higher risk-aversion parameter γ makes the weights more sensitive to consumption. Throughout the rest of the paper, we set the risk-aversion parameter $\gamma = 2$ as our baseline case, and we consider lower ($\gamma = 1$) and higher ($\gamma = 3$) values in sensitivity analyses. Following Finkelstein et al. (2019), we impose a "consumption floor" at the first percentile in forming the weights.

Using the weights, we calculate the certainty equivalent of each transfer program at each lifetime rank as $\mathrm{E}[\omega_{it} s_{it} \mid \bar{R}_i = r]$. Subtracting off expected values by rank, we obtain insurance values as $\mathrm{E}[(\omega_{it} - 1) s_{it} \mid \bar{R}_i = r]$. The calculations for the automatic counterfactual are similar.

This estimate of insurance value focuses on the risk of fluctuations in consumption around a lifetime average. It could overstate or understate the "true" insurance value for two reasons. First, toward overstatement, this approach assumes away the possibility of variation in preferences within households over time that could otherwise rationalize consumption fluctuations (e.g., state-dependent preferences). Second, toward understatement, it treats lifetime income as ex-ante certain, so that between-lifetime differences in expected transfers are not marked up.[22]

4.2 Results

Figures 2 and 3 report the results in total across our eight programs, and separately by program.

Expected Value. We find self-targeting directs transfers to the lifetime poor (see the blue lines of Figures 2 and 3), consistent with the regression results. Relative to voluntary transfers, automatic transfers would give households with the lowest lifetime incomes about 30 percent less on average. Existing transfers provide the lifetime-poorest households about \$3,800 per person per year. By comparison, automatic transfers would provide \$2,600 annually to members of this group.[23]

[22]Distinction between redistribution and insurance are fundamentally arbitrary, in that they are determined by informational assumptions. From the Rawlsian original position, in which people know nothing about their lifetime, everything is insurance, and "redistribution" is conceptually vacuous (Hendren, 2021).

[23]The bottom of the consumption distribution is similar: Average payments are \$4,000 under the voluntary transfer and \$2,300 under the automatic transfer. Appendix Table A4 reports point estimates and standard errors for expected payments by current income and lifetime income, along with payments in the automatic counterfactual. Appendix

Under budget balance, these larger expected payments to the bottom are achieved by reducing expected payments higher up the distributions of consumption and lifetime income. That is, going automatic in this way would reduce transfer progressivity with respect to consumption and lifetime income. Going automatic reduces progressivity more for transfers that are more strongly self-targeted on consumption or lifetime income, such as SNAP.

Insurance Value. The solid red lines of Figure 3 show that existing transfers provide substantial insurance value. For the lifetime poor, the average marginal transfer dollar provides approximately 25 cents of insurance value, or about $1,000 in total. Moving up the lifetime-income distribution, the insurance value of transfers comes to exceed expected payments. Said differently, the insurance benefits of the safety net are distributed less progressively than are the expected payments.

This ratio of insurance value to expected value also varies across programs. For instance, self-targeting in SSI provides little insurance value against within-lifetime shocks, which is sensible for a program that supports permanently disabled low-income adults.[24] By contrast, most of TANF's social value is insurance, sensibly so for a program with "temporary" in its name.

The red dashed lines show that automatic transfers would also provide insurance value. Overall, however, existing transfers provide more insurance value per dollar than would automatic transfers. This results follows from the finding of Figure 1 that self-targeting has both within- and between-lifetime components. Take-up of voluntary transfers thus helps the government target people facing adverse consumption shocks.

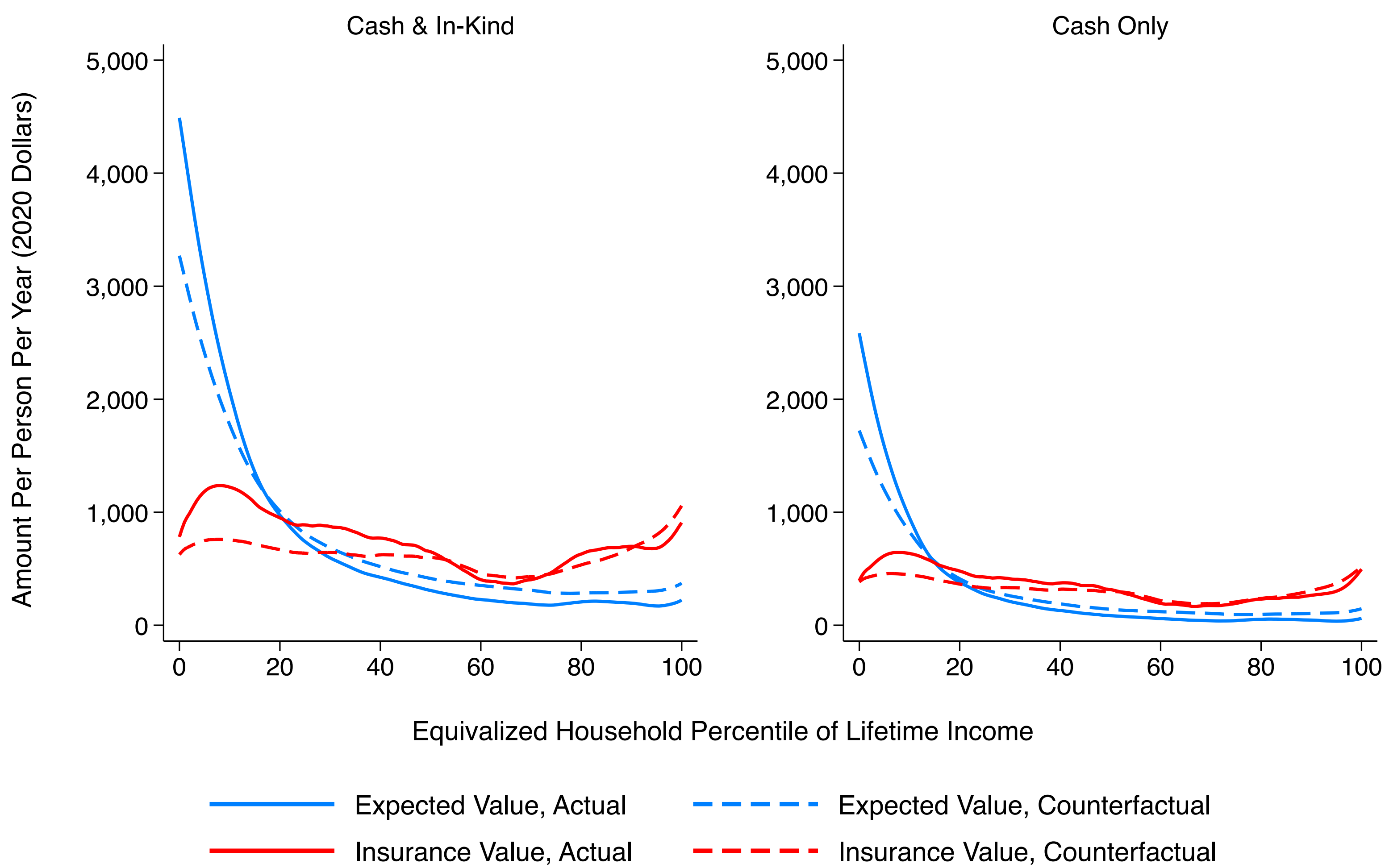
Holding the budget fixed, going automatic would reduce insurance value for the lifetime poor. Among this population, existing transfers provide about $300 more in insurance value per person per year than would our automatic counterfactual. The increases in insurance at the bottom do not come at the expense of less insurance to households with higher lifetime incomes. This is possible because the budget constraint applies to payments and not the insurance they provide. Putting together expected payments and insurance value, we can conclude that the certainty-equivalent values of automatic transfers would be less progressive in lifetime income than are existing transfers.[25]

Table A3 provides the same analysis but for consumption.

[24]The within-lifetime focus helps to explain the difference in our results from Deshpande and Lockwood (2022). Furthermore, state-dependent utility (in health) could imply that SSI provides insurance value, although not against within-lifetime shocks to consumption

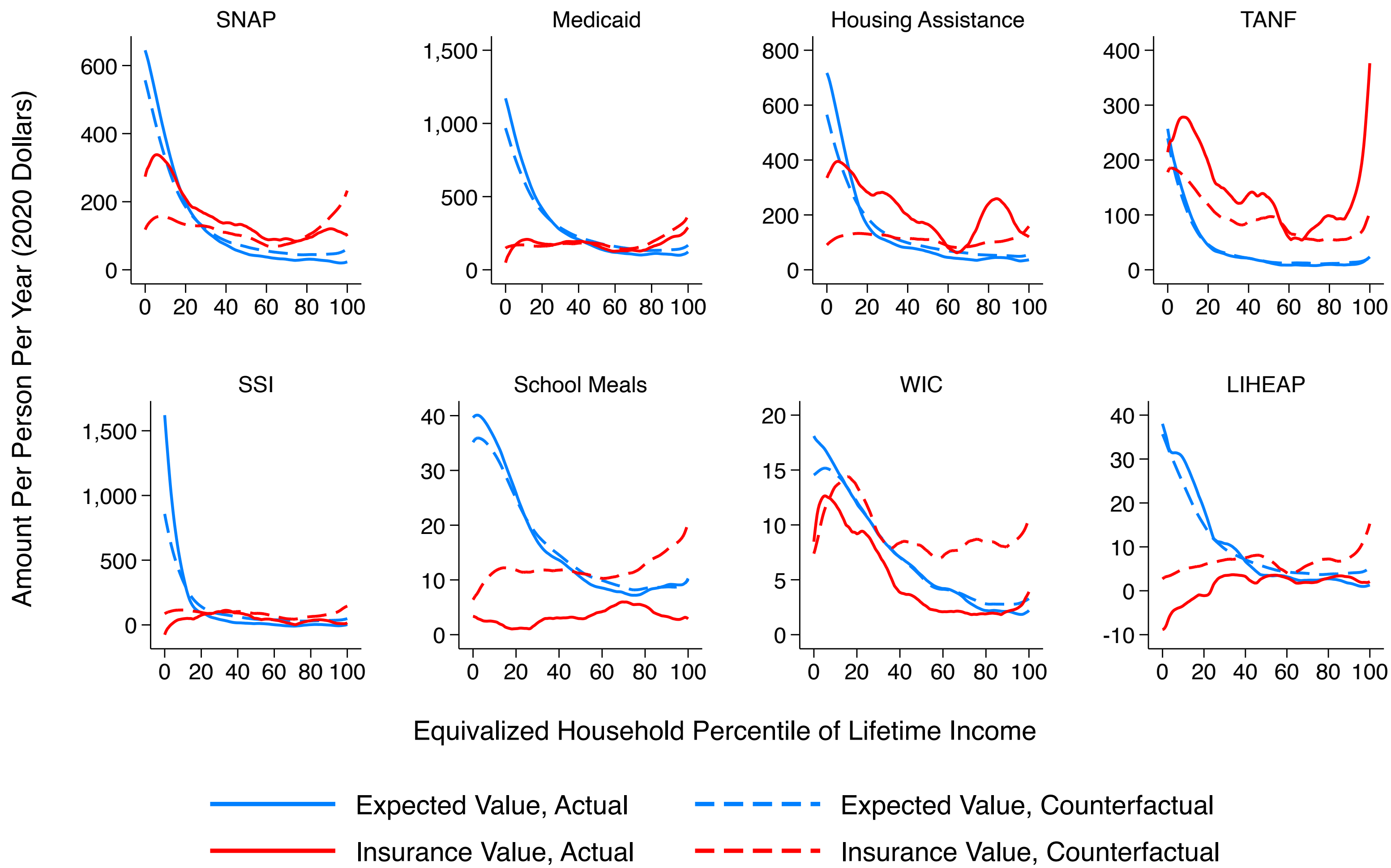
[25]Appendix Figure A16 plots the distributions of the certainty-equivalent values by program. Appendix Figure A18 presents results using different calibrations of risk aversion. Across calibrations, voluntary transfers have greater insurance value than automatic transfers for most programs. However, the levels of insurance value are inherently sensitive to the assumed level of risk aversion.

Figure 2: Expected Value and Insurance Value, Totals Across Programs



Notes: This figure displays the average annual per-capita value of transfer benefits as a functions of lifetime-income rank, equivalized for household size. Totals are reported across all eight programs and for cash transfers only. Blue lines report the expected values of transfers, and red lines report insurance values. Solid lines report existing transfers, and dashed lines report the automatic counterfactuals. The functions are estimated by local linear regressions with bandwidths of five percentiles.

Figure 3: Expected Values and Insurance Values by Program



Notes: This figure displays the average annual per-capita value of transfer benefits by program as a function of lifetime-income rank, equivalized for household size. Blue lines report the expected values of transfers, and red lines report insurance values. Solid lines report existing transfers, and dashed lines report the automatic counterfactuals. The functions are estimated by local linear regressions with bandwidths of five percentiles.

5 Welfare Analysis of Automatic Transfers

This section conducts a welfare analysis of the social costs and benefits of using voluntary take-up to target transfers. We develop a new sufficient-statistics formula for a marginal version of our reform that shifts resources from voluntary to automatic transfers. We then calibrate the formula using our data and external estimates. Proofs of all theoretical results are in Appendix D.

5.1 Basic Environment

To make the economics of the sufficient-statistics result especially clear, we first illustrate it in a basic environment without labor supply or risk aversion. Suppose there is a unit mass of people indexed by $i \in [0, 1]$. Of these, a share M choose to take up a voluntary transfer, receiving benefit s and paying a hassle cost $\kappa(i)$ to do so. The complementary share $1 - M$ do not take up because their ordeal cost strictly exceeds the benefit, $\kappa(i) > s$.

We evaluate the welfare impact of the following budget-neutral "voluntary-to-automatic" benefit reform. In the reform, the government cuts the voluntary benefit by an amount ds. Cutting the benefit causes a share $\frac{M}{s}\varepsilon_b \cdot ds$ to drop out of the transfer, where the take-up elasticity with respect to the benefit amount is $\varepsilon_b = d \log M / d \log s$. The fiscal savings from this behavioral response is $M\varepsilon_b \cdot ds$, which the government spends by giving all people $M(1 + \varepsilon_b) \cdot ds$ dollars automatically. We want to evaluate whether this reform made society better or worse off overall.

Let $\alpha(i)$ be the marginal social welfare weight of household i, setting the population average social value to unity without loss of generality. The welfare effect of the reform is

$$\begin{aligned} \frac{dW}{ds} = &\underbrace{M\mathrm{E}[\alpha(i) \mid \kappa(i) \leq s] \cdot (M(1+\varepsilon_b) - 1)}_{\text{recipients}} \\ &+ \underbrace{(1-M)\mathrm{E}[\alpha(i) \mid \kappa(i) > s] \cdot M(1+\varepsilon_b)}_{\text{nonrecipients}}. \end{aligned} \tag{5}$$

This expression follows from weighting the net changes in payments to recipients and nonrecipients by these groups' respective welfare weights. Ordeal costs do not appear, as they only change for a marginal type who was exactly indifferent to taking up. Equation 5 simplifies to

$$\frac{dW}{ds} = \beta\sigma_M^2 + M\varepsilon_b, \tag{6}$$

where the variance of take-up is $\sigma_M^2 = M(1 - M)$. The coefficient β equals $\mathrm{E}[\alpha(i) \mid \kappa(i) > s] - \mathrm{E}[\alpha(i) \mid \kappa(i) \leq s]$, which is the difference between the marginal social values of income for the recipients and the nonrecipients.[26]

Equation 6 captures our main theoretical result. The first term captures the social benefits of self-targeting. By moving toward automatic transfers, the government diminishes targeting. If take-up identifies higher-welfare-weight people on average, then making transfers automatic has undesirable redistributive properties. The targeting is more socially valuable when take-up is more informative about welfare weights $\alpha(i)$ and thus when β is larger. When all or none take up the transfer, targeting is impossible, explaining why the reform's benefits depend on the variance.

The second term represents the reduction in the social costs of ordeals. In response to the transfer cut, some people no longer take up, since the benefit no longer exceeds their ordeal cost. We apply the envelope theorem to infer these costs. With privately optimal take-up choices, the fiscal savings from people who no longer take up equals the change in the social cost of ordeals.

Example. Suppose there is a voluntary transfer with a 50-percent take-up rate. Imagine that on average, recipients consume 71 percent as much as non-recipients, and the take-up elasticity is 0.5. Society has constant-relative-risk-aversion (CRRA) preferences over consumption with parameter $\gamma = 2$. On the margin, should funds shift from this transfer to an automatic one?

The answer is no. Each dollar taken from the recipients is worth about twice ($1/0.71^2 \approx 2$) that of a dollar given to nonrecipients. Normalizing the average welfare weight to one, given 50-percent take-up, yields a regression coefficient $\beta = 0.75 - 1.5 = -0.75$. The variance of take-up is $\sigma_m^2 = (0.5)(1 - 0.5) = 0.25$. Then $dW/ds = (-0.75)(0.25) + (0.5)(0.5) = -0.0625$, so each transfer dollar ds moved reduces money-metric social welfare by about six cents.

5.2 Setup of Full Model

We next derive a similar formula in a dynamic model of optimal redistribution. The dynamics allow for a concept of consumption that is meaningfully distinct from current income. We also introduce risk aversion and endogenous labor supply, to capture the reform's impacts on insurance and efficiency. The reform is budget-neutral and distribution-neutral with respect to income, but it is generically not neutral for labor supply or insurance value.

Households. Time t is discrete, and households live forever. Each period, a household has assets $a_t \in \mathbb{R}$ and a multidimensional type (w_t, κ_t, ξ_t). The household earns a wage $w_t \in \mathbb{R}^+$, has a cost of transfer take-up $\kappa_t \in \mathbb{R}^+$, and also has a type $\xi_t \in \mathbb{R}$. The conditional distribution of κ_t is assumed to be continuous, that is, without mass points. Households apply a constant discount rate $\rho \in (0, \infty)$ in discounting utility flows over time.

Household heterogeneity beyond wages has two purposes. First, the type ξ_t encodes the

[26]Our welfare formula thus belongs to a "Baily–Chetty" class of sufficient-statistics formulas that contrast a targeting gain (in the form of a difference in social marginal utilities) with a behavioral response (in the form of an elasticity).

persistence or variability of wages and take-up costs over time. This model element allows for heterogeneous dynamics across households in types (w_t, κ_t). An important example is household income processes that vary in both a persistent component and a transitory component.[27]

Second, the take-up cost κ_t sets up the joint choice of labor supply and transfer take-up. Each period, households first choose how much labor $l_t \in \mathbb{R}^+$ to supply to generate labor income $z_t = w_t l_t$. They make this choice knowing their wage w_t but not their take-up cost κ_t beyond its conditional distribution. Let $\theta_t = (w_t, \xi_t)$ represent the household's information set in choosing labor supply. They then draw κ_t and make their take-up choice, which we denote as $\mathbb{1}_S = 1[S(z_t) \geq \kappa_t]$.

Households' current labor incomes are taxed according to the nonlinear schedule $T(z_t) : \mathbb{R}^+ \to \mathbb{R}$. Households divide their after-tax labor income, along with any assets or debts carried into the period, between consumption c_t and net assets a_{t+1}. There is also a voluntary transfer with nonlinear schedule $S(z_t) : \mathbb{R}^+ \to \mathbb{R}^+$. Households value a dollar of automatic transfer as equivalent to a cash dollar, so there is no distinction with income taxes.[28] Under the informational assumptions, the household θ views taxes and transfers as a consolidated system in the labor supply choice. Letting $M(z;\theta) = \Pr(S(z) \geq \kappa \mid \theta)$ denote its take-up probability, the household faces an expected marginal "keep" rate of $1 - T'(z) - M(z;\theta)S'(z)$ with respect to labor income z.[29] Negative taxes on labor income are possible, and capital income is untaxed.

Each period, households choose consumption, labor income, and savings to maximize its ex-ante value function, according to the Bellman equation:

$$V(a_t, \theta_t) = \max_{c_t, z_t, a_{t+1}} \left\{ u(c_t, z_t; \theta_t) + \frac{1}{1+\rho} \mathrm{E}_t [V(a_{t+1}; \theta_{t+1}) \mid a_{t+1}; \theta_t] \right\} \tag{7}$$

subject to period budget constraints

$$c_t + a_{t+1} = z_t - T(z_t) + R_t a_t + \int_0^{S(z_t)} [S(z_t) - \kappa] \, \mu(\kappa \mid \theta_t) \, d\kappa \tag{8}$$

and a borrowing constraint $a_{t+1} \geq \bar{a}_{t+1}(\theta_t)$. The variable R_t is the gross interest rate. Transversality and no-Ponzi conditions are enforced by further imposing $\lim_{t\to\infty}(1+\rho)^{-t} a_{t+1} = 0$.

Our key assumption on preferences over consumption and work hours is that they take the

[27]For instance, introducing the notation ϵ_t where $w_t = \xi_t + \epsilon_t$, we can accommodate income processes like ϵ_t as white noise and ξ_t as a persistent autoregressive process.

[28]Appendix D discusses the implications of valuing in-kind transfers differently than cash. In the model, we also do not explicitly model eligibility rules, treating them as infinite take-up costs. Households therefore do not know more than the eligibility rate at θ_t, although we relax this in our empirical implementation.

[29]These assumptions, particularly on the timing of the realization of κ_t, facilitate tractability. Otherwise, the household's labor supply would depend upon their take-up choice, which is discontinuous and would itself depend on labor supply. These assumptions ensure that the transfer system enters the labor-supply choice smoothly, and in particular through the conditional probability of take-up and the benefit amount.

form in Greenwood et al. (1988): $u(c, z; \theta) = U(c - v(z; \theta); \theta)$. These preferences allow for risk aversion and rule out income effects, as is common in the nonlinear optimal income tax literature. For each household, the choice of hours is one-to-one with labor income z, and we therefore model the household as directly choosing z.

Welfare. We define social welfare as a weighted sum of the ex-ante value functions using the welfare weights $\alpha(\theta)$:

$$W = \max_{T,S} \int_{\Theta} \alpha(\theta) V(a; \theta)\, d\mu(a, \theta),$$

where μ is the joint distribution of assets and types, and the government's instruments are the automatic and voluntary transfer schedules $T(z)$ and $S(z)$. We normalize the population-average welfare weight $\mathrm{E}[\alpha(\theta)]$ to one. When we report money-metric welfare, we rescale W by the population average social marginal value of income $\mathrm{E}[\alpha(\theta) V_a'(a; \theta)]$.

Our theoretical results on self-targeting require the following definition, which relates the social marginal value of income for a given household to the household's type parameters. It aligns with our empirical definition of self-targeting (Equation 3).

Definition 1. *We say that, at an income z, transfer take-up is* advantageously self-targeted *if the social marginal value of income declines in take-up cost:* $\frac{\partial}{\partial \kappa} \mathrm{E}[\alpha(\theta) V_a'(a; \theta) \mid z, \kappa] < 0$.

This definition embeds two motives for targeting some households with transfers: insurance and redistribution. First, risk-averse households vary in their marginal utilities of income $V_a'(a; \theta)$ over time, due to uninsurable shocks to their types (w, ξ). Second, households vary in welfare weights $\alpha(\theta)$, which we introduce following Saez and Stantcheva (2016). We use these weights to penalize inequality in lifetime incomes that goes beyond household risk aversion alone.

For either motive, social benefits may arise when take-up costs κ lead households to make take-up choices that are correlated with the type ξ, over which society may have preferences. A specific example would be if households can smooth consumption, and ξ encodes a household's long-run average wage and w encodes their current wage. At any w, low-ξ households would consume less than high-ξ households. If, all else equal, take-up costs κ are rising in ξ, then selection into take-up on ξ would be advantageous, consistent with our results in Section 3. Intuitively, self-targeting is advantageous when, within sets of households that otherwise look identical to the government, the first to take up transfers are the "most deserving" in the society's judgment.

5.3 Theoretical Result and Interpretation

"Voluntary-to-Automatic" Reform. We consider the following reform, which is a marginal version of the simulation in Section 4. The transfer $S(z)$ is cut by ds at every income z. Taxes at z are cut by $\tau(z) = \bar{M}(z) ds$, which is equivalent to providing an automatic transfer of $\bar{M}(z) ds$ valued at par with cash.[30] By consequence, people at each income level are compensated on average for the transfer cut. However, marginal rates have changed by $\tau'(z) = \frac{d}{dz}\bar{M}(z)ds$ at z. Fiscal savings from marginal transfer recipients are redistributed as a lump sum to all households, and the revenue cost of the labor supply response is financed by lump-sum taxes.[31]

We see this specific reform as the natural way to quantify the welfare gains from self-targeting. First, by fixing the transfer's budget, the reform focuses attention on optimal design within a program, rather than the program's merits versus another use of funds. Second, as in Kaplow (2011), requiring neutrality with respect to the income distribution removes incidental impacts of the policy change on the overall progressivity of taxes and transfers. Our welfare calculations thus do not reflect changes in progressivity that could, in principle, be achieved by income taxes alone. Third, a flat change in the transfer has an intuitive real-world analog: changing a fully-voluntary transfer into one with a small automatic transfer with a large top-up provided upon application.

The next proposition presents the welfare effect of this marginal shift toward automatic transfers.

Proposition 1. *The change in welfare from the reform is*

$$\begin{aligned}\frac{dW}{ds} = &\underbrace{\beta\sigma_M^2}_{\text{lost value of self-targeting}} + \underbrace{\bar{M}\bar{\varepsilon}_b}_{\text{fiscal savings from marginals}} \\ &+ \underbrace{\int_z \frac{\bar{M}'(z)z\bar{\varepsilon}_\tau(z)}{1-T'(z)}\frac{d}{dz}\Big(S(z)\bar{M}(z) - T(z)\Big)\,dH(z)}_{\text{labor-supply effect}},\end{aligned} \tag{9}$$

where β is the coefficient on take-up from a regression of welfare weights on take-up controlling for income, $H(z)$ is the income distribution, $\bar{M}(z) = \int_\xi M(z;\theta)\,d\mu(\xi|z)$ is the take-up rate at income z, $\bar{M} = \int_z \bar{M}(z)dH(z)$ is the overall take-up rate, $\sigma_M^2 = \int_z \bar{M}(z)(1-\bar{M}(z))dH(z)$ is the within-income variance in take-up, $\bar{\varepsilon}_b$ is an average take-up elasticity with respect to benefit size, and $\bar{\varepsilon}_\tau(z) = \int_\xi \frac{1-T'(z)}{z}\frac{\partial z}{\partial(1-\tau)}(\theta)\,d\mu(\xi|z)$ is the elasticity of income with respect to a small change τ' in the marginal tax rate of those with initial income z, as in Jacquet and Lehmann (2014).

Equation 9 shows the welfare result from the basic environment carries over into the Mirrleesian setting with a labor-supply choice. A regression coefficient still summarizes the targeting benefits, and a take-up elasticity still summarizes the ordeal costs. We show formally in Appendix D that, when self-targeting is advantageous and $S(z)$ is positive, the first term in Equation 9 is negative. That is, society loses some benefits of self-targeting to move toward automatic transfers.

[30] We discuss the parity assumption below, and Appendix D relaxes it formally.

[31] In a more general class of marginal transfer reforms, one would have to account for redistribution both between and across incomes, fiscal savings from marginal recipients, and labor-supply effects. With non-marginal changes to voluntary transfers, one cannot apply the envelope theorem to reveal ordeal costs.

The welfare-relevant coefficient β is a weighted comparison of marginal social value of income of recipients and similar nonrecipients. In particular, $\beta = \Delta\mathrm{E}[\alpha(\theta)V_a'(a;\theta)]$, where α is the social welfare weight, V_a' is the marginal utility of income, r is the lifetime-income rank, and z is the current-income rank. The difference operator is defined to be $\Delta\mathrm{E}[y] = \int_z w(z)(\mathrm{E}[y \mid z, D = 1] - \mathrm{E}[y \mid z, D = 0])\, dH(z)$ for some outcome y, income z, and transfer receipt D. The weights are according to the variance of receipt: $w(z) = \bar{M}(z)[1 - \bar{M}(z)] / \int_z \bar{M}(z)[1 - \bar{M}(z)]\, dH(z)$.

Yet there are some differences with the basic environment. First, we incorporate insurance value. Second, the formula adds a term to account for the fiscal impact of the labor supply response to the reform. As we prove in Appendix D, the term is negative when the tax system is optimal and take-up decreases in income ($\bar{M}'(z) < 0$). Under these assumptions, the automatic reform requires higher marginal tax rates to offset the cut to the voluntary transfer, reducing labor supply.

Interpretation. We make four observations about how we understand this theoretical result, related to program scale, insurance value, ordeal costs, and its application to in-kind transfers.

First, the welfare impact of our reform varies in an intriguing way with the take-up rate. Up to the labor-supply effect, the relationship is U-shaped. Holding fixed the self-targeting coefficient β and the take-up elasticity $\bar{\varepsilon}_b$, there is a unique take-up rate ($M^* \approx \frac{1}{2}(1 + \bar{\varepsilon}_b/\beta)$) at which dW/ds hits a minimum (i.e., in which shifting toward an automatic transfer has the worst welfare impact). Therefore, at a lower or higher take-up rate than M^*, the reform would be more desirable. The easier intuition for this point is that universal take-up has no targeting benefits, but take-up is costly. The harder intuition is that take-up can also be "too low": Inducing the very first person to take up has no social value, as by the envelope theorem, their take-up cost exactly equals the benefit level at which they take up. That is, only after the first person takes up are there any inframarginal households and thus is any first-order welfare gain possible.

Second, Proposition 1 facilitates distinctions between the redistribution-related and insurance-related welfare benefits. Holding fixed the other sufficient statistics, risk aversion enters only through the self-targeting coefficient β. Our decomposition is as follows:

$$\beta = \underbrace{\Delta\mathrm{E}\left[\alpha\mathrm{E}[V_a' \mid r]\right]}_{\text{redistribution}} + \underbrace{\Delta\mathrm{E}\left[\alpha\mathrm{E}[V_a' \mid r]\left(\frac{V_a'}{\mathrm{E}[V_a' \mid r]} - 1\right)\right]}_{\text{insurance}}, \tag{10}$$

where r is the lifetime income rank. This decomposition follows our approach in Section 4 of distinguishing expected payments and insurance, but it further incorporates welfare weights, as the purpose is welfare analysis. That is, households can be currently needy (high $\alpha V_a'$) either because they have high need in all states of life (high $\alpha\mathrm{E}[V_a'|r]$) or they are high current need relative other states (high $V_a'/E[V_a'|r]$).

Third, our reform differs fundamentally from changes to ordeals. We instead take the ordeal as given and reallocate resources between voluntary and automatic transfers. Our welfare formula therefore weighs the value of transfers to inframarginal recipients against ordeal costs to *marginal* recipients that disenroll when the voluntary transfer is cut. By contrast, the welfare analysis of ordeal reforms weighs the change in ordeal costs to *inframarginal* recipients against the fiscal externalities from changes in take-up (e.g., Finkelstein and Notowidigdo, 2019).

A virtue of our reform is that the welfare-relevant measure of ordeal costs is obtained by the envelope theorem. These are otherwise difficult to measure. However, envelope-based estimates of ordeal costs require households to make transfer take-up decisions optimally. This assumption might be seen as problematic, since research has found non-optimizing behavior in take-up (Bhargava and Manoli, 2015; Finkelstein and Notowidigdo, 2019; Anders and Rafkin, 2022). Yet the optimizing assumption works against our conclusion, as it yields upper bounds on ordeal costs. If households do not take-up because of mis-optimization or a lack of information, then ordeal costs would be smaller than what is implied by equating them to marginal benefits. Transfers would then achieve advantageous self-targeting at a lower real resource cost than if households were optimizing.

Fourth, our framework immediately extends to in-kind transfers, or other transfers that households may value differently from their cost of provision. Suppose households are willing to pay $\lambda < 1$ for each transfer dollar. Our expression for dW/ds from Proposition 1 would then be multiplied by λ, as we show in Proposition 2 in Appendix D. The welfare effect dW/ds remains correct in units of households' willingness-to-pay for the transfer, rather than in dollars. The parameter λ simply rescales the welfare effect and cannot reverse its sign.

5.4 Quantification

Due to the trade-off between self-targeting and ordeal costs, the welfare effect of shifting between voluntary and automatic transfers is ambiguous. To estimate the welfare effect, we calibrate Equation 9 using our results and several external inputs (see Appendix D for calibration details). We discuss where our welfare calculations are more and less sensitive to assumptions, as there is the room for reasonable disagreement in the calibration of the external inputs and some simplifications.

Calibration. From the PSID, we obtain receipt rates $\bar{M}(z)$, average benefits $S(z)$, and the income distribution $h(z)$. The first sufficient statistic for targeting, the receipt-rate variance σ_M^2, follows from our estimates of $\bar{M}(z)$.

The second sufficient statistic, the regression coefficient β, requires us to set welfare weights and the household's period utility function. In our primary estimates, we assume the planner has CRRA preferences over consumption with parameter $\gamma_{\text{sp}} = 1$. For the household, we follow Greenwood et al. (1988) and use a CRRA form for household period utility: $u(c, l; \theta) = \frac{1}{1-\gamma}(c - \psi \frac{l^{1+1/\eta}}{1+1/\eta})^{1-\gamma}$ for Frisch elasticity η. We continue use a risk-aversion parameter $\gamma = 2$ (Chetty and Finkelstein, 2013). We also set the labor supply elasticity $\eta = 0.3$ (Saez et al., 2012), and we draw on the PSID microdata for household equivalized consumption as c and labor hours as l. These choices allow us to internally calibrate each household's labor disutility parameter as $\psi = w/l^{1/\eta}$.

A higher γ_{sp} or γ would respectively strengthen the redistributive and insurance motives, favoring voluntary transfers if they improve targeting. Having chosen these parameters, we use the joint distribution of income and consumption to compute the welfare weight for each household. We then estimate differences in welfare weights between transfer recipients and non-recipients conditional on income, similar to the self-targeting analysis in Section 3.

We set the take-up elasticity $\bar{\varepsilon}_b$ to 0.6, the upper end of the range in Krueger and Meyer (2002)'s review of take-up elasticities for unemployment insurance.[32] We also assume a constant elasticity of taxable income $\bar{\varepsilon}_\tau(z) = 0.3$, again following Saez et al. (2012). We calibrate a piecewise-linear tax schedule $T(z)$ using effective average marginal tax rates that incorporate federal and state taxes on income and payroll (Congressional Budget Office, 2015).

Results. Panel A of Table 3 reports our primary estimates of the welfare effects of reallocating resources from a given voluntary transfer to an automatic one. Columns 1 and 2 show the social costs from giving up some self-targeting, decomposing the first term in Equation 9 into redistribution and insurance via Equation 10. Column 3 shows the fiscal savings on marginal households who exit the transfer (the second term in Equation 9). Due to the envelope argument explained above, Column 3 can also be interpreted as the social savings on ordeal costs among marginal households. Column 4 shows the labor-supply effect (the sum of the third and fourth terms in Equation 9). Column 5 shows the total effect of the reform.

To take SNAP as an example, we find that making SNAP more automatic forgoes some of the social benefits of self-targeting. On the margin, self-targeting is worth about 16.3 cents per dollar of SNAP. About 4.5 cents of this benefit is attributed to between-lifetime redistribution and 11.8 cents to within-lifetime insurance. In per-recipient terms, self-targeting yields a social benefit of about \$480 per year, an amount that we see as obviously unlikely to be offset by ordeal costs. By comparison, the government saves 5.9 cents per SNAP dollar from marginal households who exit when the voluntary benefit is cut. Finally, the automatic transfer increases marginal tax rates, which reduces labor supply and imposes a 1.1 cent fiscal externality. Together, the net effect of making SNAP more automatic on the margin is a net social loss of 11.5 cents per dollar. The magnitudes of welfare effects seem consequential, especially for a budget-neutral reform.

Overall, we find a stark trade-off between the social benefits of self-targeting and the social costs of ordeals. Looking across transfers, social benefits are often equal to or greater than our

[32] The upper end of this range is conservative for our analysis in the sense that it favors automatic transfers. McGarry (1996) likewise finds a take-up elasticity for SSI benefits of 0.5. We are not aware of estimates of take-up elasticities with respect to benefit level for other U.S. transfers, or newer estimates for the same transfers.

upper-bound estimates of social costs. By consequence, the net social gains from making transfers automatic tend to be be negative or small, and they are not well-approximated by the social savings on ordeal costs alone. That result is reflected in the dollar-weighted average, which shows that on the margin the forgone social benefits of self-targeting actually exceed the social costs of ordeals. In summary, the value of self-targeting appears to be a credible argument for the status quo of voluntary transfers and against automatic transfers.

Across programs, we attribute most of the social gains from self-targeting to better insurance rather than deeper redistribution. This may be surprising in light of Section 4, where differences between voluntary and automatic transfers seem mostly about expected payments and not insurance. Insurance value greatly shapes the social value of the reform mostly due to the budget constraint on expected payments. That is, welfare gains from more transfers to the lifetime-poorest are undone by welfare losses from transfer cuts elsewhere. This logic does not apply to insurance value.

There is also considerable heterogeneity in welfare effects across programs. Ordeals in some transfers seem ineffectual: that is, they have social costs but do not induce socially valuable self-targeting. For example, our results suggest potential welfare gains from universal free school meals. Automatic benefits, by contrast, appear highly costly in housing-assistance programs. Importantly, these programs have severe ordeals: low-quality and constrained choices, as well as long waiting lists. Our framework is thus not uniformly favorable towards ordeals but makes finer distinctions according to how effective an ordeal is in causing self-targeting.

Table 3: Welfare Effects of Making Transfers Automatic (Cents per Transfer Dollar)

	Self-Targeting Gains		Other Forces		Total
	Redistribution	Insurance	Upper Bound on Ordeals	Labor-Supply Effects	
	(1)	(2)	(3)	(4)	(5)
Panel A: Primary Estimates					
Dollar-Weighted Average	**-3.0**	**-5.7**	**6.3**	**-1.2**	**-3.6**
SNAP	-4.5	-11.8	5.9	-1.1	-11.5
Medicaid	-3.3	-4.8	8.6	-1.6	-1.1
Housing Assistance	-3.2	-10.2	3.1	-0.7	-11.0
TANF	-1.1	-5.9	0.6	-0.1	-6.4
SSI	-1.1	2.0	3.5	-0.6	3.7
School Lunch	-2.1	-2.8	6.1	-1.4	-0.2
WIC	-1.0	-3.0	2.4	-0.5	-2.0
LIHEAP	-0.7	0.7	2.3	-0.4	1.9
Panel B: Sensitivity (Dollar-Weighted Average)					
Utilitarian social preferences ($\gamma_{\text{sp}} = 0$)	-2.5	-4.4	6.3	-1.2	-1.9
Less progressive social preferences ($\gamma_{\text{sp}} = \frac{1}{2}$)	-3.1	-4.9	6.3	-1.2	-3.0
More progressive social preferences ($\gamma_{\text{sp}} = 2$)	-2.4	-6.8	6.3	-1.2	-4.2
Negishi social preferences	-0.0	-3.4	6.3	-1.2	1.7
Restrict to eligibles only	-9.1	-21.0	23.0	-3.4	-10.5
Less risk averse ($\gamma = 1$)	-3.3	-3.6	6.3	-1.2	-1.9
More risk averse ($\gamma = 4$)	-3.1	-6.3	6.3	-1.2	-4.4
Take-up elasticity $\eta = 0.3$	-3.0	-5.7	3.1	-1.2	-6.8
Take-up elasticity $\eta = 1.2$	-3.0	-5.7	12.5	-1.2	2.6
Elasticity of taxable income $\varepsilon = 0.15$	-3.1	-5.7	6.3	-0.6	-3.1
Elasticity of taxable income $\varepsilon = 0.6$	-3.0	-5.6	6.3	-2.4	-4.7

Notes: This table reports estimates of the welfare effects of the reform, which marginally reduces the voluntary transfer to make it automatic. We calibrate the welfare weights by assuming a CES social welfare function with curvature parameter $\gamma = 2$. We calibrate the fiscal cost of marginals by assuming the takeup elasticity is $\bar{\varepsilon}_b = 0.6$. We calibrate the elasticity of taxable income at $\varepsilon_\tau = 0.3$. All columns report the money-metric welfare gains in cents per transfer dollar. Columns correspond to the terms of Equation 9, where we divide each term by the average social marginal utility, which yields a money-metric interpretation.

Sensitivity Analysis and Limitations. Panel B examines the sensitivity of our results to the calibrated parameters.

The more risk-averse are households, and the more society cares about redistribution, the larger are the welfare losses from forgoing self-targeting in transfers. Put another way, automating transfers is likely to be desirable only when people care about consumption fluctuations or when society cares less about lifetime inequality. A society with "Negishi" preferences (no gains from redistributing lifetime income) would automate transfers.[33]

The take-up elasticity is critically important to fiscal costs and thus the implied ordeal costs. If take-up is more highly responsive to the benefit level than we expect, this would imply larger ordeal costs on the margin and thus could motivate automatic transfers. Results are less sensitive to the elasticity of taxable income. Across these permutations of our analysis, self-targeting remains a quantitatively important advantage of voluntary transfers. Indeed, self-targeting typically eliminates most if not all of the social savings on ordeals, even at upper-bound values for ordeal costs.

We also estimate the welfare impacts of our reform exclusively among the simulated-eligible. Both the social benefits and costs grow. The social benefits grow because the eligible have high average social welfare weights, so moving dollars within this population is more consequential. The costs grow because take-up rates are far higher than receipt rates unconditional on eligibility, which raises both ordeal costs and the labor-supply impacts.

Valuing transfer dollars differently from cash, such as for in-kind benefits, would scale all entries in Table 3 by the willingness to pay for the transfer per dollar cost (Appendix D). For example, if people value a dollar of Section 8 housing vouchers at 80 cents (as in Reeder, 1985), then the overall welfare loss per transfer dollar in our reform is 8.8 cents (= 11.0×0.8).

This welfare analysis has several limitations. First, the above reform is fictitious, as transfer reforms do not need to be marginal or budget-balanced by cutting spending within the program. We consciously ignore many practicalities of implementing automatic transfers, which may be quite important (see, e.g., Wu and Meyer, 2024). Thus, our analysis can only say so much about the welfare impact of real-world policy proposals, like fully converting SNAP to be automatic and financing the reform via the deficit. These policies involve additional parameters including inframarginals' ordeal costs, reduced stigma from fully-automatic programs, and the marginal cost of public funds. Second, the analysis does not account for differences in the government's administrative costs between voluntary and automatic transfers. Little is known about the appropriate values for these costs (Isaacs, 2008), but they likely favor automatic transfers. Third, we ignore behavioral responses to transfers beyond take-up and labor supply, such as cross-program take-up spillovers or dynamic

[33]For each lifetime-income rank, the Negishi weight is the inverse of the average marginal utility of income. The comparative static with social preferences runs opposite to current U.S. political debates, where left-leaning people often argue for automatic transfers, perhaps because they ignore budget balance or distributional neutrality.

incentives for human-capital investment. Fourth, we assume homogeneous labor supply elasticities, which precludes consideration of self-targeting on elasticities rather than levels.

6 Conclusion

A large body of empirical research has studied many specific ordeals in transfer programs. It finds mixed evidence that, on the margin, these ordeals have favorable selection properties. Taken as a whole, this literature would seem to have radical implications for the design of social safety nets: Why do governments hassle people by making them ask for help, if those who do not ask are no less in need? Why not just send help automatically?

What is true among the complier population for the studied ordeals does not, we find, generalize to always- and never-takers of transfer programs. We show transfer recipients are, relative to eligible non-recipients with similar incomes, highly negatively selected on consumption and lifetime income. This self-targeting raises not only the redistributive impact, but also the insurance value, of voluntary transfers by comparison to automatic versions of the same programs.

Self-targeting might rescue the case for voluntary take-up. To determine if it does, we quantify the social trade-off between self-targeting and ordeal costs. In particular, we examine reforms that incrementally shift redistribution from voluntary to automatic transfers. Calibrating a welfare-effect formula using our empirical estimates, we find that the social benefits from self-targeting generally equal or exceed upper-bound estimates of the social costs of ordeals. There is, by consequence, a credible case for imposing take-up costs on eligible people. However, some transfers impose costs but achieve minimal self-targeting, and in these programs, the U.S. could possibly achieve welfare gains by shifting away from take-up.

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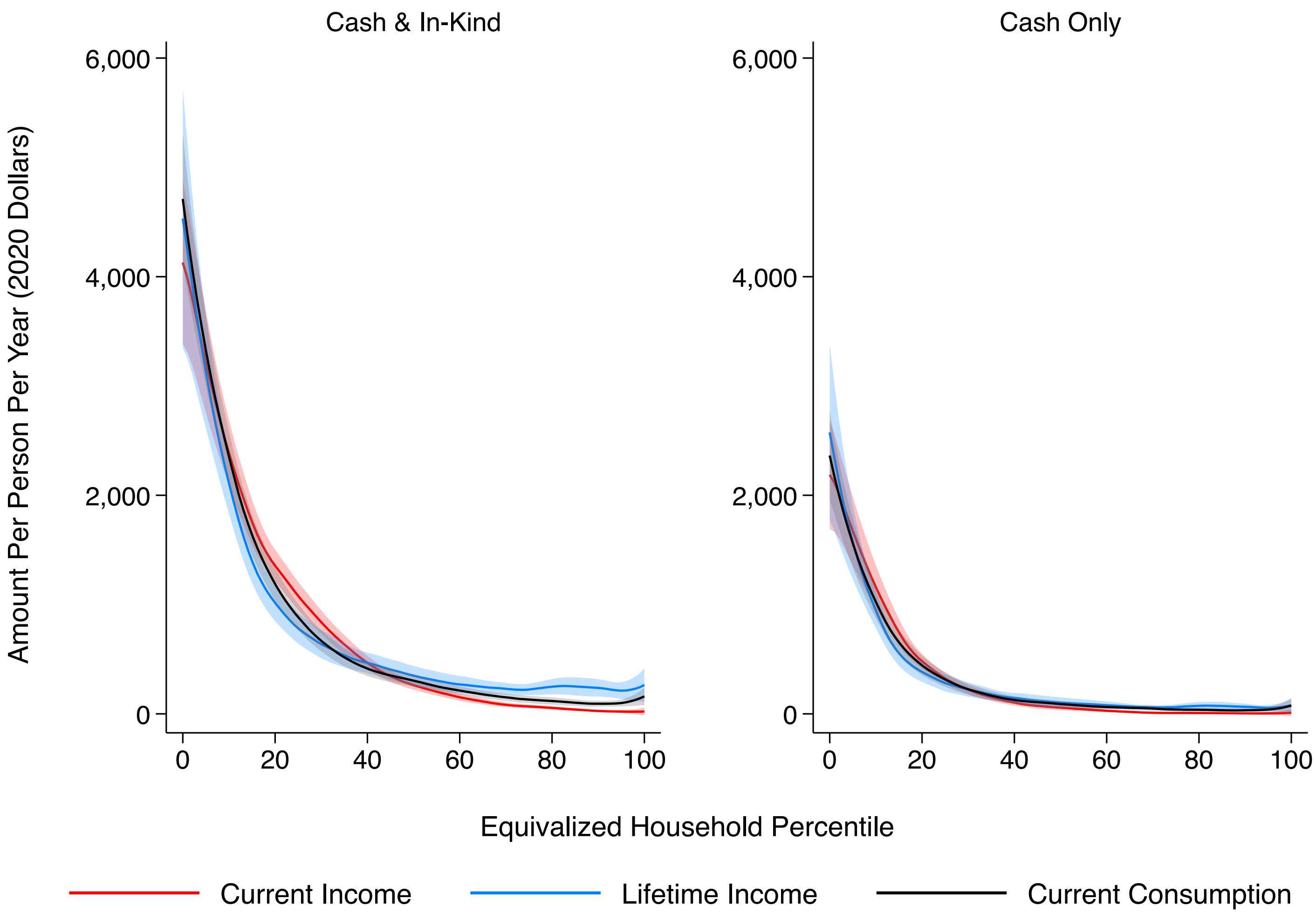
Appendices for Online Publication

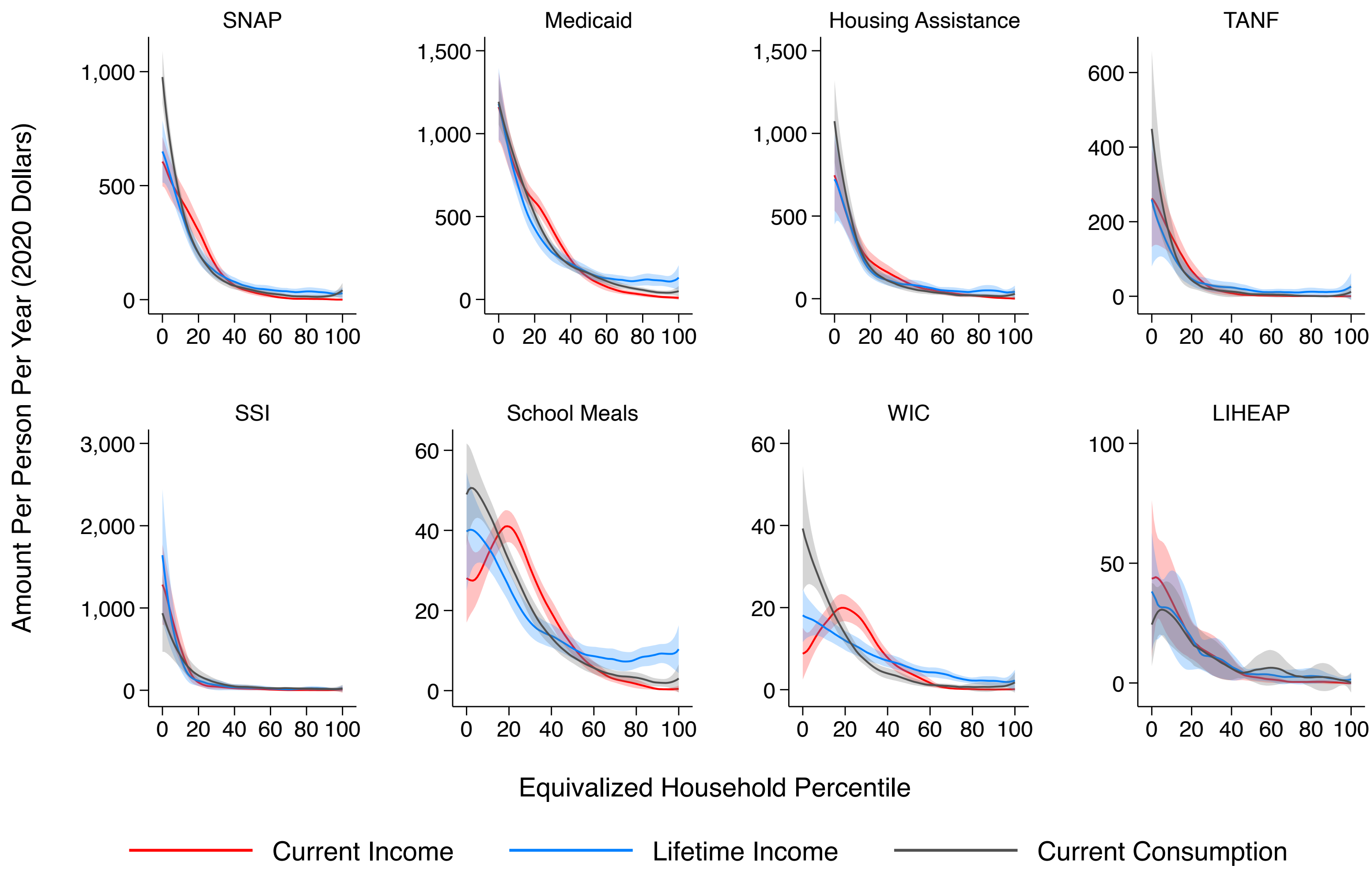
A Additional Tables and Figures

Figure A1: Average Total Annual Per-Capita Benefits



Notes: This figure displays average total annual per-capita dollar amounts as functions of household rank in the distributions of equivalized current income, lifetime income, and consumption. The functions are estimated by local linear regressions with bandwidths of five percentiles. Shaded regions reflect bootstrapped 95-percent simultaneous confidence bands, as in Chernozhukov et al. (2013), with clustering by household.

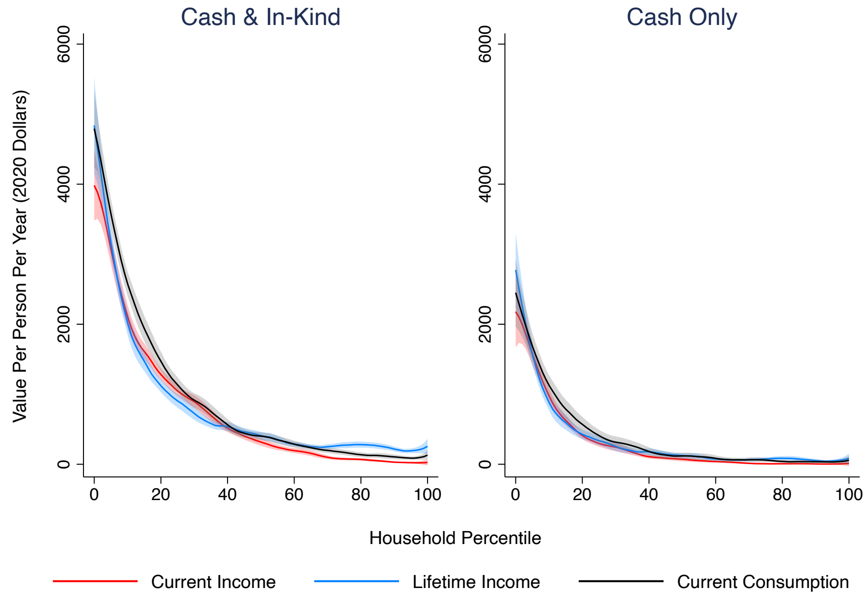
Figure A2: Average Annual Per-Capita Benefits by Program



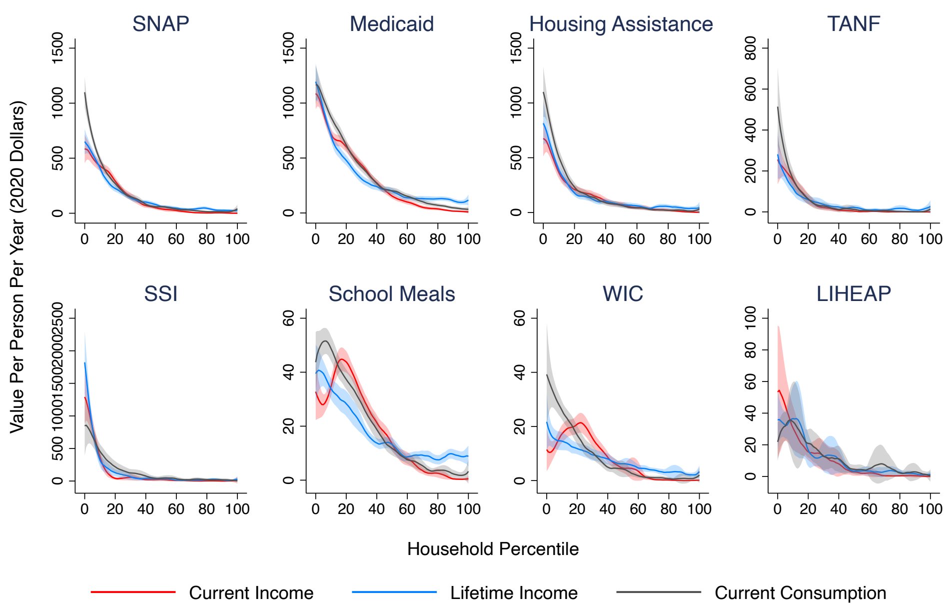
Notes: This figure displays the average annual per-capita value of transfer benefits by program as functions of household rank in the distributions of equivalized current income, lifetime income, and current consumption. The functions are estimated by local linear regressions with bandwidths of five percentiles. Shaded regions reflect bootstrapped 95-percent simultaneous confidence bands, as in Chernozhukov et al. (2013), with clustering by household.

Figure A3: Receipt and Value of Transfer Benefits as a Function of Household Rank

Panel A: Average Total Annual Per-Capita Transfer

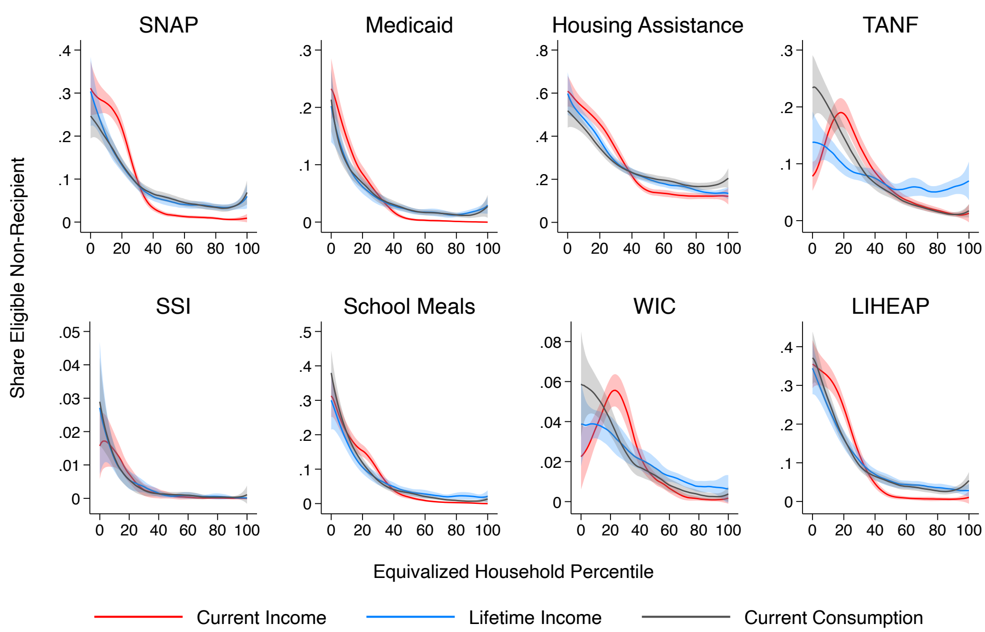


Panel B: Average Annual Per-Capita Transfer by Program



Notes: This figure displays average annual per-capita values of benefits, in total and by program, as functions of household ranks, in the distributions of household current income, lifetime income, and current consumption. There is no equivalence scale applied to household income. The functions are estimated by local linear regressions with bandwidths of five percentiles. Shaded regions reflect bootstrapped 95-percent simultaneous confidence bands, as in Chernozhukov et al. (2013), with clustering by household.

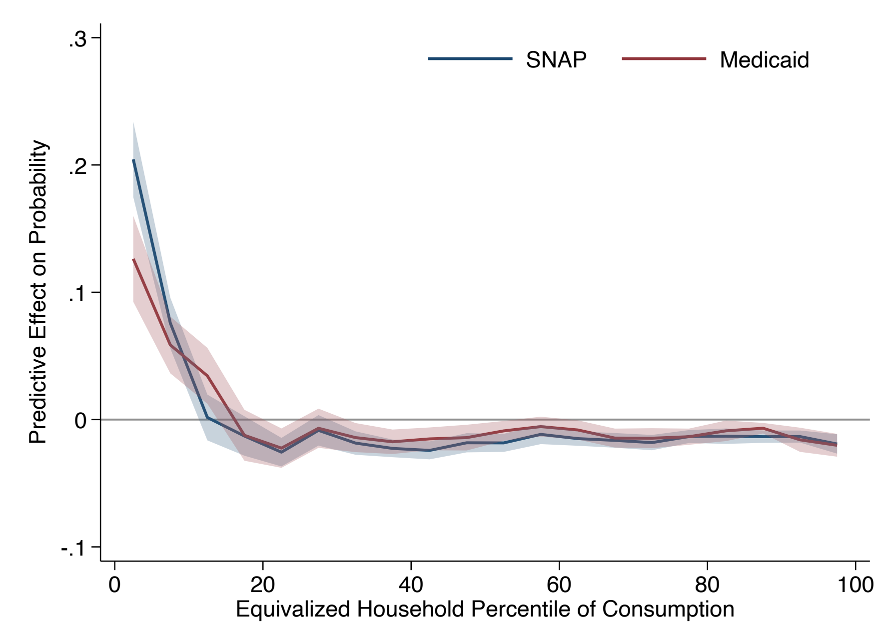
Figure A4: Eligible Non-Receipt Rates by Program



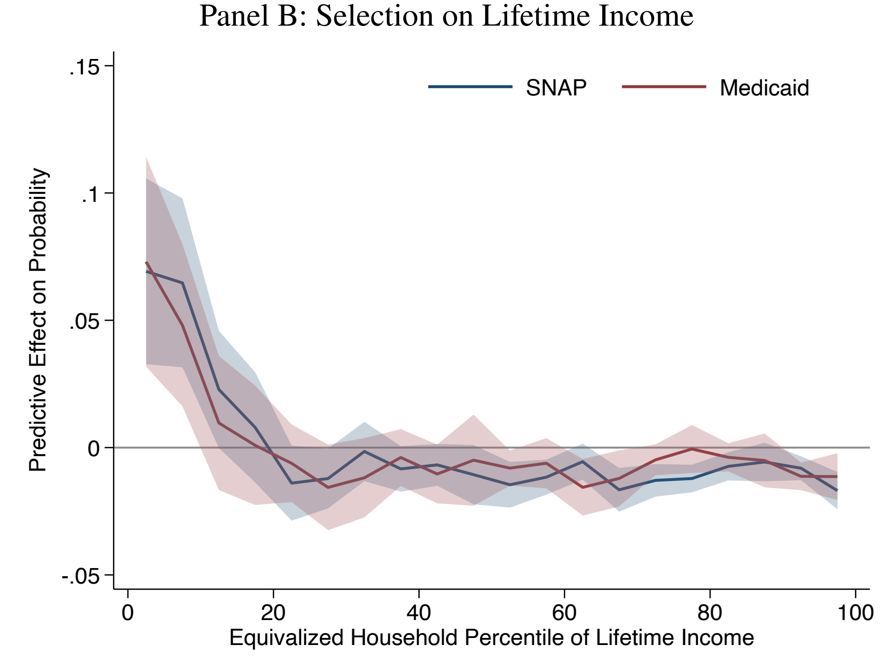
Notes: This figure displays the shares of households that are eligible nonrecipients of a transfer as functions of their ranks in the distributions of equivalized current income, lifetime income, and consumption. The functions are estimated by local linear regressions with a bandwidth of five percentiles. Shaded regions reflect bootstrapped 95-percent simultaneous confidence bands, with clustering by household.

Figure A5: Distributional Decompositions of Self-Targeting by Consumption Ventile

Panel A: Selection on Consumption



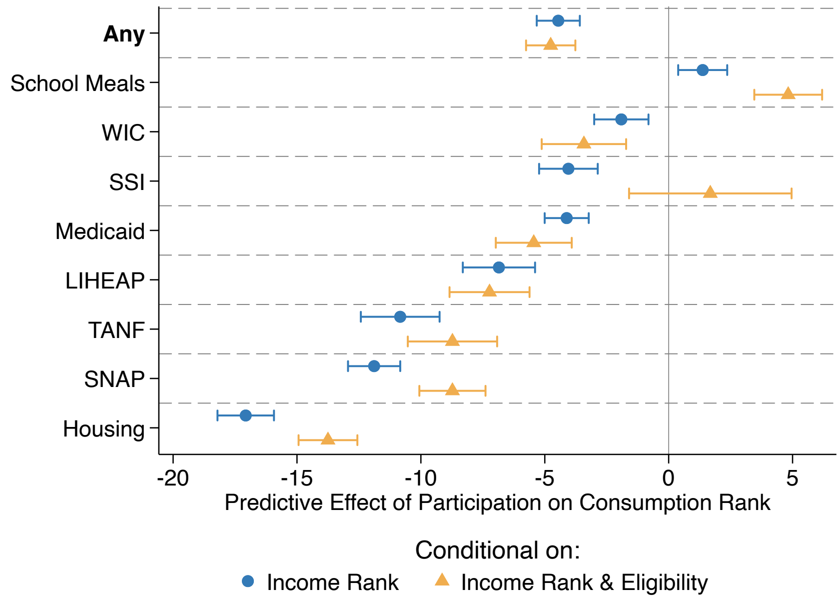
Panel B: Selection on Lifetime Income



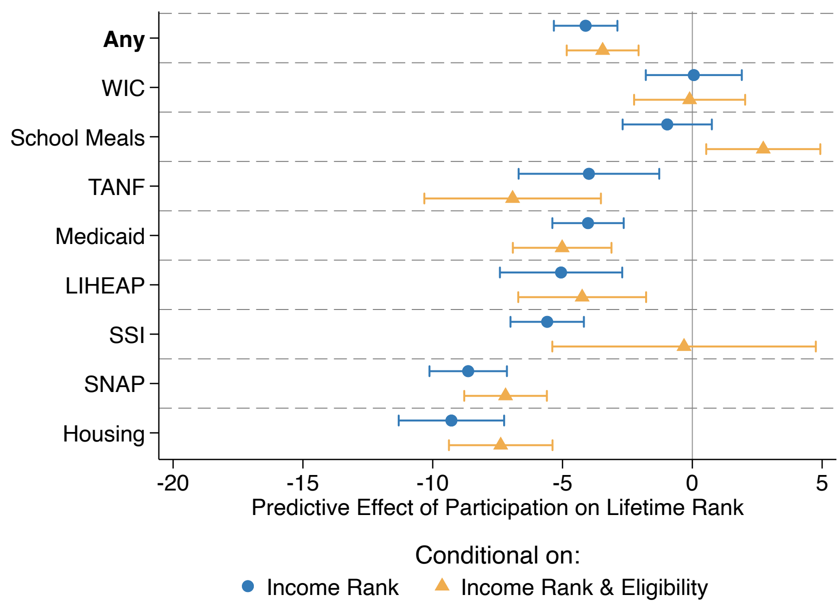
Notes: This figure displays estimates of the predictive effect of transfer receipt on consumption rank (Panel A) or lifetime-income rank (Panel B), conditional on current-income rank (coefficient γ from Equation 4). We estimate this specification using ventiles of consumption or lifetime-income ranks as outcomes—that is, each point is a coefficient from a separate regression with an outcome $1(\bar{R}_i \in [a, b))$. In both panels, 95-percent confidence intervals reflect clustered standard errors by household.

Figure A6: Self-Targeting in Transfer Programs (Non-Equivalized Households)

Panel A: Selection on Consumption



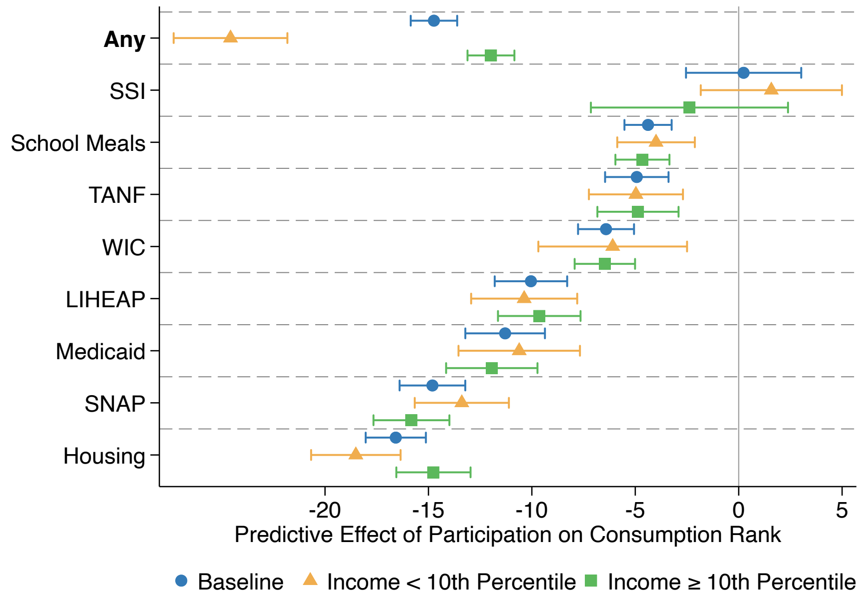
Panel B: Selection on Lifetime Income



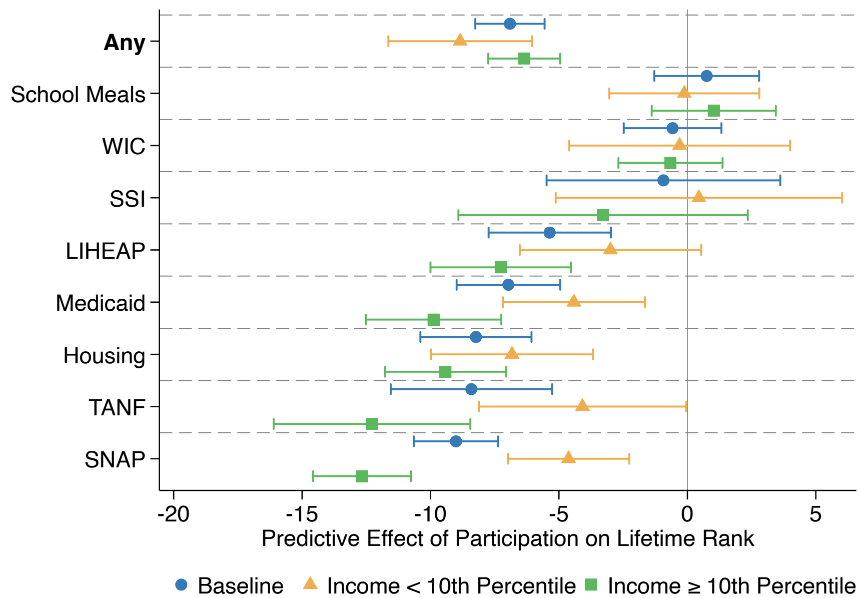
Notes: This figure displays estimates of the predictive effect of transfer receipt on consumption rank (Panel A) or lifetime-income rank (Panel B), conditional on current-income rank (coefficient γ from Equation 4). No equivalence-scale adjustment is applied in ranking households on consumption and lifetime income. For the yellow diamonds, we estimate the regression only on people whom we simulate to be eligible. Panel C augments the specification with person-level fixed effects. The “any” row of Panels A–C is an indicator for receipt of at least one of the eight transfers. Panel D adapts Equation 4 by replacing the transfer indicator with indicators for the number of transfers received in that year. In all panels, 95-percent confidence intervals reflect clustered standard errors by household.

Figure A7: Heterogeneous Self-Targeting by Current Income Rank

Panel A: Selection on Consumption



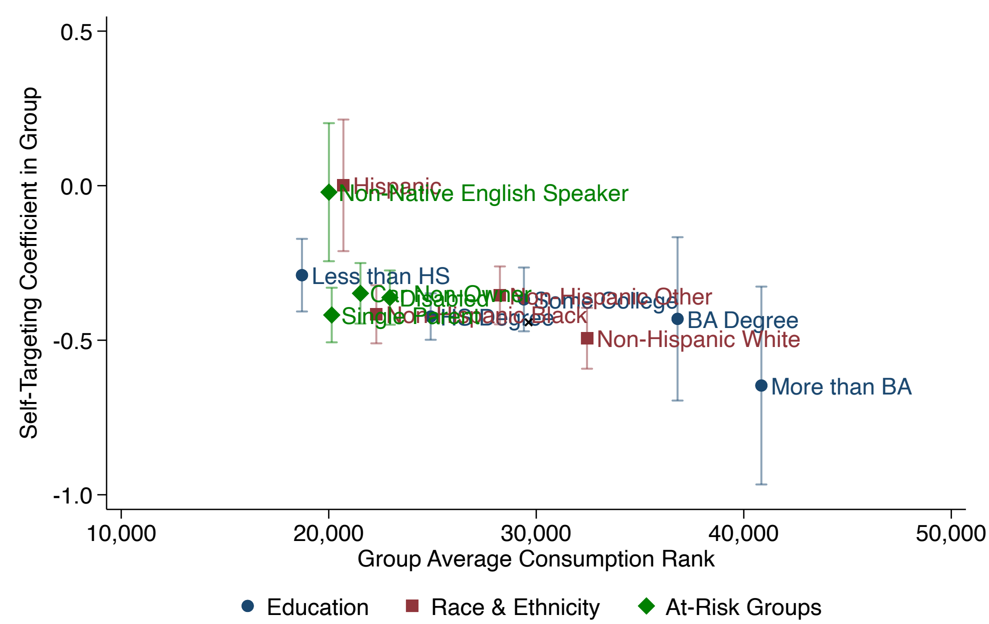
Panel B: Selection on Lifetime Income



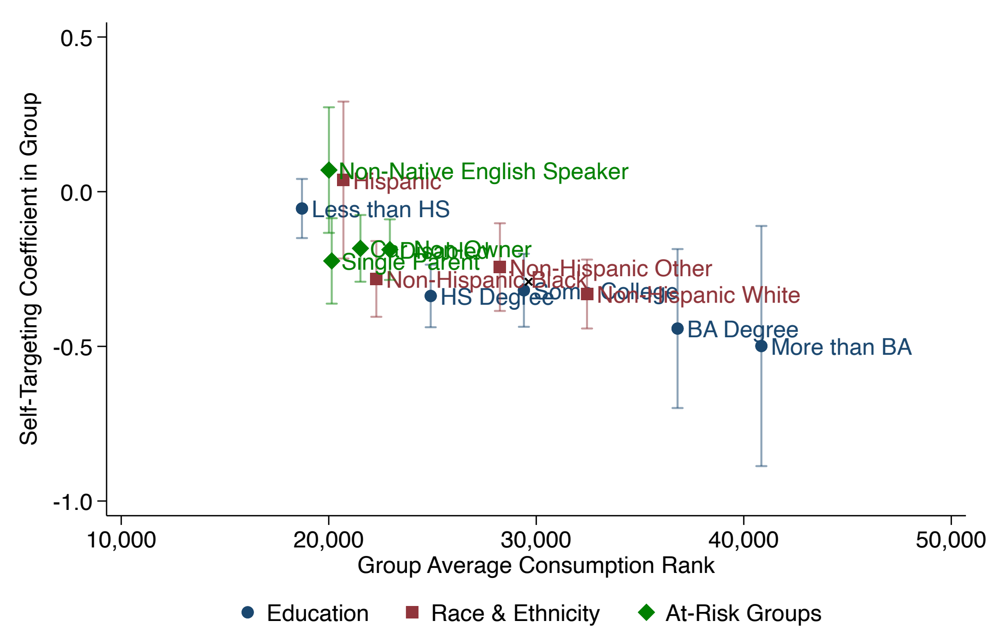
Notes: This figure displays estimates of the predictive effect of transfer benefit receipt on consumption rank or lifetime-income rank, conditional on current-income rank (coefficient β from Equation 4). Estimates in blue circles repeat our main estimates in Figure 1. For estimates in yellow triangles or green squares, we respectively split the sample at the tenth percentile of the distribution of equivalized current household income. The confidence intervals are at the 95-percent level with clustering by household.

Figure A8: Demographic Heterogeneity in Self-Targeting

(a) SNAP



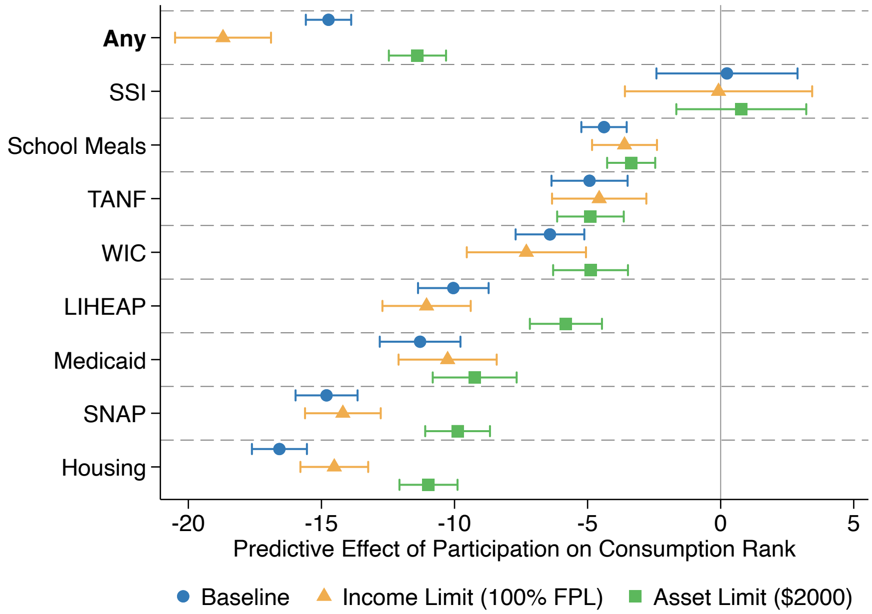
(b) Medicaid



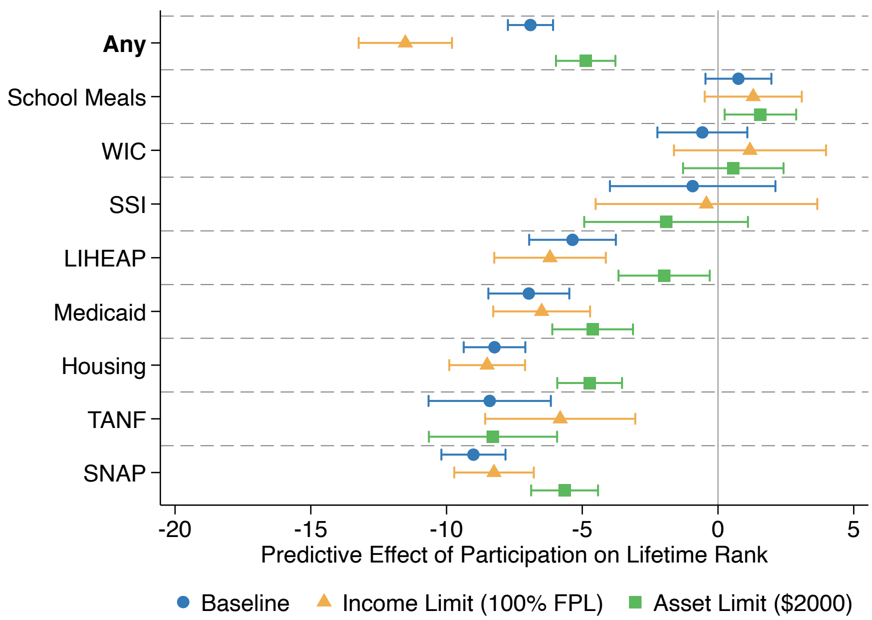
Notes: This figure displays estimates of the predictive effects of transfer receipt on the level of consumption, conditional on current-income rank, that are specific by demographic group. We split the sample by education (less than HS, HS degree, some college, BA only, more than BA), race/ethnicity (non-Hispanic white, non-Hispanic black, Hispanic, non-Hispanic other), and by "at-risk" group (single parent, disabled, non-native English speaker, car non-owner) We use a Poisson regression version of Equation 4, estimated on the simulated eligible only. The motivation for this specification is that we found our rank–rank version of otherwise variation generates spurious heterogeneity in self-targeting due to variation in the baseline consumption rank across groups and the resultant "boundary effects" near the zeroth percentile. Panels A and B are for SNAP and Medicaid respectively. The confidence intervals are for the 95-percent level and reflect clustered standard errors by household. The black "x" coordinate indicates the population-average self-targeting coefficient and consumption rank.

Figure A9: Adjustments to Eligibility Simulations

Panel A: Selection on Consumption



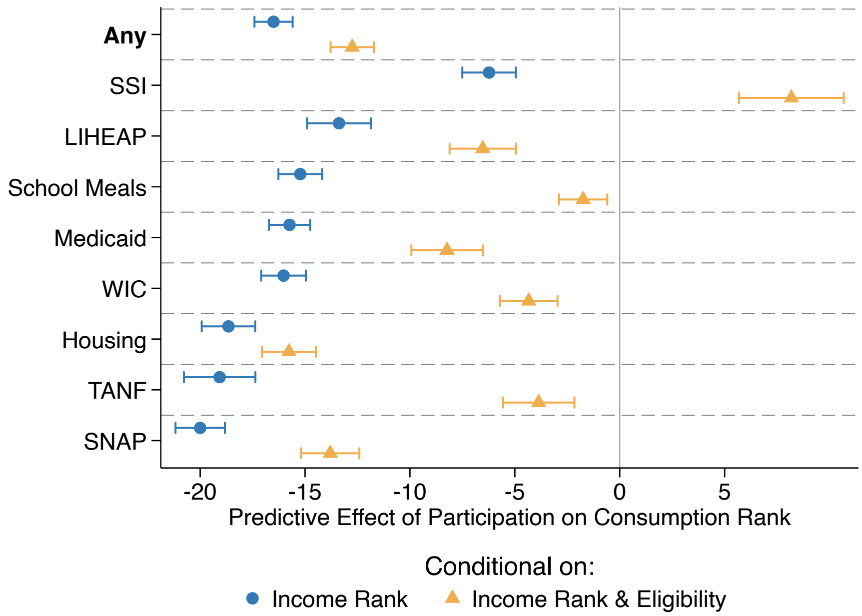
Panel B: Selection on Lifetime Income



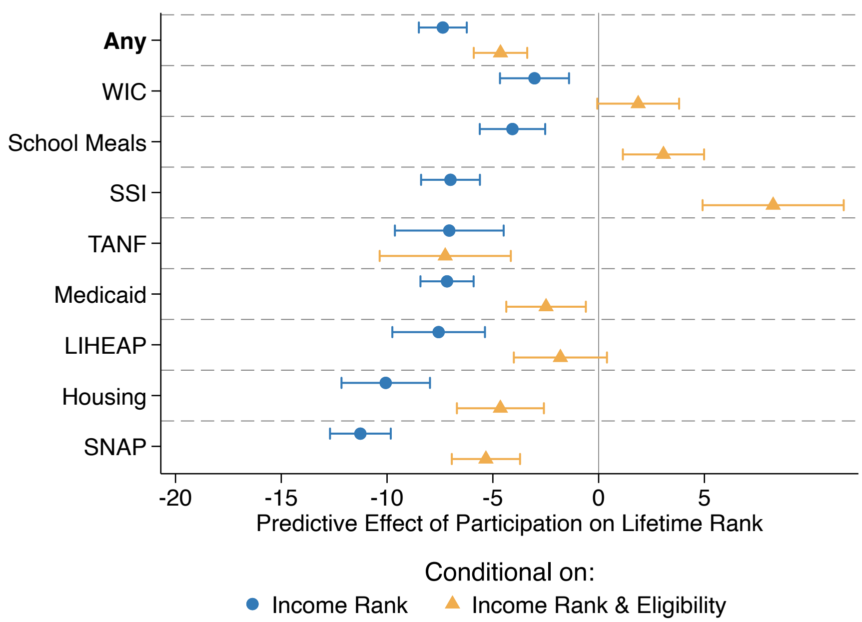
Notes: This figure displays estimates of the predictive effect of transfer benefit receipt on consumption rank or lifetime-income rank, conditional on current-income rank (coefficient β from Equation 4). For estimates represented by yellow triangles or green squares, we respectively impose an income limit (at 100% of the federal poverty level) or an asset limit (at $2,000 in liquid assets, in 2020 dollars adjusted for the Consumer Price Index) on top of our eligibility simulations. The confidence intervals are at the 95-percent level with clustering by household.

Figure A10: Reclassifying Simulated Ineligible Recipients

Panel A: Selection on Consumption



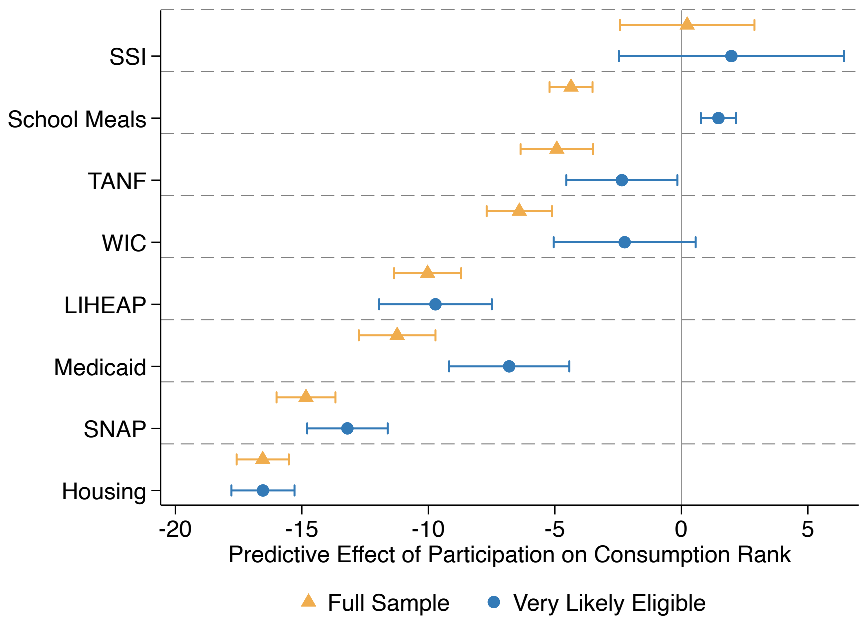
Panel B: Selection on Lifetime Income



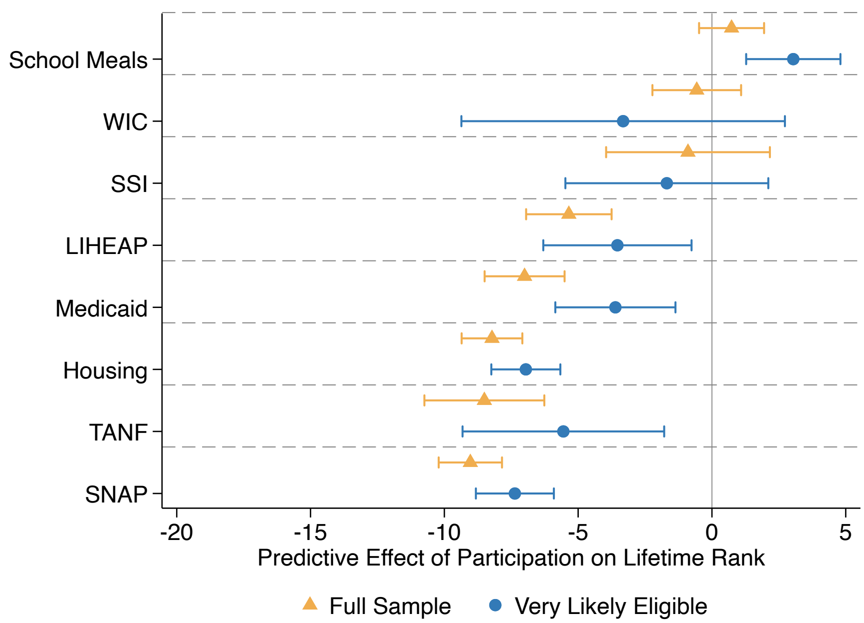
Notes: This figure displays estimates of the predictive effect of transfer benefit receipt on consumption rank or lifetime-income rank, conditional on current-income rank (coefficient β from Equation 4). For estimates represented by blue circles, we add no additional control variables to the specification, whereas for the yellow diamonds, we add program eligibility. The confidence intervals are at the 95-percent level with clustering by household. Eligibility samples in both panels reclassify all recipients as eligible, even if we initially simulate them to be ineligible.

Figure A11: Selection into Transfer Receipt: Very Likely Eligible Subsample

Panel A: Selection on Consumption



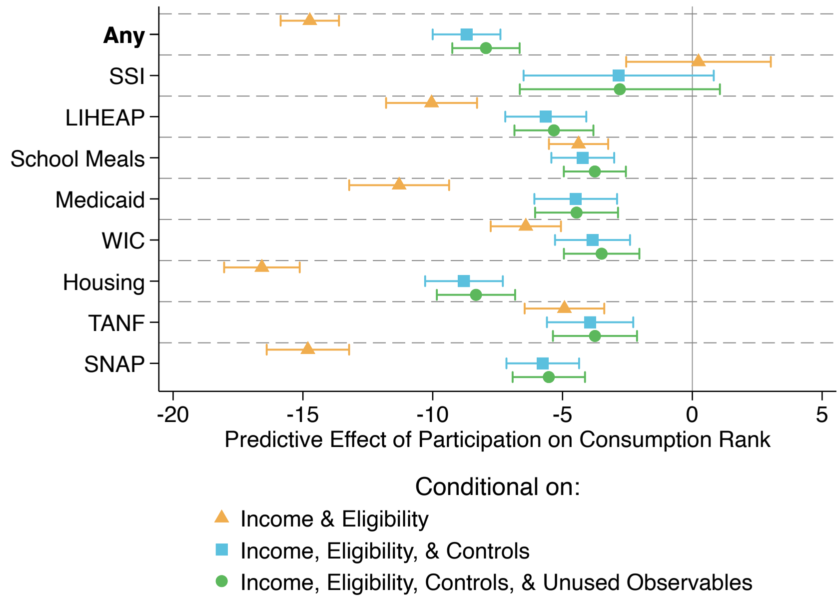
Panel B: Selection on Lifetime Income



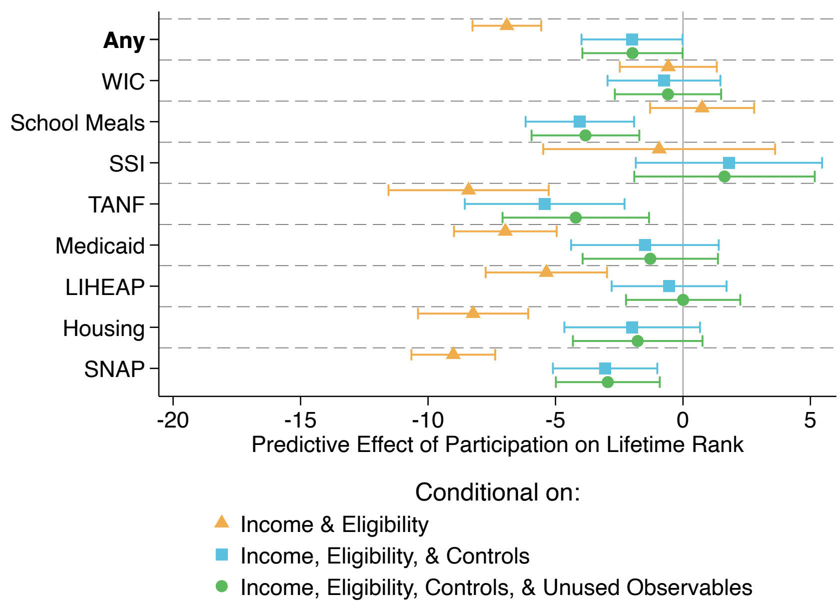
Notes: This figure displays estimates of the predictive effect of transfer receipt on consumption rank or lifetime-income rank, conditional on current-income rank (coefficient γ from Equation 4). Both yellow diamonds and blue circles restrict the sample to simulated eligibles. For estimates represented by blue circles, we further limit the sample to people who, in a logistic regression of simulated eligibility status on demographic observables, have a predicted probability of eligibility above 75 percent. The eligibility logit uses the following demographic variables: age (as a quadratic), sex, marital status, race/ethnicity (white, black, Hispanic, other), education (less than high school, high school, some college, BA, more than BA), household size, homeownership, disability, presence of a child in the household, income as a share of the federal povety level, and rank-transformed current income, lifetime income, and consumption. The confidence intervals are for the 95-percent level and clustering by household.

Figure A12: What Explains Selection into Transfer Receipt? With Controls

Panel A: Selection on Consumption



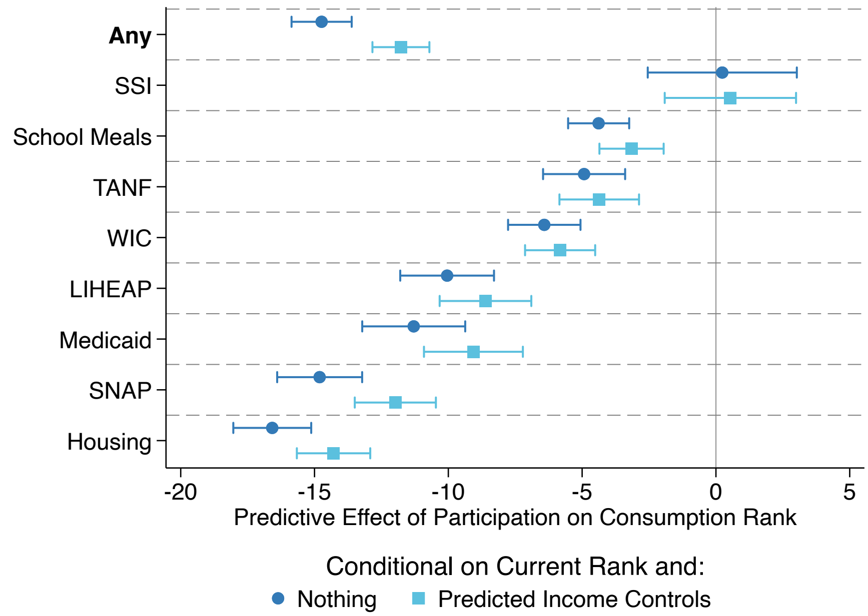
Panel B: Selection on Lifetime Income



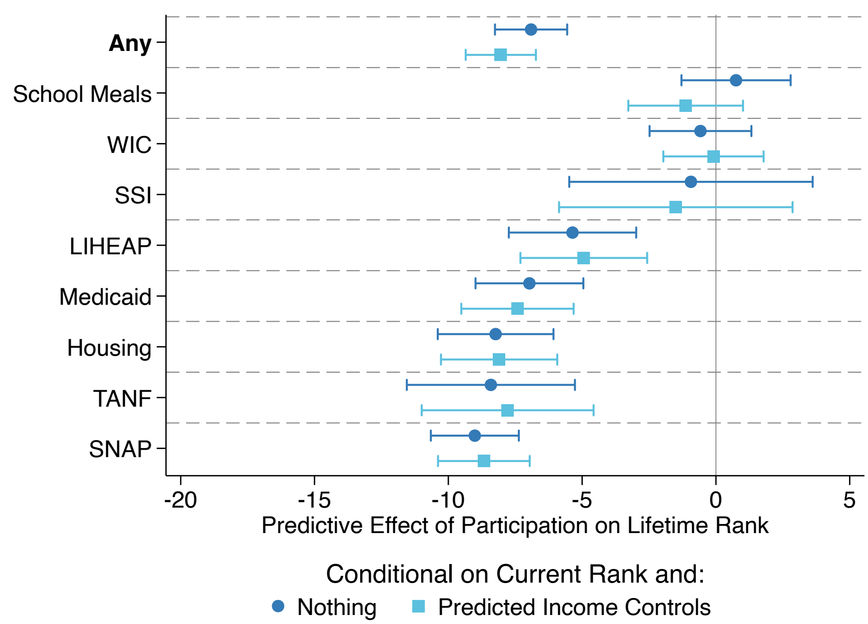
Notes: This figure displays estimates of the predictive effect of transfer receipt on consumption rank (Panel A) or lifetime-income rank (Panel B), conditional on current-income rank (coefficient β from Equation 4). For estimates represented by the yellow diamonds, we estimate the regression only on people whom we simulate to be eligible. For estimates represented by the teal squares, we condition on eligibility and characteristics that enter any eligibility rule. For estimates represented by green circles, we condition on eligibility, eligibility characteristics, and several demographic characteristics (race, education, and marital status). The "any" row is an indicator for receipt of at least one of the eight transfers. The confidence intervals are for the 95-percent level and reflect clustered standard errors by household.

Figure A13: Selection into Transfers, with Predicted-Income Control

Panel A: Consumption

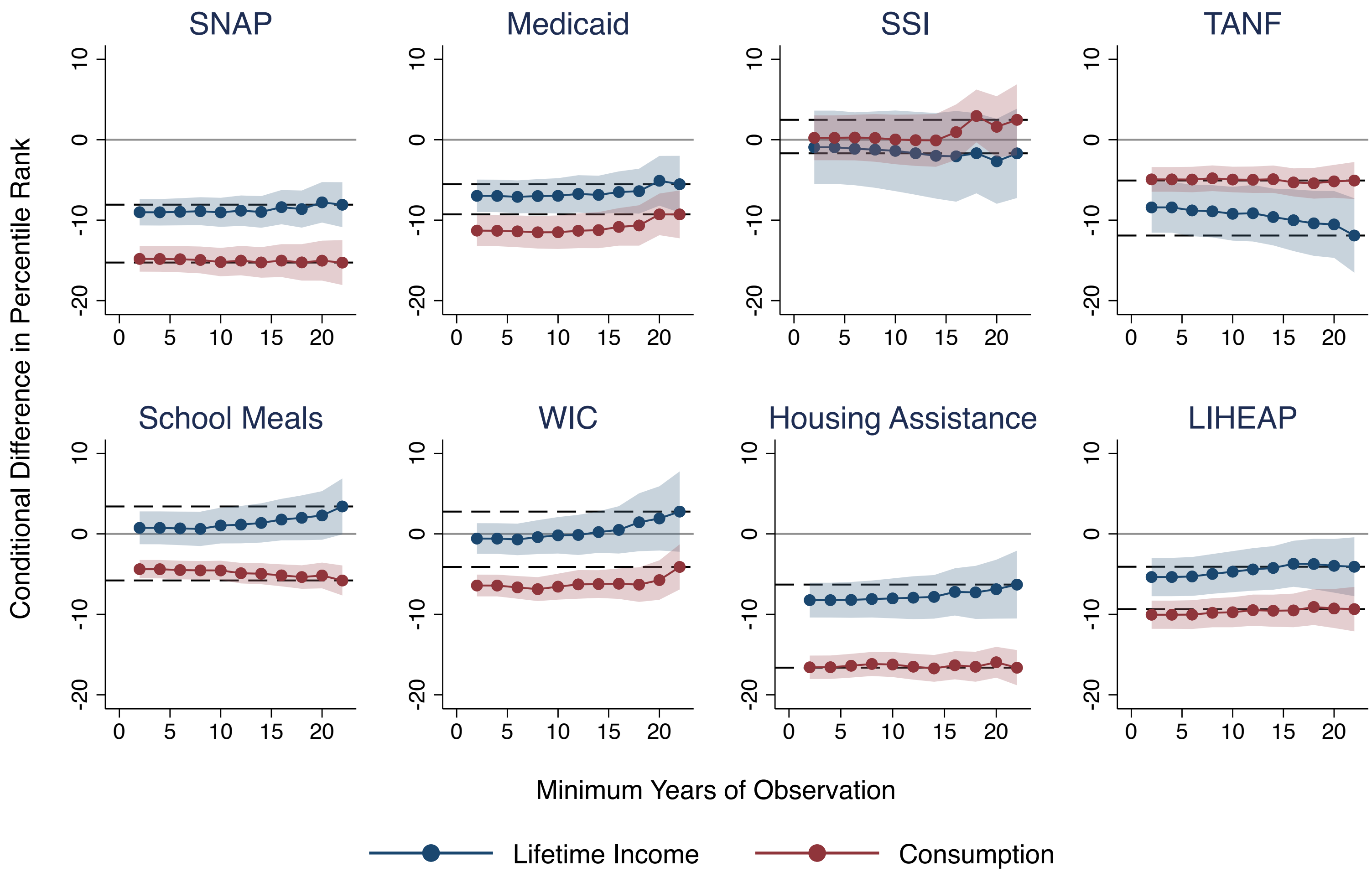


Panel B: Lifetime Income



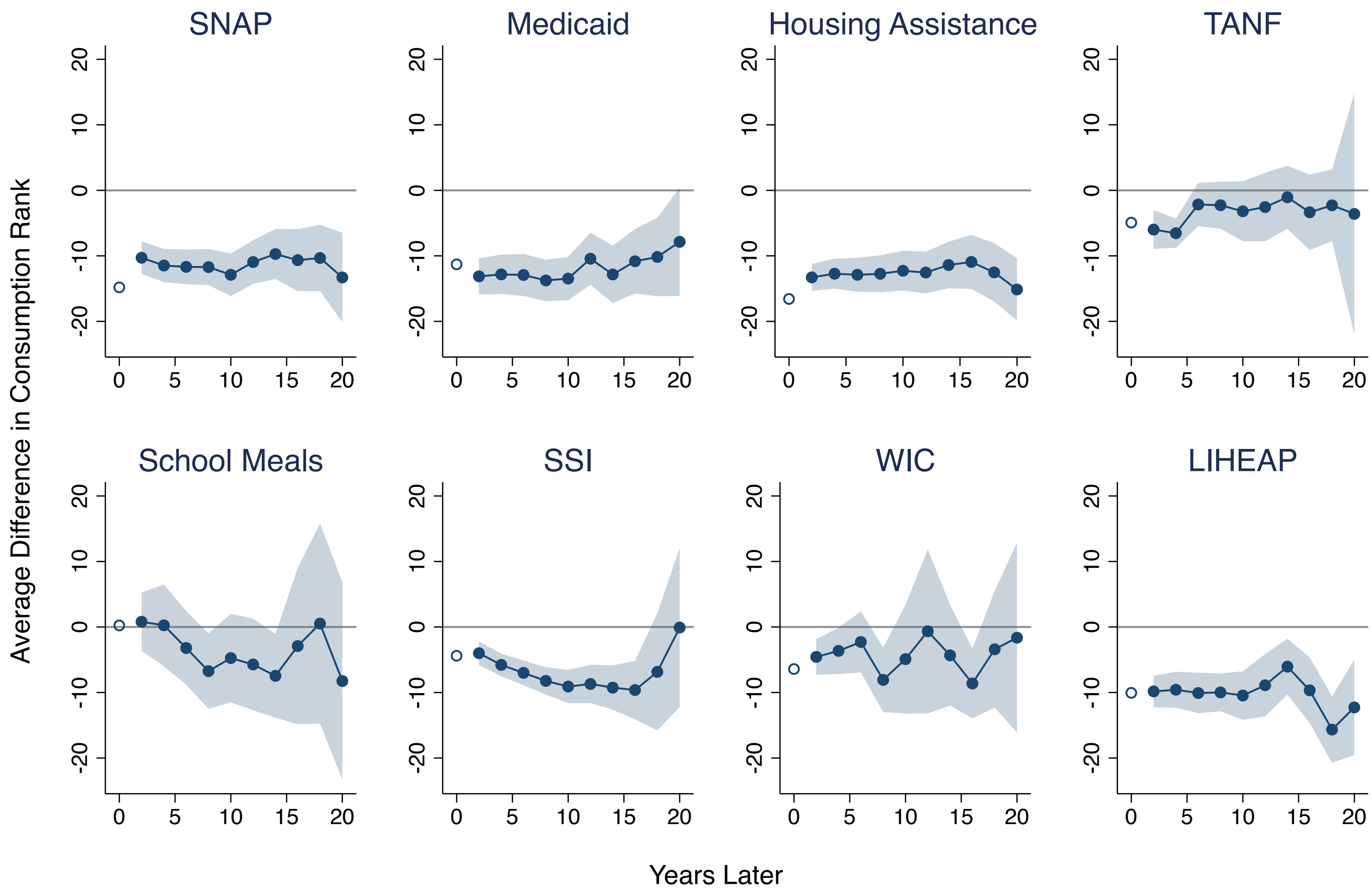
Notes: This figure displays estimates of the predictive effect of transfer receipt on equivalized household consumption rank, conditional on current-income rank as well as predicted-income rank. Income prediction uses a Poisson regression as explained in Section 3. For estimates represented by blue circles, we do not add additional control variables to the specification, whereas for the teal squares, we include the predicted-income control. Both samples are limited to simulated-eligible people. The "any" row of Panel A is an indicator for receipt of at least one of the eight transfers. The confidence intervals are at the 95-percent level with clustering by household.

Figure A14: Self-Targeting by Minimum Years of Observation



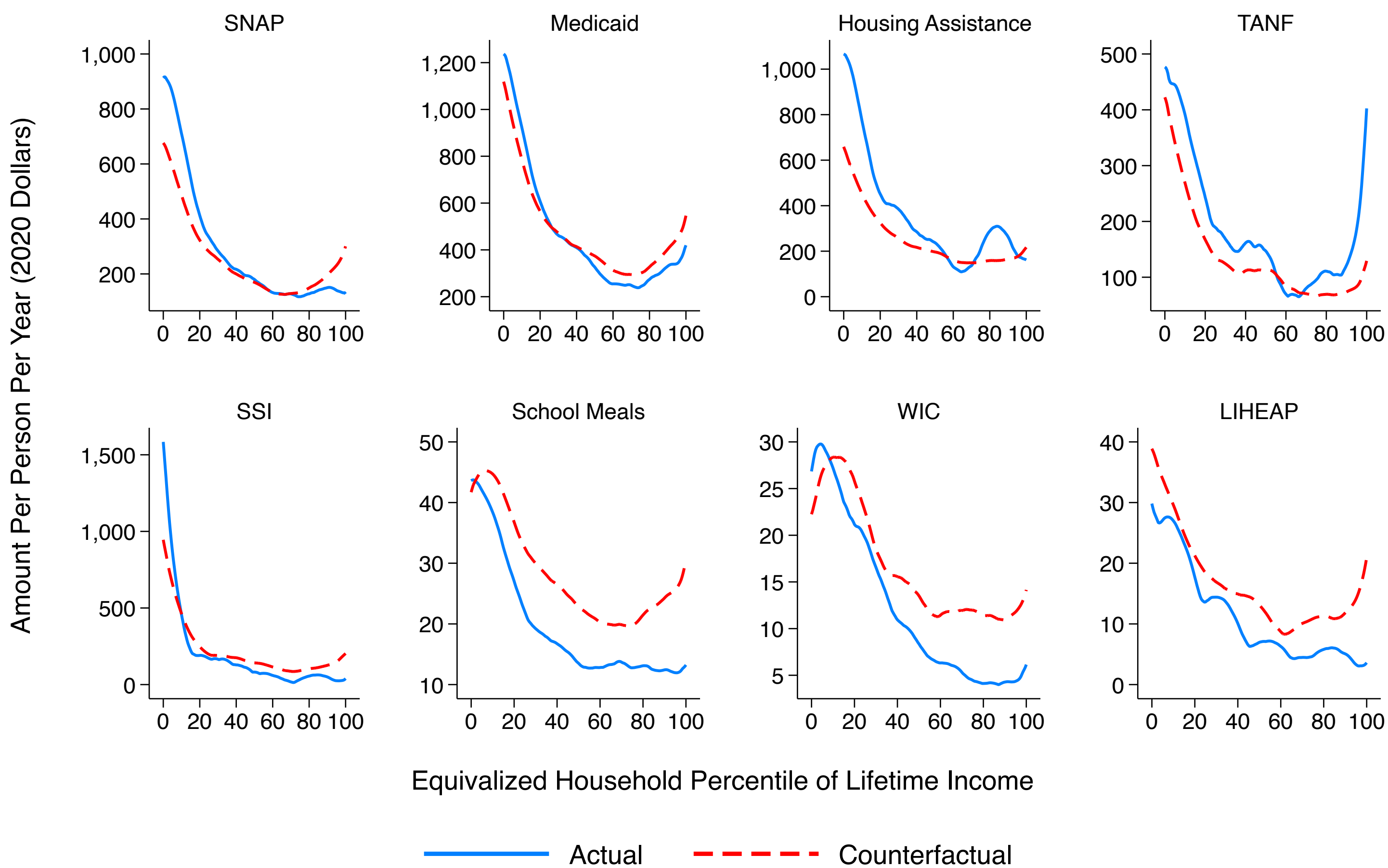
Notes: This figure displays estimates of the predictive effect of transfer benefit receipt on consumption rank or lifetime-income rank, conditional on current-income rank (coefficient γ from Equation 4). Moving to the right of each panel, we restrict the sample to eligible people with progressively more years of observation in the PSID. Dashed lines indicate our baseline estimates. Shaded regions reflect confidence intervals at the 95-percent level with clustering by household.

Figure A15: Self-Targeting on Transfer Receipt in the Distant Future



Notes: This figure displays the predictive effect of transfer receipt k years ahead on consumption rank this year conditional on current income rank. The regression equation is $\bar{R}_{it} = \alpha_{ct} + \beta D_{i,t+k} + f(R_{it}) + u_{it}$, where we plot β for each horizon k. The estimation sample is always restricted to current eligible nonrecipients, $D_{it} = 0$. Shaded regions reflect bootstrapped 95-percent pointwise confidence intervals, with clustering by household.

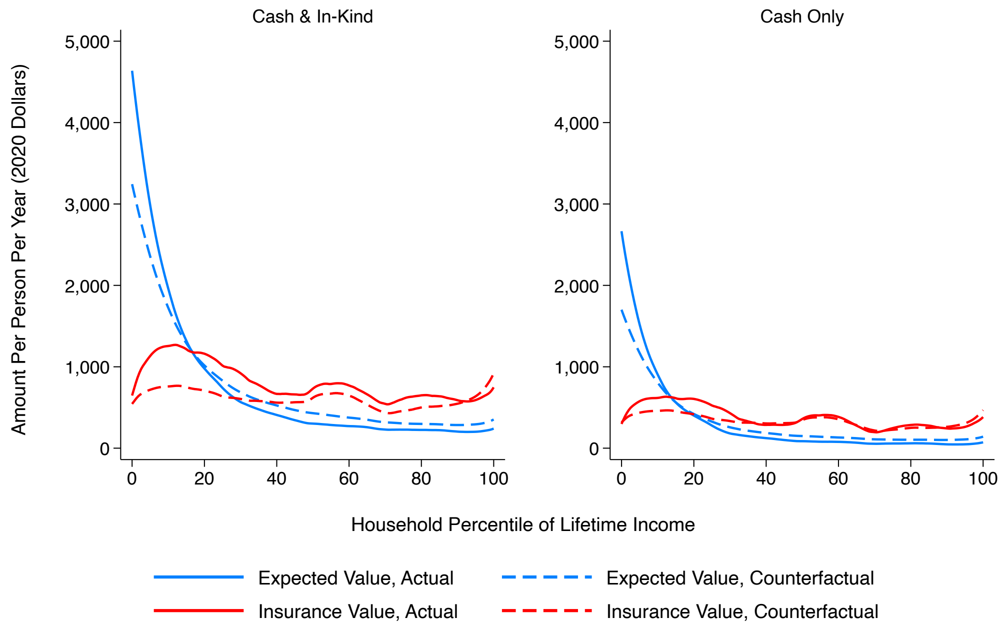
Figure A16: Certainty-Equivalent Values by Program



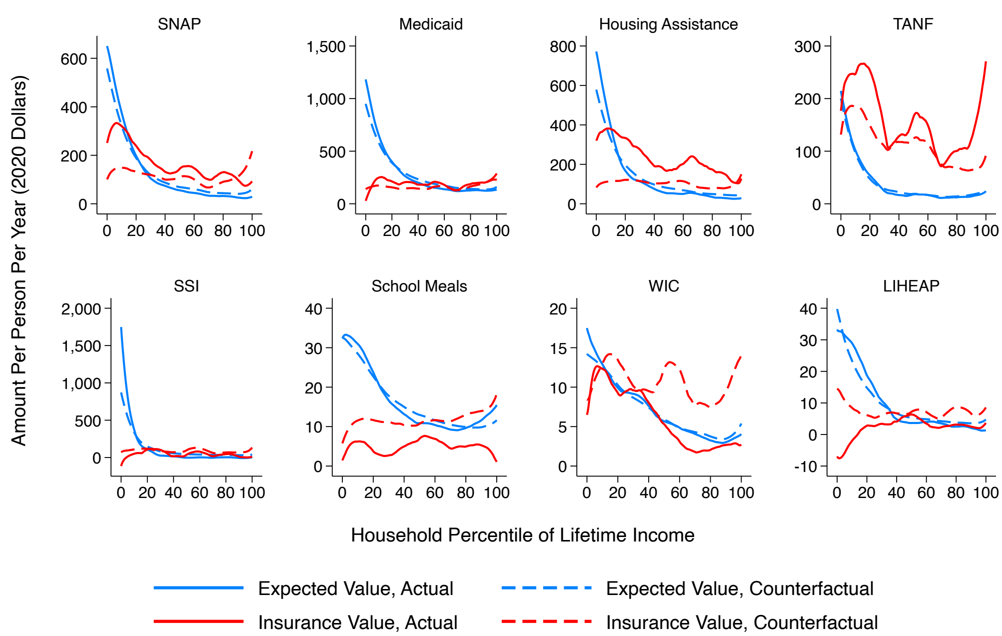
Notes: This figure displays the average annual per-capita certainty-equivalent value of transfer benefits by program as a function of lifetime-income rank, equivalized for household size. Certainty equivalents include both expected value and insurance value. Blue solid lines report existing voluntary transfers, and red dashed lines report the automatic counterfactual. Solid lines report existing transfers, and dashed lines report the automatic counterfactuals. The functions are estimated by local linear regressions with bandwidths of five percentiles.

Figure A17: Expected Value and Insurance Value for Non-Equivalized Households

Panel A: Totals Across Programs



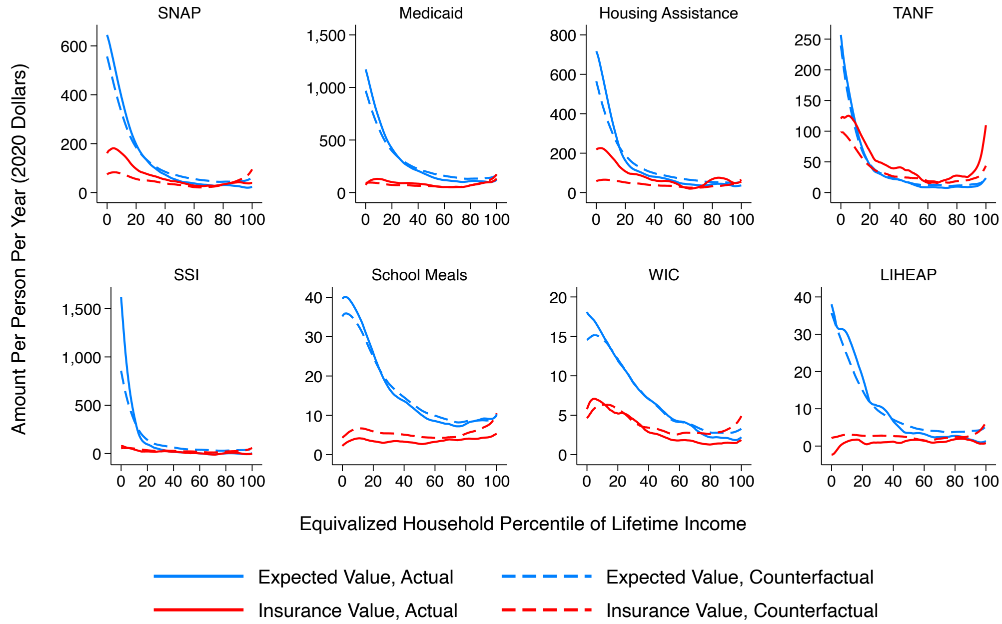
Panel B: By Program



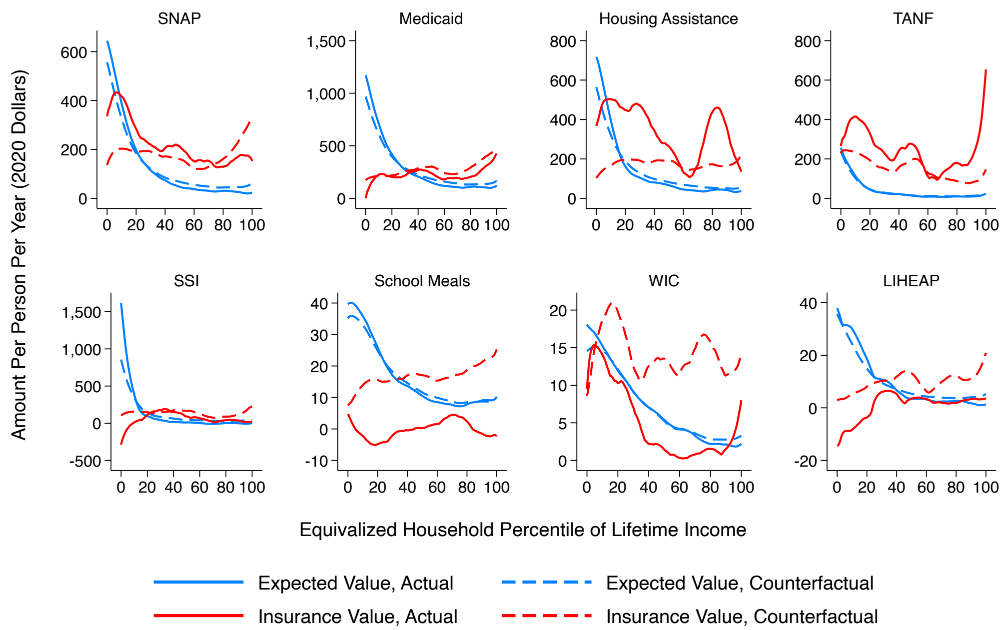
Notes: This figure displays the average annual per-capita values of transfer benefits as a functions of lifetime-income rank, not equivalized for household size. In Panel A, totals are reported across all eight programs and for cash transfers only. In Panel B, values are reported by program. Blue lines report the expected values of transfers, and red lines report insurance values. Solid lines report existing transfers, and dashed lines report the automatic counterfactuals. The functions are estimated by local linear regressions with bandwidths of five percentiles.

Figure A18: Insurance Values Under Alternative Calibrations of Risk Aversion

Panel A: Low Risk Aversion (γ = 1)

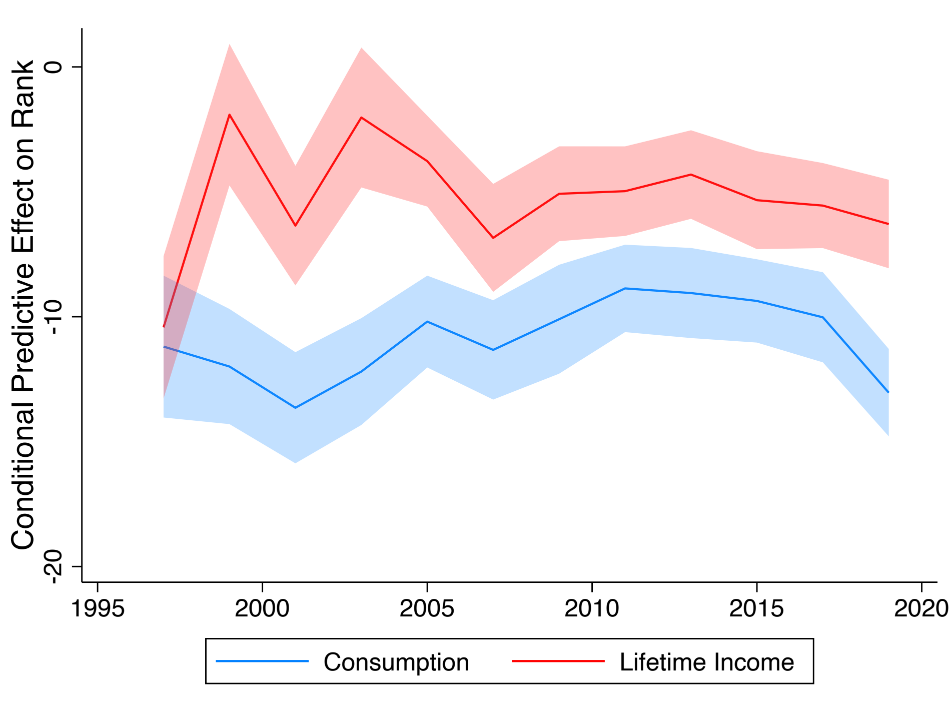


Panel B: High Risk Aversion (γ = 3)



Notes: This figure displays the average annual per-capita value of transfer benefits by program as a function of lifetime-income rank, equivalized for household size. Blue lines report the expected values of transfers, and red lines report insurance values. Panel A uses a risk-aversion parameter $\gamma = 1$, and Panel B uses $\gamma = 3$. Solid lines report existing transfers, and dashed lines report the automatic counterfactuals. The functions are estimated by local linear regressions with bandwidths of five percentiles.

Figure A19: Selection into Transfer Receipt Over Time

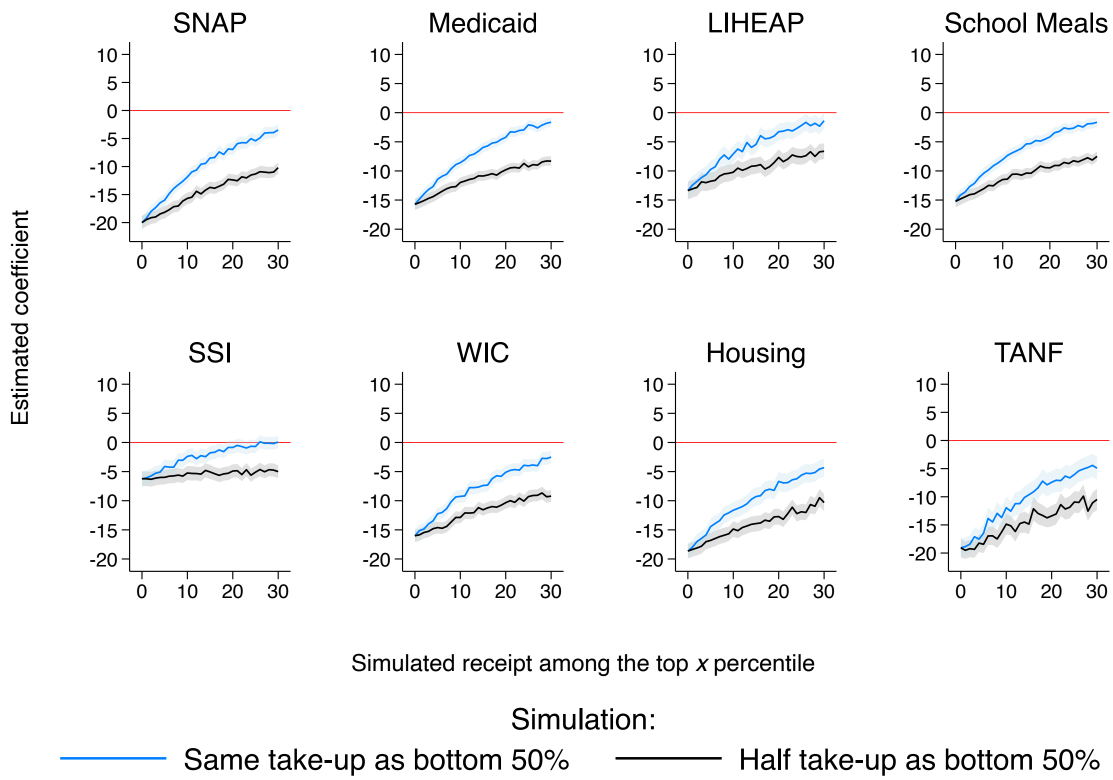


Notes: This figure displays coefficients from the following regression specification:

$$\bar{R}_{its} = \alpha_{cts} + \beta_t D_{its} + f_s(R_{it}) + u_{its},$$

where i denotes households, t denotes years, and s denotes transfer programs. The outcome $\bar{R}_{its}$ is equivalized household consumption rank in the blue line and equivalized household lifetime income rank in red. The data are stacked across programs, so that each individual–year appears eight times, once for each transfer program s. Each sample is limited to that transfer's simulated-eligible population. We thus include cohort-year effects α_{cts} specific to each transfer, as well as transfer-specific spline controls $f_s(\cdot)$ for current-income rank R_{it}. The coefficients β_t thus report an average selection effect across transfer programs in a given year t. Shaded regions reflect 95-percent pointwise confidence intervals, with clustering by household. All regressions use PSID sample weights but are not otherwise adjust to account for variation in transfer program size.

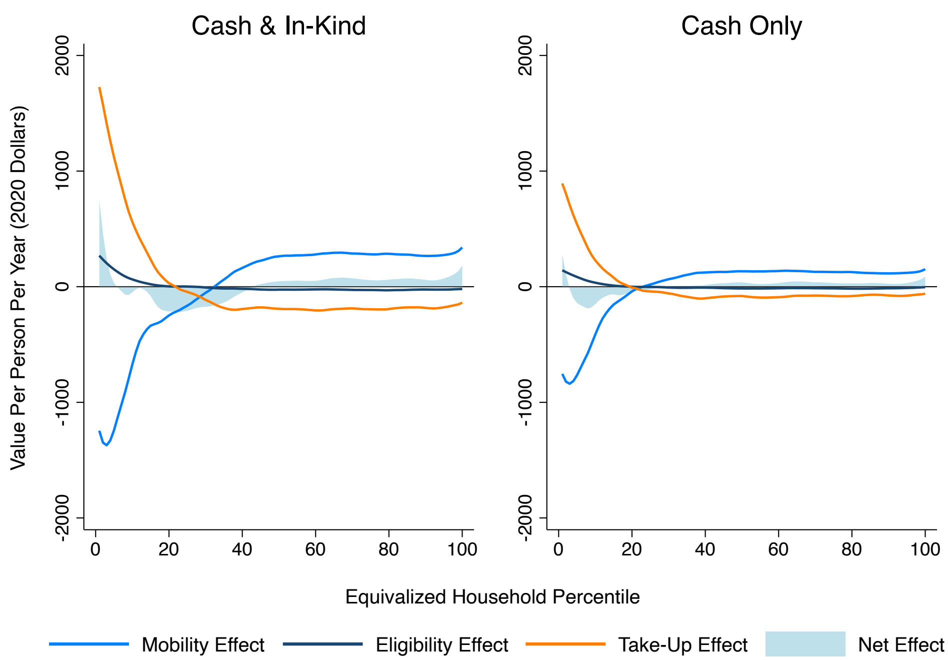
Figure A20: Measurement Error Simulations



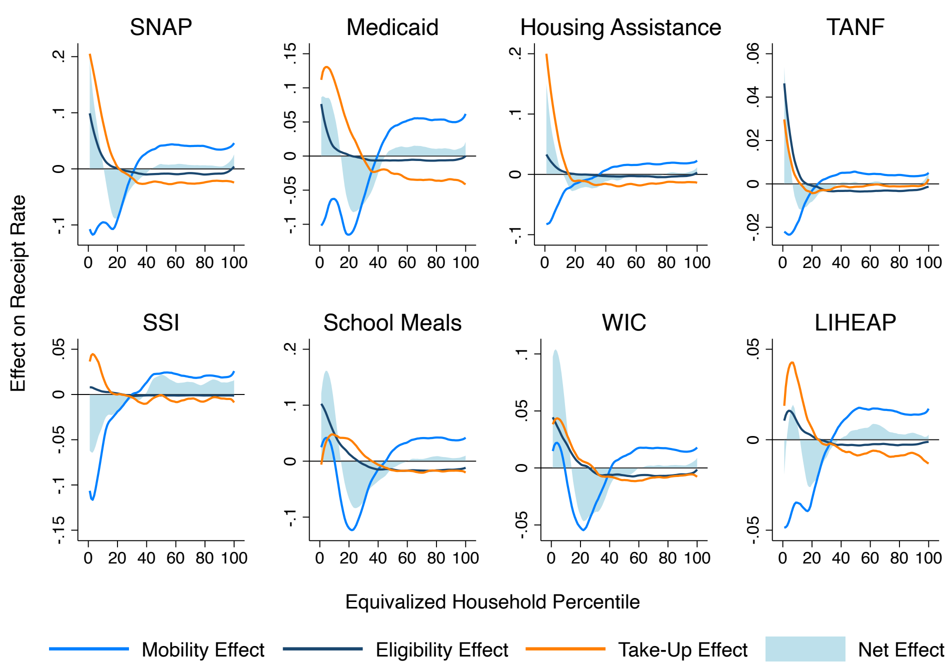
Notes: This figure displays the predictive effect of transfer receipt on consumption rank given income rank (Equation 4) when we assume that take-up is underreported for the top x consumption percentiles. In blue, we assume that the top x percentiles actually have the same take-up rate as the bottom half of the consumption distribution. In black we assume that top take-up rate is half that of the bottom half of the consumption distribution. Shaded regions reflect 95-percent pointwise confidence intervals, with clustering by household.

Figure A21: Decomposition of Difference Between Consumption and Current Incidence

Panel A: Average Total Annual Per-Capita Benefits

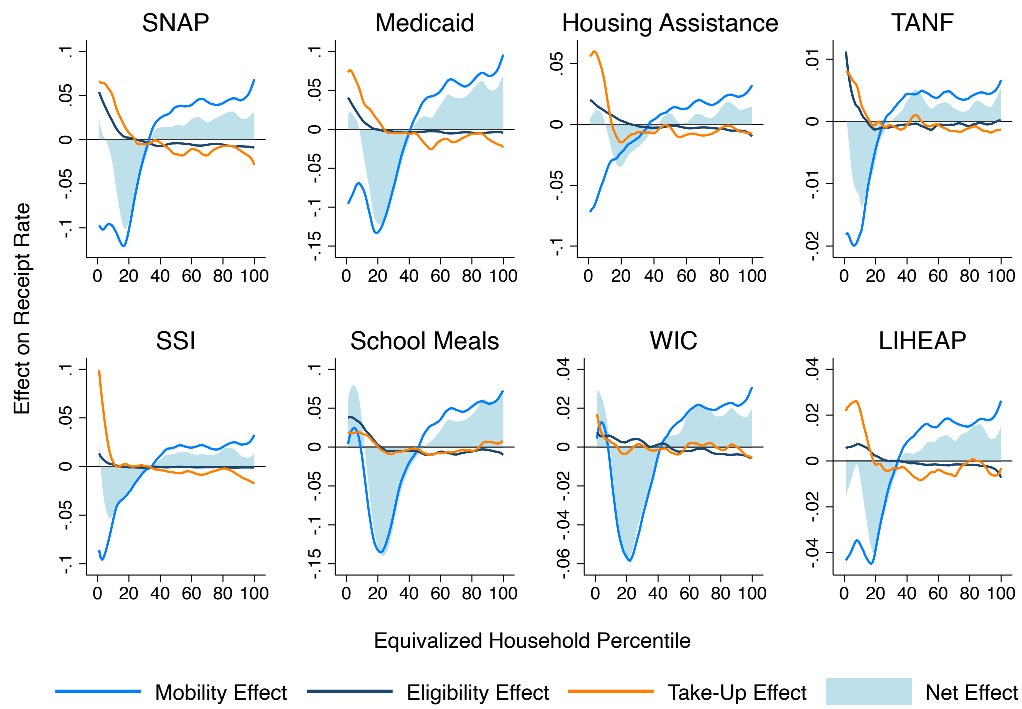


Panel B: Receipt Rate by Program



Notes: This figure displays decompositions of differences between current-income and consumption incidence into mobility, eligibility, and take-up effects. The underlying functions are estimated using local linear regressions with bandwidths of five percentiles.

Figure A22: Decomposition of Difference Between Lifetime and Current Incidence



Notes: This figure displays a decomposition of the difference between current and lifetime incidence into mobility and take-up effects. The original functions are estimated using a local linear regression with a bandwidth of five percentiles.

Table A1: SNAP Receipt, Eligibility, and Take-Up Rates by Income and Lifetime Income Quintile

Panel A: Receipt Rate

		Income Quintile					
		1	2	3	4	5	Avg.
	1	42.8	20.1	6.7	1.5	0.1	35.3
	2	27.3	11.7	2.9	1.0	0.1	8.5
Lifetime Income Quintile	3	20.5	9.8	2.0	0.4	0.2	3.3
	4	19.5	6.2	1.9	0.4	0.2	1.3
	5	20.2	5.2	2.3	0.4	0.2	0.9
	Avg.	33.6	12.2	2.8	0.5	0.2	

Panel B: Simulated Eligibility Rate

		Income Quintile					
		1	2	3	4	5	Avg.
	1	80.5	24.0	0.0	0.0	0.0	49.7
	2	91.3	31.6	3.8	2.0	2.2	58.3
Lifetime Income Quintile	2	78.2	21.2	2.0	1.5	1.1	21.3
	3	72.9	17.5	1.4	0.7	0.5	11.4
	4	65.3	16.6	1.7	0.9	0.6	8.6
	5	58.6	14.4	2.2	1.4	0.6	8.0
	Avg.	81.7	22.0	2.0	1.1	0.6	

Panel C: Take-Up Rate Among Simulated Eligibles

		Income Quintile					
		1	2	3	4	5	Avg.
	1	45.2	34.5	.	.	.	43.4
	2	30.6	24.8	.	.	.	28.0
Lifetime Income Quintile	3	22.7	21.4	.	.	.	21.2
	4	19.3	15.6	.	.	.	17.0
	5	22.9	9.1	.	.	.	17.7
	Avg.	37.3	26.1	.	.	.	

Notes: This table reports the shares of households that receive SNAP (Panel A), are simulated to be eligible for SNAP (Panel B), and take up SNAP conditional on being simulated eligible (Panel C). Households are split by quintiles of equivalized household current and lifetime income. Due to low rates of simulated eligibility, we do not report take-up rates for the top three income quintiles.

Table A2: Dollar and Percentage Differences Between Recipients and Similar-Income Non-Recipients

	Proportion Difference		Difference in 2020 Constant Dollars	
	Consumption (1)	Lifetime Income (2)	Consumption (3)	Lifetime Income (4)
SNAP	-0.522***	-0.646***	-7,867***	-20,370**
	(0.038)	(0.225)	(631)	(8,677)
Medicaid	-0.462***	-0.458***	-6,039***	-10,783***
	(0.044)	(0.077)	(650)	(2,216)
Housing Assistance	-0.445***	-0.443***	-7,386***	-16,176***
	(0.035)	(0.104)	(613)	(4,320)
TANF	-0.258***	-0.170	-2,463***	-4,487
	(0.044)	(0.171)	(419)	(4,532)
SSI	0.036	0.193	334	1,708
	(0.071)	(0.165)	(655)	(1,536)
School Meals	-0.350***	-0.375***	-3,883***	-8,673***
	(0.027)	(0.119)	(327)	(3,321)
WIC	-0.238***	-0.144**	-2,565***	-4,042**
	(0.026)	(0.058)	(294)	(1,629)
LIHEAP	-0.365***	-0.316***	-5,177***	-6,887***
	(0.037)	(0.091)	(589)	(2,023)

Notes: This table reports estimates of differences in consumption and lifetime income between transfer receipients and nonrecipients, conditional on current income. All columns report estimates obtained via Poisson regression. Columns 1 and 2 report exponentiated coefficients ($\exp(\beta) - 1$) from these regressions. Columns 3 and 4 report the dollar effects. Each cell is its own regression. All specifications control flexibly for the logarithm of equivalized current household income using cubic basis splines. Standard errors are clustered by household. $^{*} = p < 0.10$, $^{**} = p < 0.05$, $^{***} = p < 0.01$.

Table A3: Transfer Amounts and Receipt Rates at the Bottom of the Distributions of Income and Consumption

	Income				Consumption				Counterfactual for Consumption			
Percentiles	0–5	0–10	0–25	0–50	0–5	0–10	0–25	0–50	0–5	0–10	0–25	0–50
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Panel A: Average Total Dollars Per Person Per Year												
Total	3,662	3,217	2,184	1,380	3,978	3,289	2,133	1,316	2,288	2,025	1,535	1,102
	(168)	(144)	(85)	(55)	(149)	(116)	(80)	(56)	(36)	(33)	(28)	(23)
Cash	1,929	1,677	1,014	577	1,924	1,542	925	545	1,093	936	666	456
	(117)	(97)	(54)	(32)	(98)	(77)	(49)	(32)	(23)	(20)	(16)	(12)
Panel B: Average Amounts by Program												
SNAP	554	510	409	247	788	641	396	236	403	362	276	195
	(26)	(23)	(16)	(11)	(29)	(22)	(15)	(11)	(5)	(5)	(5)	(4)
Medicaid	1,054	940	746	518	1,093	978	739	488	746	687	556	415
	(41)	(35)	(21)	(15)	(32)	(26)	(19)	(15)	(8)	(7)	(7)	(7)
Housing Assistance	644	563	374	246	882	693	411	245	404	357	273	199
	(52)	(41)	(26)	(17)	(59)	(45)	(27)	(17)	(7)	(6)	(5)	(4)
TANF	235	219	138	75	344	252	132	73	144	124	87	59
	(29)	(23)	(12)	(7)	(32)	(21)	(11)	(6)	(3)	(3)	(2)	(2)
SSI	1,104	909	441	238	777	626	376	222	521	428	285	189
	(106)	(84)	(43)	(24)	(94)	(68)	(40)	(23)	(17)	(13)	(9)	(7)
School Meals	29	28	36	29	50	50	41	28	33	33	30	23
	(2)	(2)	(1)	(1)	(3)	(2)	(1)	(1)	(0)	(0)	(0)	(0)
WIC	10	12	17	13	35	30	22	13	15	15	14	11
	(1)	(1)	(1)	(1)	(3)	(2)	(1)	(1)	(0)	(0)	(0)	(0)
LIHEAP	42	45	30	19	30	34	24	16	30	27	20	15
	(5)	(6)	(3)	(2)	(3)	(5)	(3)	(2)	(1)	(1)	(0)	(0)

Notes: This table reports sample means of average transfer payments per person per year (in 2020 constant dollars), in total and by transfer program. Each column is for a different range of percentiles in a distribution. Columns 1–4 are with respect to household equivalized current income, and Columns 5–8 are with respect to household equivalized consumption. Columns 9–12 calculate the consumption incidence under the counterfactual in which receipt is a function of income rank. Parentheses report standard errors clustered by household.

Table A4: Transfer Amounts and Receipt Rates at the Bottom of the Distributions of Current and Lifetime Income

	Current Income				Lifetime Income				Counterfactual for Lifetime Income			
Percentiles	0–5	0–10	0–25	0–50	0–5	0–10	0–25	0–50	0–5	0–10	0–25	0–50
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Panel A: Average Dollars Per Person Per Year												
Total	3,662	3,217	2,184	1,380	3,812	3,090	1,921	1,221	2,557	2,185	1,547	1,082
	(168)	(144)	(85)	(55)	(241)	(151)	(87)	(56)	(45)	(37)	(31)	(24)
Cash	1,929	1,677	1,014	577	2,049	1,534	875	522	1,273	1,049	690	457
	(117)	(97)	(54)	(32)	(179)	(104)	(53)	(32)	(28)	(22)	(17)	(13)
Panel B: Receipt Rates												
SNAP	554	510	409	247	586	527	350	219	426	377	274	190
	(26)	(23)	(16)	(11)	(35)	(28)	(17)	(11)	(6)	(5)	(5)	(4)
Medicaid	1,054	940	746	518	1,071	926	655	445	794	712	546	401
	(41)	(35)	(21)	(15)	(46)	(33)	(22)	(16)	(11)	(9)	(8)	(7)
Housing Assistance	644	563	374	246	639	577	347	222	450	383	275	195
	(52)	(41)	(26)	(17)	(60)	(49)	(26)	(16)	(8)	(7)	(5)	(4)
TANF	235	219	138	75	216	172	104	64	166	137	90	59
	(29)	(23)	(12)	(7)	(34)	(21)	(11)	(6)	(4)	(3)	(2)	(2)
SSI	1,104	909	441	238	1,224	812	398	223	654	511	309	195
	(106)	(84)	(43)	(24)	(172)	(96)	(43)	(23)	(19)	(14)	(10)	(7)
School Meals	29	28	36	29	40	40	33	24	30	30	27	22
	(2)	(2)	(1)	(1)	(3)	(2)	(1)	(1)	(0)	(0)	(0)	(0)
WIC	10	12	17	13	17	17	14	11	13	13	12	10
	(1)	(1)	(1)	(1)	(2)	(1)	(1)	(1)	(0)	(0)	(0)	(0)
LIHEAP	42	45	30	19	33	31	25	17	33	29	21	15
	(5)	(6)	(3)	(2)	(4)	(3)	(3)	(2)	(1)	(1)	(1)	(0)

Notes: This table reports sample means of average transfer payments per person per year (in 2020 constant dollars), in total and by transfer program. Each column is for a different range of percentiles in a distribution. Columns 1–4 are with respect to household equivalized current income, and Columns 5–8 are with respect to household equivalized lifetime income. Columns 9–12 calculate the lifetime incidence under the counterfactual in which receipt is a function of current-income rank. Parentheses report standard errors clustered by household.

Table A5: Selection into Transfers, Adjusted for Misreporting of Transfer Receipt

	Baseline		Adjusted for Misreporting	
	Consumption (1)	Lifetime Income (2)	Consumption (3)	Lifetime Income (4)
Panel A: SNAP				
Receives Transfer	-17.6***	-11.1***	-26.4***	-14.3***
	(0.6)	(0.6)	(0.8)	(0.9)
Panel B: Medicaid				
Receives Transfer	-14.4***	-7.0***	-23.4***	-12.2***
	(0.5)	(0.5)	(0.7)	(0.8)

Notes: This table examines the effect of corrections for misreporting of transfer receipt on estimates of selection into transfers by consumption rank and lifetime-income rank, conditional on current-income rank. The estimating equation is Equation 4. In Columns 3 and 4, we replace reported receipt with the adjusted measures from Mittag (2019) for SNAP and Davern et al. (2019) for Medicaid. All specifications control flexibly for current-income rank using cubic basis splines. Standard errors are clustered by household. * = $p < 0.10$, ** = $p < 0.05$, *** = $p < 0.01$.

Table A6: Well-Measured Consumption and Transfer Receipt

	SNAP (1)	Medicaid (2)	Housing Assistance (3)	TANF (4)	SSI (5)	School Meals (6)	WIC (7)	LIHEAP (8)
Panel A: Rent and Owner's Equivalent Rent (1997–2019)								
Receives Transfer	-0.482***	-0.376***	-0.820***	-0.324***	0.058	-0.228***	-0.197***	-0.279***
	(0.037)	(0.035)	(0.037)	(0.057)	(0.090)	(0.029)	(0.028)	(0.045)
Panel B: Vehicle Lease Cost and Equivalent Lease Cost (1999–2019)								
Receives Transfer	-0.303***	-0.320***	-0.174***	-0.070**	0.005	-0.255***	-0.081***	-0.153***
	(0.020)	(0.024)	(0.026)	(0.030)	(0.056)	(0.015)	(0.020)	(0.026)
Panel C: Food at Home Expenditure (1999–2019)								
Receives Transfer	-0.585***	-0.322***	-0.238***	-0.382***	0.038	-0.054**	-0.288***	-0.301***
	(0.026)	(0.031)	(0.031)	(0.049)	(0.093)	(0.026)	(0.029)	(0.034)
Panel D: Utility Expenditure (1999–2019)								
Receives Transfer	-0.085***	-0.176***	-0.339***	-0.072*	0.153**	-0.151***	-0.122***	0.020
	(0.029)	(0.036)	(0.048)	(0.042)	(0.068)	(0.024)	(0.026)	(0.033)
Panel E: Gasoline Expenditure (1999–2019)								
Receives Transfer	-0.109***	-0.178***	-0.135***	-0.067	-0.254***	-0.113***	-0.069**	-0.173***
	(0.031)	(0.040)	(0.038)	(0.051)	(0.092)	(0.026)	(0.029)	(0.040)

Notes: This table reports estimates of the predictive effect of transfer receipt on (log) levels of reported consumption, conditional on current-income rank and being simulated-eligible for the transfer. Each panel row is for a different consumption outcome, and each column is for a different transfer. The year ranges in parentheses indicate data coverage for the outcome of interest. All specifications control flexibly for current-income rank using cubic basis splines. Standard errors are clustered by household. $^{*} = p < 0.10$, $^{**} = p < 0.05$, $^{***} = p < 0.01$.

Table A7: Durable-Goods Ownership and Transfer Receipt

	SNAP (1)	Medicaid (2)	Housing Assistance (3)	TANF (4)	SSI (5)	School Meals (6)	WIC (7)	LIHEAP (8)
Panel A: HH Owns Primary Residence (1997–2019)								
Receives Transfer	-0.124***	-0.088***	-0.390***	-0.160***	0.075	0.042**	0.007	-0.044*
	(0.015)	(0.019)	(0.012)	(0.021)	(0.056)	(0.019)	(0.017)	(0.026)
Panel B: Number of Rooms in Home (1997–2019)								
Receives Transfer	-0.386***	-0.252***	-0.723***	-0.052	0.103	-0.160**	-0.420***	0.149*
	(0.066)	(0.075)	(0.067)	(0.110)	(0.157)	(0.063)	(0.072)	(0.079)
Panel C: Central Air Conditioning at Home (1997–2009)								
Receives Transfer	-0.017	0.001	0.017	-0.153***	-0.045	0.000	-0.043*	-0.024
	(0.021)	(0.026)	(0.029)	(0.029)	(0.068)	(0.019)	(0.026)	(0.026)
Panel D: HH Owns a Car (1999–2019)								
Receives Transfer	-0.083***	-0.033	-0.189***	-0.106***	0.047	0.101***	0.041**	-0.017
	(0.019)	(0.021)	(0.023)	(0.025)	(0.049)	(0.017)	(0.020)	(0.021)
Panel D: HH Owns a Computer (2003–2019)								
Receives Transfer	-0.085***	0.030	-0.076***	-0.040	0.120**	0.066***	-0.040	-0.041*
	(0.019)	(0.025)	(0.024)	(0.031)	(0.050)	(0.018)	(0.025)	(0.024)
Panel E: HH Owns a Smartphone (2015–2019)								
Receives Transfer	-0.007	0.067**	-0.023	0.028	0.031	0.090***	0.028	-0.058
	(0.024)	(0.028)	(0.032)	(0.038)	(0.085)	(0.019)	(0.017)	(0.038)

Notes: This table reports estimates of the predictive effect of transfer receipt on measures of household durable-goods ownership, conditional on current-income rank and being simulated-eligible for the transfer. Each panel row is for a different consumption outcome, and each column is for a different transfer. The year ranges in parentheses indicate data coverage for the outcome of interest. All specifications control flexibly for current-income rank using cubic basis splines. Standard errors are clustered by household. $^{*} = p < 0.10$, $^{**} = p < 0.05$, $^{***} = p < 0.01$.

Table A8: Simulated Eligibility and Receipt Rates

Program	P(Receive | Simulated Eligible) (1)	P(Receive | Not Sim. Elig.) (2)	P(Sim. Elig. | Receive) (3)	P(Sim. Elig. | Not Receive) (4)
Housing Assistance	0.11	0.02	0.78	0.34
LIHEAP	0.14	0.01	0.72	0.17
Medicaid	0.51	0.10	0.43	0.08
SNAP	0.34	0.03	0.74	0.16
SSI	0.29	0.05	0.11	0.02
TANF	0.07	0.00	0.76	0.11
School Meals	0.46	0.04	0.64	0.09
WIC	0.46	0.02	0.49	0.02

Notes: This table presents the share of households who do or do not receive transfers, conditional on our simulated eligibility measures. See Appendix B for details on forming the measures of simulated eligibility.

Table A9: Explaining Differences in Take-Up Rates

	PSID	Official
Receipt Rate	9.8%	7% – 17%
Sim. Elig. Rate	14.4%	14% –17%
Take-Up		
Ratio of Rates	68.1%	48% – 84%
Among Sim. Elig.	42.2%	n.a.

Notes: This table reports household rates of receipt, simulated eligibility, and take-up for SNAP. PSID estimates reflect an average over PSID samples for 1997–2019. Official estimates are ranges over annual averages for the same years in U.S. Department of Agriculture (2022b).

Table A10: Review of Estimated Take-Up Rates

Program	PSID	Benchmarks	References
SNAP	42.2%	48% – 84%	U.S. Department of Agriculture (2022b)
Medicaid	54.6%	52% – 70%	Sommers et al. (2012)
Housing Assistance	10.1%	10%, 18.2%	Olsen (2003), Congressional Research Service (2015)
TANF	10.4%	28.4%	Congressional Research Service (2015)
SSI	59.8%	57.5%, 66.6%	McGarry and Schoeni (2015), Congressional Research Service (2015)
WIC	41.4%	39.1% – 65.3%	Guan et al. (2023), U.S. Department of Agriculture (2022a), Congressional Research Service (2015)
LIHEAP	17.0%	22.2%	Congressional Research Service (2015)

Notes: This table reports estimates of take-up rates from other research.

B Data Appendix

This appendix first explains measurement details for consumption, lifetime income, and simulated eligibility. Next, it discusses the estimation of take-up rates.

B.1 Consumption Ranks

We compute households' equivalized consumption ranks using the expenditure data available in the PSID in a given year. Not all consumption categories are available in each year. In particular, we observe expenditures on clothing, furniture, travel, and recreation starting in 2005 and computer expenditures starting in 2017. We observe housing rents (actual and imputed) starting in 1997, and all other expenditures starting in 1999. These expenditure categories are childcare, education, food, health, transportation, and utilities (energy and water starting in 1999, phone/cable/internet starting in 2005).

As noted in Section 2, we follow Meyer and Sullivan (2023) in making two adjustments so that we more closely measure consumption rather than expenditure. Broadly, these adjustments estimate consumption flows from households' two key durable goods, homes and vehicles.

For renters, we take their paid rents as their housing consumption. For owner-occupiers, we obtain imputed rents in several steps. In 2017 and 2019, owner-occupier households were asked "If someone were to rent this (apartment/mobile home/home) today, how much do you think it would rent for per month, unfurnished and without utilities?" We take these values as housing consumption for such households. For all years in our sample period, households who report that their housing is free are asked "How much would it rent for if it were rented?", which we use as their housing consumption. Finally, we construct a mapping from home values to owners' equivalent rents using the cross-sectional relationship in 2017 and 2019 between households' estimates of their home's value and its equivalent rent.[34]

Transportation consumption is constructed as follows. We count any expenditures on gasoline, parking, public transportation, taxis, other transportation toward the household's transportation consumption. Due to PSID data limitations, we also count as consumption any expenditures on vehicles other than the household's three reported primary vehicles. For households that lease any of their three primary vehicles, we count their lease costs toward transportation consumption. For households that own any of their three primary vehicles, we impute the equivalent lease cost from a hedonic regression.

To estimate this hedonic regression, we restructure our data into a vehicle-level dataset. House-

[34]For households that do not report an exact home value, we use the midpoint of the elicited range. For households who say their homes are worth more than $400,000, but do not report an exact value, we impute it as the sample mean conditional on exceeding $400,000 among households who report exact home values.

holds that lease or own a vehicle report the vehicle's manufacturer (e.g., Toyota), its make (e.g., Lexus), its age at acquisition (year of purchase or lease minus model year), and its "type" (car, pickup/truck, van, utility, or motor home). We estimate Poisson regression models of all two-way interactions of these variables, along with indicator variables for calendar year and the rank (1/2/3) of the vehicle in the household's list, The outcomes are purchase price or lease cost, winsorizing values at the first and 99th percentiles. We then collapse these predicted values for purchase price and lease cost to the level of manufacturer, make, age, and type. This procedure yields an estimated lease cost equivalent for owned vehicles.[35]

B.2 Lifetime Income Ranks

Step 1: Estimate lifecycle regression parameters. Letting i index individuals, t index calendar years, and a index age in years, we estimate Poisson regression models of the following form:

$$\mathrm{E}[y_{it} \mid X_{it}] = \exp(\alpha_i \lambda_a + X_{it}' \beta_a), \tag{11}$$

where α_i is an individual fixed effect, α_t is a calendar-year fixed effect, X_{it} is a matrix of time-varying demographic characteristics, and λ_a and β_a are vectors of age-specific coefficients. The outcome y_{it} is individual income. For individuals with zero income in all observed years, we impute a constant annual income of \$100.[36] The age-specific coefficients are initialized to $\lambda_a = 1$ for all a but will be estimated in an outer loop discussed below. We make several adjustments before using the regression results to estimate lifetime-income ranks.

Step 2: Shrink fixed effects. First, we apply the empirical Bayes methods in Morris (1983) to shrink the estimated individual fixed effects $\widehat{\alpha_i}$ toward a conditional expectation fit from several time-invariant individual characteristics.[37] These methods accommodate both unequal individual means and unequal sampling variances in the fixed effects by iteratively re-estimating the extent of true heterogeneity among individuals and the conditional expectation function using weighted least squares. Our baseline specification uses sex, race, and ethnicity to fit this conditional expectation.

Step 3: Outer loop. Haider and Solon (2006) emphasize that the "error-in-variables" model of lifetime income is misspecified, as the predictive effect of individual fixed effects grows over the

[35] For missing values, we impute using the cross-sectional relationship between fitted purchase prices and fitted lease values from these Poisson regressions.

[36] In an unadjusted Poisson regression, estimates of the individual fixed effects α_i diverge to negative infinity for any individual i who earns $y_{it} = 0$ for all observed periods t. By setting their y_{it} to a very low positive value, we obtain convergent fixed effects and rank these individuals at the bottom of the lifetime-income distribution. Importantly, this procedure does affect our estimates of β, as the fixed effects α_i perfectly explain the income of these individuals.

[37] Recent applications of these methods in economics include Chandra et al. (2016) and Sorkin (2018). We refer interested readers to their appendices for detailed expositions. One key modification we make to their approach is to use a within-individual Bayesian bootstrap (Rubin, 1981) instead of actual resampling.

lifecycle. To account for this, we estimate the λ_a terms in Equation 1 through the following outer loop. Consider the first loop, in which we have initialized $\lambda_a = 1$ and have shrunken estimates of α_i. We can estimate the following Poisson model:

$$\mathrm{E}[y_{it} \mid X_{it}] = \exp(\widehat{\alpha}_i \lambda_a + X'_{it}\beta_a), \tag{12}$$

importantly treating $\{\widehat{\alpha}_i\}$ as data rather than as parameters. We then obtain coefficient estimates $\{\widehat{\lambda}_a\}$, and with these in hand, we return to step 1 and iterate until convergence of $\{\widehat{\alpha}_i, \widehat{\lambda}_a, \widehat{\beta}_a\}$. In practice, we find that convergence is fast; three runs of the outer loop are sufficient.

Step 4: Balance the panel. Having estimated the model in Equation 1, we use it to predict income from ages 18 to 65, irrespective of the years in which we observe an individual's actual income. An individual's predicted income in year t is $\widehat{y}_{it} = \exp(\widehat{\alpha}^*_i \widehat{\lambda}_a + X'_{it}\widehat{\beta}_a)$, where $\widehat{\alpha}^*_i$ are the shrunken estimates of the individual fixed effects. Lifetime income are then

$$\overline{y}_i = \sum_t \widehat{y}_{it}, \tag{13}$$

where the summation over t is for the years $\{\underline{T}_i, \ldots, \overline{T}_i\}$ in which individual i is between the ages of 18 and 65. Importantly, however, we do not observe individual characteristics X_{it} in all years and therefore must impute them. In our baseline specification, we assume these characteristics are unchanged from the nearest period of observation, except for age.

Step 5: Construct ranks. We define an individual's lifetime income percentile rank as $\Pr(y \leq \overline{y}_i \mid c_i = c)$, where $\overline{y}_i$ is their estimated lifetime income and c_i is their birth-year cohort. We define an individual's current income percentile rank as $\Pr(y \leq y_{it} \mid c_i = c)$, again ranking individuals each year within their birth cohorts. Appendix A presents figures of our main results when we do not rank current income within cohorts.

We construct current and lifetime household income percentile ranks as follows. Let $j(i,t)$ indicate i's spouse in year t, and let $h(i,t)$ indicate the household of which i is a part at t. As explained above, current household income is the sum of the head's and spouse's individual current income: $y^h_{i,t} = y_{it} + y_{j(i,t),t}$. Our lifetime concept of household income follows each individual through the sequence of households during their adult life, again using individuals' income fitted from Equation 1 and the subsequent adjustments. That is, the lifetime household income of individual i is

$$\overline{y}^h_i = \sum_t e(\widehat{y}^h_{it}) = \sum_t e(\widehat{y}_{it} + \widehat{y}_{j(i,t),t}) \tag{14}$$

where t is again summed over the years in which i is between ages 18 and 65. The function $e(\cdot)$ equivalizes household income for differences in household size in each year. If we were to restrict our sample to stable households over time (as in, e.g., Fullerton and Lim Rogers, 1993), our definition of household income would coincide exactly with the natural concept. However, it accommodates unstable households in a way that is meaningful as a measure of living standards.

B.3 Simulated Eligibility

Supplemental Nutrition Assistance Program (SNAP). SNAP eligibility is determined on the basis of three tests: (1) a gross-income test, (2) a net-income test, and (3) an asset test. Recipients of TANF and SSI are always categorically SNAP-eligible.

We use state-level gross-income tests from 1996 to 2016 from SNAP Policy Database, maintained by the Economic Research Service of the U.S. Department of Agriculture.[38] We assume these thresholds are unchanged from 2016 through 2019. Until 2000, all U.S. states had a SNAP gross-income test at 130 percent of the Federal Poverty Level (FPL). Under "broad-based categorical eligibility" (BBCE), states raised gross-income limits.

The net-income test requires that income net of specific deductions is less than 100 percent of the FPL. Starting from gross income, all households take a standard deduction as a function of their household size; they also deduct 20 percent of household earnings from gross income. There are four further deductions that may be applied to gross income. We focus on the most important, the "excess shelter deduction." This deduction subtracts housing costs, inclusive of utilities, that exceed half of net income after accounting for all other deductions. The excess-shelter deduction is capped at a level that depends on household size. Standard deductions and excess-shelter deduction caps vary by year but are different for Alaska and Hawaii; we collected these policy parameters from Federal Register notices. The three other deductions—for child support, medical expenses, and dependent care—appear rarely used in eligibility determinations, and we ignore them.[39]

We use asset-test thresholds from the SNAP Policy Database. We apply the asset test rules to household liquid savings, due to the exemption of most relevant other categories of wealth. The asset limit for nonelderly households was $2,000 from the 1980s until 2014, when it was raised to $2,250. The asset test is eliminated under BBCE.

There are special eligibility rules covering households with elderly or disabled adults. In particular, these households are only subject to the net-income test (no gross-income test). They also face higher asset-test threshold of $4,250, unless the threshold has been raised under BBCE. We assume the asset-test threshold for such households is the maximum of $4,250 and their BBCE asset-test threshold for all other households.

Special eligibility rules also apply to non-citizen immigrants according to their date of arrival

[38] See https://www.ers.usda.gov/data-products/snap-policy-data-sets/.

[39] For further details, see Center on Budget and Policy Priorities, "A Quick Guide to SNAP Eligibility and Benefits."

and current immigration status, pursuant to the Personal Responsibility and Work Opportunity Reconciliation Act of 1996 (PRWORA) and subsequent state and federal laws. We code these from Zimmermann and Tumlin (1999) and capture subsequent updates in an April 2022 report by the Food Research and Access Center, "State Food Assistance Programs: Addressing Gaps in SNAP Eligibility for Immigrants."

Medicaid. Medicaid eligibility is determined by income and asset tests that vary by state and with household characteristics. After 2013, interstate variation in eligibility follows largely from the Affordable Care Act's Medicaid expansion. Before 2013, interstate variation is driven by state use of Medicaid waiver programs. In most states, SSI recipients are categorically Medicaid-eligible; we also apply this to states which, under the "209(b)" rules, in principle have some Medicaid eligibility rules that are more stringent than for SSI.

Income eligibility thresholds come primarily from the Kaiser Family Foundation (KFF), with our supplementation to fill gaps in the data. We imputed that thresholds did not change when there are data gaps but we know thresholds on both ends of the gap were the same. Different income tests apply to non-disabled adults, parents, and pregnant women. Income eligibility is most complicated for disabled adults, who may become eligible under a number of pathways, including Medicaid buy-in and being "medically needy." We determine whether a household qualifies as medically needy using reported health expenditures.

We hand-collected Medicaid asset-test thresholds from state-agency websites and policy reports that will be included in our replication files. The thresholds vary for singles and couples, and for the Medicaid buy-in and medically-needy pathways. When we were unable to find state asset-test thresholds in a given year, we imputed it from surrounding years or used the federal thresholds.

Special eligibility rules also apply to non-citizen immigrants according to their date of arrival and current immigration status, pursuant to the Personal Responsibility and Work Opportunity Reconciliation Act of 1996 (PRWORA) and subsequent state and federal laws. We code these using Zimmermann and Tumlin (1999), as well as periodic reports of the National Immigration Law Center and the U.S. Department of Health and Human Services (Assistant Secretary for Planning and Evaluation).

Housing Assistance. Eligibility for housing assistance (Section 8 and public housing) is determined by income. Income is measured relative to Area Median Income (AMI) at the level of metropolitan area or non-metropolitan county. As we do not have sub-state geographic identifiers, we use state-level AMIs by household size. Public housing authorities may set their income thresholds between 50 percent and 80 percent of AMI. We assume an eligibility threshold of 50 percent of AMI, as large-city public housing authorities typically impose this threshold at voucher take-up or occupancy of the public-housing unit.

There is no asset test for housing assistance. Until 2014, however, households with no actual asset income but significant wealth could be excluded from housing assistance on the basis of imputed asset income. This imputation used a "passbook savings rate" of two percent until 2014. In 2014, HUD Notice H 2014-15 set this rate to almost zero, essentially eliminating the treatment of assets as income.

Special eligibility rules also apply to non-citizen immigrants according to their date of arrival and current immigration status. Approximating PRWORA, we code a household as ineligible if it contains no citizens or "qualified" non-citizen immigrants (permanent residents, asylees, refugees, and Temporary Protected Status).

Supplemental Security Income (SSI). SSI eligibility is determined by disability of an adult or child member of the household, an income test, and an asset test.

Households are ineligible if their income exceeds a federal "substantial gainful activity" (SGA) threshold. This SGA threshold rose gradually from \$500 per month in 1997 to \$1,220 in 2019. We also label households ineligible if their countable income exceeds the Federal Benefit Rate (FBR), which implies they would not be eligible for a positive SSI benefit amount. Monthly countable income for SSI is defined by the following formula:

$$y_{\text{countable}} = \max\{0, y_{\text{earned}} + y_{\text{unearned}} - 0.5 \cdot \max\{0, y_{\text{earned}} - 65\} - 20\},$$

where y_{earned} and y_{unearned} are monthly earned and monthly unearned income respectively.

Single-adult households are ineligible for SSI if they possess more than \$2,000 in countable assets. The asset threshold is \$3,000 for couples. Countable assets are financial assets only after 2005 and financial assets plus the excess of vehicle wealth above \$4,500 before 2005.

Special eligibility rules also apply to non-citizen immigrants according to their date of arrival and current immigration status, due to PRWORA and subsequent state and federal laws. We code these using Zimmermann and Tumlin (1999), as well as periodic reports of the National Immigration Law Center.

Women, Infants, and Children (WIC). WIC eligibility is determined by the presence of a child under age five in the household and an income test. The income test is that their income is no greater than 185 percent of the FPL. Households are also categorically WIC-eligible if they have such a child and receive SNAP, TANF, or Medicaid.

Low-Income Heating and Energy Assistance Program (LIHEAP). A household is LIHEAP-eligible if they pay utilities, satisfy an income test, and satisfy an asset test. We determine whether a household pays utilities based on reported utility expenditures.

States set their own income-test thresholds, and these differ by LIHEAP sub-program. Our eligibility simulation focuses on non-crisis heating assistance, the largest sub-program. For 1997–2007 and 2015–2019, we obtain these from the LIHEAP Clearinghouse website, using Internet Archive to obtain the first interval. We obtained the intermediate years from LIHEAP Reports to Congress.

Information was more limited on LIHEAP asset tests. From the Clearinghouse, Reports to Congress, and state-agency websites, we were able to determine whether states had asset tests for all years. The levels of the asset threshold, however, we have only beginning in 2015. We assume these thresholds were unchanged from 1997 to 2015 if the state always had an asset test. For states that had an asset test but eliminated it before 2015, we impute a limit of $5,000. We assume the assets covered by the test are liquid savings, although definitions appear to vary somewhat by state.

States may also make SNAP, SSI, and TANF recipients categorically eligible for LIHEAP. We obtained states' categorical-eligibility rules for fiscal year 2019 from the "Detailed Model Plan" submissions included in their SF-424 grant applications for federal LIHEAP funds. We assume that categorical-eligibility rules are unchanged over the entire period.

Special eligibility rules also apply to non-citizen immigrants according to their date of arrival and current immigration status. Approximating PRWORA, we code a person as ineligible if they are neither a citizen nor a "qualified" non-citizen immigrant (permanent residents, asylees, refugees, and Temporary Protected Status).

School Meals. A household is eligible for the National School Lunch Program and the School Breakfast Program if they have a school-age child (ages 5 to 18) and have an income less than 185 percent of the FPL. We use the threshold to qualify for reduced-price meals. The threshold is 150 percent of the FPL for free meals. Households can also be categorically eligible if they receive SNAP, TANF, or other means-tested transfers.[40]

The Healthy Hunger-Free Kids Act of 2010 established the Community Eligibility Provision (CEP), which offers free school meals universally in high-poverty areas. We do not account for school-meals eligibility via the CEP, as we lack sub-state geographic identifiers.

Temporary Aid for Needy Families (TANF). We heavily rely on data and eligibility simulations from the Urban Institute's TRIM3 model. We use the following variables from TRIM3: state-year-household size gross income eligibility thresholds; state-year "standard of need" data, which scale the income eligibility thresholds; state-year "adjustment variables," which adjust the standard of need; state-year TANF asset thresholds; state-year earnings disregards (fixed levels and shares of income). Some states use a net asset test; in those cases, we impute their gross threshold as the median non-missing gross standard of need.

[40]See Rebecca R. Skinner and Randy Alison Aussenberg, "Overview of ESEA Title I-A and the School Meals' Community Eligibility Provision," Congressional Research Service Report R44568, 2016.

A household's income is below the TANF eligibility threshold if their income, less the disregard, is less than the standard of need times the eligibility threshold. A household is eligible for TANF if they have a child, are below the eligibility threshold, and are below the asset threshold.

This eligibility simulation neglects the following forces. First, we do not incorporate the TANF net income thresholds. Second, we assume that the vehicle asset test deducts the full value of the vehicle, which occurs in 39 states of 50 states. Third, some states do not require children in the household to be eligible.

Special eligibility rules also apply to non-citizen immigrants according to their date of arrival and current immigration status, pursuant to the Personal Responsibility and Work Opportunity Reconciliation Act of 1996 (PRWORA) and subsequent state and federal laws. We code these using Zimmermann and Tumlin (1999) and a 2014 report for the U.S. Department of Health and Human Services (Assistant Secretary for Planning and Evaluation).

B.4 Data Sources on Budgetary Cost

- *SNAP*: Laura Tiehen, "The Food Assistance Landscape: Fiscal Year 2019 Annual Report," Economic Research Service, U.S. Department of Agriculture, July 2020.
- *Medicaid*: U.S. Centers for Medicare & Medicaid Services, "CMS Office of the Actuary Releases 2019 National Health Expenditures," 16 December 2020.
- *Housing Assistance*: Donna Kimura, "Fiscal 2019 HUD Budget Approved," *Affordable Housing Finance*, 20 February 2019.
- *SSI*: Office of Research, Evaluation, and Statistics and Office of Retirement and Disability Policy, Social Security Administration, "SSI Annual Statistical Report, 2019," SSA Publication No. 13-11827, August 2020.
- *TANF*: Office of Family Assistance, Administration for Children & Families, U.S. Department of Health and Human Services, "TANF and MOE Spending and Transfers by Activity, FY 2019," 22 October 2020.
- *WIC*: Food and Nutrition Service, U.S. Department of Agriculture, "WIC Program Participation and Costs," 10 February 2023.
- *LIHEAP*: Office of Community Services, Administration for Children & Families, U.S. Department of Health and Human Services, "LIHEAP DCL Funding Release FY 2019," 26 October 2018.
- *School Meals*: Economic Research Service, U.S. Department of Agriculture, "National School Lunch Program" and "School Breakfast Program," 3 August 2020.

B.5 Take-Up and Receipt Rates

In this section, we examine why take-up rates in Table 1 are smaller than official estimates. Here, we define take-up rates to be the share of households that take up, among households we simulate to be eligible. Receipt rates are the share of households that take up, among all households.

Summary Statistics. Appendix Table A8 reports sample-average estimates of take-up rates as well as three other quantities: (1) receipt rates among the simulated-ineligible, (2) rates of simulated eligibility among receipients, and (3) rates of simulated eligibility among nonrecipients.

Take-Up Rates in SNAP. Appendix Table A9 compares our estimates of take-up rates to USDA official estimates. Ours are clearly lower than the USDA estimates.

One explanation for the discrepancy is that, without conditioning on simulated eligibility, a lower share of households report that they receive SNAP in the PSID than are known to receive SNAP in administrative data. The implied rate of underreporting of SNAP benefits is roughly consistent with prior research (e.g., Meyer et al., 2009). Underreporting of receipt can thus explain some of the difference in take-up rates. We see this as encouraging: A substantial literature has developed around this form of measurement error in take-up rates, and we use corrections for underreporting developed in Mittag (2019) as a robustness check (see Section 3).

A second explanation for low take-up rates in the PSID relates to how eligibility enters official estimates. Appendix Table A9 shows two methods of computing SNAP take-up rates: (1) the ratio of total receipt to total simulated eligibility and (2) the receipt rate among the simulated eligible. The first approach is accurate only if every recipient is simulated ineligible. We avoid that assumption by taking the second approach in the main text.

By applying both approaches in the PSID, we can offer novel evidence on this potential contributor to the discrepancy in estimates of take-up rates. Appendix Table A9 shows that, when we follow the first approach, we move toward the USDA take-up rates. The second approach, however, yields a lower take-up rate due to what we call "misalignment": Many recipient households in administrative data are likely to be simulated as ineligible if they were to appear in the survey data. Such households would enter the numerator but not the denominator of the USDA estimate, and this appears to be a key factor in the discrepancy. For this reason, official estimates of take-up rates —which divide the administrative data in the numerator by simulated eligibility estimates, without adjusting for false negatives in eligibility—likely overstate true rates.

The most striking example of this upward bias is when the first approach yields estimates exceeding 100-percent take-up. For instance, in fiscal year 2018, the USDA reported take-up statistics with such issues in four states: DE, IL, OR, WA.[41] Furthermore, in all of these states in that year, there were specific demographic cells with official reported take-up rates of less than 100

[41] See https://www.fns.usda.gov/usamap/2018.

percent, which is mathematically inconsistent with 100-percent take-up overall. In our view, these challenges suggest that survey-data measures of take-up are not obviously dominated by approaches leveraging administrative data.

Benchmarking Other Programs. Appendix Table A10 reports a selection of estimated take-up rates in other research. These estimates come from a variety of data sources (administrative and survey) and time periods. Overall, our estimates of take-up rates are lower than administrative estimates, but they appear generally consistent with other survey-based estimates of self-reported receipt among simulated eligibles. For these programs, the most plausible explanation for low take-up rates is survey underreporting of transfer receipt, as in, e.g., Meyer et al. (2009).

C Further Results

This appendix presents several analyses left out of the main paper due to space constraints.

C.1 Distributional Analysis

Using the distributional analysis framework in Section 4, we report descriptive statistics about the expected value of transfer payments over three rankings of households: current income, current consumption, and lifetime income. All three of these ranking variables are equivalized for household size and composition following Citro and Michael (1995).

Appendix Figure A1 plots the average total annual per-capita value of transfer benefits as functions of these ranks. In the left sub-panel, we combine transfers across the eight transfer programs we study. The right sub-panel excludes in-kind programs, for which we impute cash equivalents. Appendix Figure A2 shows results separately by program.

Both figures also show that transfer incidence with respect to consumption, lifetime income, and current income are remarkably similar, overall and for most individual transfer programs. That is, average annual per-capita benefits are about as high at the bottom of the distributions of consumption and lifetime income as they are at the bottom of the current-income distribution. Households in the bottom five percentiles of the income distribution, for instance, receive approximately $3,700 in transfers per person per year, as compared to $4,300 for the lowest-consumption households.

These results reflect the approximate balance of two forces—self-targeting and within-lifetime income fluctuations—as we show through a formal decomposition in this appendix. Within-lifetime income volatility moves the lifetime or consumption incidence of transfers higher up into these distributions, as some amount of benefits goes to households in their lower-income spells who ultimately have high consumption and lifetime earnings (as in, e.g., Poterba, 1989). Yet our results suggest the compressive effect of income fluctuations is fully undone by self-targeting. That is, transfers have so little incidence at high lifetime ranks because high-lifetime-income households rarely take up transfers even in low-income spells of temporary eligibility eligible. By contrast, low-consumption households do take up transfers when eligible.

Although eligible nonrecipients of most transfers are poor in current-income terms, many have significant consumption or lifetime resources, as we show in Appendix Figure A4. This result captures the two-sided implications of self-targeting: If recipients are advantageously selected, then the complementary population of eligible non-recipients must be adversely selected. For instance, one third of eligible nonrecipients of SNAP and Medicaid have above-median consumption, and a similar fraction has above-median lifetime income. The share of households who are eligible but do not take-up is therefore much lower among the consumption-poor and lifetime-poor than among the current poor. For instance in Medicaid, about one fifth of the consumption- and lifetime-poorest are eligible but do not take up, as compared to one third of current-poorest.

C.2 A Decomposition of Distributional Incidence

Section 3 uses regressions to show that take-up, not only eligibility, explains why transfer receipt identifies people with low consumption given income. Here we extend the decomposition approach of Brewer et al. (2020) to carefully distinguish between selection via eligibility rules and selective take-up among the eligible.

We will use this decomposition to answer question such as: Why do a greater or smaller share of people at consumption rank r receive a transfer than do people at current-income rank r? Our decomposition is

$$\overline{p}(r) - p(r) = \underbrace{[\overline{p}(r) - s(r)]}_{\text{take-up effect}} + \underbrace{[s(r) - q(r)]}_{\text{eligibility effect}} + \underbrace{[q(r) - p(r)]}_{\text{mobility effect}}. \tag{15}$$

We define each of these three terms in what follows.

Mobility Effect. The mobility effect is the component of the difference between incidence with respect to consumption and incidence with respect to income that results from year-to-year income mobility. To measure the mobility effect, we estimate the share $q(r)$ of people with consumption rank $\bar{R}_i = r$ who would have received a given transfer in a given year if, counterfactually, transfer receipt were only a function of current-income rank. That is, our mobility-only counterfactual is

$$q(r) = \int \Pr(D_{it} = 1 \mid R_{it})\, dF(R_{it} \mid \bar{R}_i = r), \tag{16}$$

where D_{it} indicates transfer receipt and $F(R_{it} \mid \bar{R}_i = r)$ is the conditional distribution of current-income ranks R_{it} at consumption rank r. The mobility effect at rank r is the difference between $q(r)$ and the empirical receipt rate at a current-income rank r, $p(r) = \Pr(D_{it} = 1 \mid R_{it} = 1)$.

Intuitively, the mobility effect applies the consumption–income "transition matrix" to the receipt rate by income rank, yielding the counterfactual receipt rates at each consumption rank.

Eligibility Effect. The eligibility effect is the component of the difference in incidence due to eligibility rules that, among low-income households, tag those with low consumption and low lifetime income. To measure the eligibility effect, we define the receipt rate at consumption rank r under a second counterfactual, $s(r)$. Here the probability of take-up conditional on eligibility is a function of current income alone, whereas we let eligibility depend on both current income and consumption (or current income and lifetime income). This counterfactual receipt rate $s(r)$ thus depends only on eligibility and current-income rank. The difference between these two counterfactuals, $q(r)$ and $s(r)$, is the eligibility effect.

When eligibility can be measured perfectly—that is, $E_i = 0$ implies $D_i = 0$—our second counterfactual is defined as

$$s(r) = \int \Pr(D_{it} = 1 \mid R_{it}, E_{it} = 1) \Pr(E_{it} = 1 \mid R_{it}, \bar{R}_i = r) \, dF(R_{it} \mid \bar{R}_i = r),$$

where E_{it} indicates eligibility. This expression emerges from the following reasoning. The probability $\Pr(D_{it} = 1 \mid R_{it}, E_{it} = 1)$ is the transfer take-up rate among the eligible at current-income rank R_{it}. The probability $\Pr(E_{it} = 1 \mid R_{it}, \bar{R}_i = r)$ is the eligibility rate at current-income rank R_{it} and consumption rank r. Multiplying these two probabilities yields a predicted receipt rate for people with current-income rank R_{it} that embeds the desired independence assumption: Given current income, take-up among the eligible is uninformative about consumption or lifetime income.[42] Integrating over the conditional distribution of current-income ranks given consumption, $F(R_{it} \mid \bar{R}_i = r)$, we have a counterfactual receipt rate $s(r)$ at consumption rank r that permits a role for eligibility rules while shutting down selective take-up.

Imperfect measurement of eligibility requires a more complex expression for the counterfactual. We now take the counterfactual to be:

$$s(r) = \int \Big[\Pr(D_{it} = 1 \mid R_{it}, E_{it} = 0) \Pr(E_{it} = 0 \mid R_{it}, \bar{R}_i = r) + \Pr(D_{it} = 1 \mid R_{it}, E_{it} = 1) \Pr(E_{it} = 1 \mid R_{it}, \bar{R}_i = r) \Big] \, dF(R_{it} \mid \bar{R}_i = r),$$

where all terms are as defined above. We use $s(r)$ as the "automatic counterfactual" in the main text of the paper.

[42]To better understand this counterfactual, note that the law of conditional probability implies that: $\Pr(D_{it} = 1 \mid \bar{R}_i = r) = \int \Pr(D_{it} = 1 \mid R_{it}, \bar{R}_i = r) \, dF(R_{it} \mid \bar{R}_i = r)$. By consequence, the counterfactual receipt rate $s(r)$ equals the empirical receipt rate at a given consumption rank if and only if take-up among eligibles is conditionally independent of consumption given current income at all consumption ranks.

Take-Up Effect. The take-up effect is the residual component after accounting for mobility and eligibility. Intuitively, there is some take-up effect when consumption and lifetime income matter for take-up rates among the eligible. We define the take-up effect as the residual difference between the counterfactual receipt rate $s(r)$ and the empirical receipt rate at the same consumption rank, $\overline{p}(r) = \Pr(D_{it} = 1 \mid \bar{R}_i = 1)$.

C.3 Results of Decomposition

Appendix Figure A21 displays decomposition results for the difference between current-income and consumption incidence. Panel A displays results for the average total annual per-capita value of benefits, and Panel B displays results for receipt rates by program. The blue, black, and yellow lines plot the mobility, eligibility, and take-up effects, respectively; each represents the difference in consumption incidence with respect to current-income incidence if only this effect were present. The blue shaded region indicates the net effect, equal to the difference between the consumption and current-income incidence at the indicated rank; this region integrates to zero.

Absent eligibility and take-up effects, the incidence of transfer programs would be markedly less progressive at the bottom of the distribution. Consistent with our prior results, we find a primary role for selective take-up among eligible households, and a secondary role for eligibility rules, in shaping the consumption incidence of transfer programs.

In total over transfer programs, selective take-up increases the average annual value of benefits received by the consumption-poorest people by about \$1700 per person relative to a counterfactual in which transfer receipt is a function of current income and eligibility alone. Eligibility rules contribute about \$250 per person at the bottom of the consumption distribution. Together these amounts represent about one quarter of the average transfer per capita per year to consumption-poorest people. Mobility tends to shift the incidence of transfers up the consumption distribution, but across programs, the eligibility and take-up effects largely offset the mobility effect.

The primacy of take-up applies broadly across the eight programs we study. Eligibility rules appear most important in TANF, school meals, and WIC, whereas they appear essentially irrelevant for housing assistance. While mobility alone would reduce the receipt rate among the consumption poor by about 10 percentage points for SNAP, Medicaid, and housing assistance, selective take-up increases receipt rates in these programs at the bottom of the consumption distribution by 10 to 20 percentage points. These are economically large differences relative to underlying rates of transfer receipt as well as relative to the effects of mobility.

C.4 Selection Over Time

Our data span 1997 to 2019, allowing us to address how the U.S. safety net has evolved over this period. We estimate a version of Equation 4 that allows for year-specific coefficients on transfer receipt. To allow us to describe broad trends, we "stack" the data over programs and include program-specific controls for current income. Across our eight transfer programs, Appendix Figure A19 shows little detectable change in either kind of self-targeting over time.

C.5 Measurement Error: Simulation

How much measurement error is required to overturn our results? We conduct an adversarial simulation to probe the robustness of Figure 1 to extreme amounts of measurement error. We consider the coefficient γ in Equation 4, which represents the marginal effect of take up of a given transfer program on lifetime rank, controlling for current rank.[43] We simulate measurement error as follows:

1. Obtain the take-up rate among the bottom 50% of households ranked in consumption terms, $\hat{D} \in [0, 1]$.
2. Assign the top x% in consumption ranks to have some constant $c \in [0, 1]$ times the take-up rate of the bottom 50%: $c\hat{D}$.
3. Estimate Equation 4 using the simulated data.

This exercise generates a large amount of measurement error at the top of the distribution. The take-up rate in the bottom 50% of the current consumption distribution is a natural bound on the take-up rate of the top x% of the consumption distribution, unless the programs' targeting properties are very perverse. The parameter c governs whether the measurement error is as severe as possible ($c = 1$).

We find measurement error would need to be very severe to overturn our results. Appendix Figure A20 shows the estimated coefficient $\hat{\beta}$ as a function of the share of the top of the consumption distribution that has severe measurement error. In black, we present the estimates if the true take-up rate is half the bottom 50%'s take-up rate. The blue lines show the estimates if the true take-up rate is the same as the bottom 50%'s take-up rate. When $x = 0$, the estimates coincide with Figure 1. As long as true take-up at the top is half the poorest's take-up rate, we continue to reject $\beta = 0$. If true take-up is equal, then we can no longer reject $\beta = 0$ for $x \geq 15$ or so. These results are logical: if we impute take-up rates that are the same as at the bottom of the distribution for much of the top

[43] We do not condition on simulated eligibility in these specifications, to isolate the magnitude of take-up measurement error without controlling for a potentially contaminated confound.

of the distribution, we no longer find evidence of selection. But as long as measurement error does not exceed half the take-up rate, we decisively reject the null.

D Theory Appendix

D.1 Proof of Proposition 1

Proof. We begin by analyzing the household's problem. Let η_t and λ_t denote, respectively, the Lagrange multipliers on the household's period-t borrowing and budget constraints. The first-order conditions of the household's problem (Equation 7) are

$$[c_t] \quad U'(c_t - v(z_t;\theta_t);\theta_t) - \eta_t = 0 \tag{17}$$

$$[z_t] \quad U'(c_t - v(z_t;\theta_t);\theta_t)v'(z_t;\theta_t) + \eta_t[1 - T'(z_t) + S'(z_t)M(z_t;\theta_t)] = 0 \tag{18}$$

$$[a_{t+1}] \quad (1+\rho)^{-1}\mathrm{E}[V'(a_{t+1};\theta_{t+1}) \mid a_{t+1},\theta_t] - \eta_t - \lambda_t = 0. \tag{19}$$

We note that, for the first-order condition with respect to z_t, there is no term $S(z)M'(z;\theta)$ coming from a reduction in the take-up probability when the household chooses a higher z_t. This is by the Leibniz rule:

$$\int_0^{S(z)} [S(z) - \kappa]\, \mu(\kappa \mid \theta)\, d\kappa \tag{20}$$

$$= [S(z) - S(z)] \cdot S'(z) - [S(z) - 0] \cdot 0 + \int_0^{S(z)} S'(z)\, \mu(\kappa \mid \theta)\, d\kappa \tag{21}$$

$$= S'(z) \int_0^{S(z)} \mu(\kappa \mid \theta)\, d\kappa \tag{22}$$

$$= S'(z)M(z;\theta). \tag{23}$$

Then, combining the first-order conditions with respect to c_t and z_t, we obtain

$$1 - T'(z_t) + S'(z_t)M(z_t;\theta_t) = -v(z_t;\theta_t). \tag{24}$$

This equation highlights the key property of GHH preferences in our context, which is that the $U'(\cdot)$ terms cancel out. By consequence, there are no income effects in labor supply.

Combining the first-order conditions for c_t and a_{t+1}, we derive the Euler equation:

$$U'(c_t - v(z_t;\theta_t);\theta_t) = (1+\rho)^{-1}\mathrm{E}[V'(a_{t+1};\theta_{t+1}) \mid a_{t+1},\theta_t] - \rho_t. \tag{25}$$

Next, we calculate elasticities of labor supply with respect to tax and transfer changes in terms of primitives. Following Jacquet and Lehmann (2014), we apply perturbations to the tax and transfer system about z_0 of the form $\hat{T} = T + \tau(z - z_0) - \nu$ and $\hat{S} = S + \varsigma(s - s_0) - \vartheta$. This reform increases the marginal tax rate or marginal transfer rate by τ or ς and decreases the tax level or transfer level by ν or ϑ.

The key change is to Equation 24, which becomes

$$\mathcal{F}_1 = 1 - T'(z_t) - \tau + [S'(z_t) - \varsigma] \Pr\left(S(z_t) + \varsigma(z_t - z_0) \leq \kappa_t \,|\, z_t; \theta_t\right) - v'(z_t; \theta_t). \tag{26}$$

To use the implicit function theorem, we calculate the derivatives:

$$\mathcal{F}_{1,z}|_{\tau=\varsigma=0} = -T''(z_t) + S''(z_t)M(z_t; \theta_t) + [S'(z_t)]^2 M(z_t; \theta_t) - v''(z_t; \theta_t) \tag{27}$$

$$\mathcal{F}_{1,\tau}|_{\tau=\varsigma=0} = -1 \tag{28}$$

$$\mathcal{F}_{1,\varsigma}|_{\tau=\varsigma=0} = M(z_t; \theta_t). \tag{29}$$

We note that $z_t \to z_0$ as $\tau, \varsigma, \vartheta \to 0$. By comparison, there is no impact of the reform on the Euler equation (Equation 25):

$$\mathcal{F}_2 = U'(c_t - v(z_t; \theta_t); \theta_t) - (1 + \rho)^{-1}\mathrm{E}[V'(a_{t+1}; \theta_{t+1}) \,|\, a_{t+1}, \theta_t] + \rho_t \tag{30}$$

As a result, we have that

$$\mathcal{F}_{2,a}|_{\tau=\varsigma=0} = -U''(c_t - v(z_t; \theta_t); \theta_t) - (1 + \rho)^{-1}\mathrm{E}[V''(a_{t+1}; \theta_{t+1}) \,|\, a_{t+1}, \theta_t] + \frac{\partial \rho_t}{\partial a_{t+1}} \tag{31}$$

$$\mathcal{F}_{2,\tau}|_{\tau=\varsigma=0} = 0 \tag{32}$$

$$\mathcal{F}_{2,\varsigma}|_{\tau=\varsigma=0} = 0. \tag{33}$$

Hence, by the implicit function theorem, we have that

$$\frac{\partial z}{\partial \tau} = \frac{-1}{\mathcal{F}_{1,z}(z; \theta)}, \quad \frac{\partial z}{\partial \varsigma} = \frac{M(z; \theta)}{\mathcal{F}_{1,z}(z; \theta)}, \quad \frac{\partial a}{\partial \tau} = \frac{\partial a}{\partial \varsigma} = 0. \tag{34}$$

For later use, we define the compensated labor supply elasticity as:

$$\varepsilon_z(z; \theta) = \frac{1 - T'(z)}{z} \frac{\partial z(\theta)}{\partial(1 - \tau)}\bigg|_{\tau=0} \tag{35}$$

We now wish to be more precise about the specific perturbation reform. The reform is local to an income z_0, and thus $\hat{T} = T + \tau(z - z_0) - \nu$ and $\hat{S} = S + \varsigma(s - s_0) - \vartheta$. We set the level shift in $S(z)$ to $\vartheta = ds$, the change in the marginal tax rate to $\tau = \frac{d}{dz}\bar{M}(z)ds$, and the decrease in everyone's level of taxes to $\nu = \mathrm{E}[S(z)\bar{m}(z)] + \bar{M}(z)ds$ due to the reduced voluntary transfer expenditure.[44] Here, the term $\bar{m}(z)$ is the density of households with income z who are indifferent to take-up. Finally, any changes in revenue due to changes in labor supply are redistributed as a lump sum.

We proceed in deriving Equation 9 term by term.

First Term (Self-Targeting). The total direct tax change accruing to pre-reform income level z is a reduction of $\bar{M}(z)ds$, which has a welfare effect of $\mathrm{E}\left[\alpha(\theta)V'_a(a;\theta) \mid z = z(\theta)\right] \bar{M}(z)ds$ at z.

This welfare effect is weighed against the reduction of the transfer by ds at all incomes. Again applying the Leibniz rule, this has a welfare effect of

$$ds \int_z \mathrm{E}_{\kappa \leq S(z)} \left[\alpha(\theta)V'_a(a;\theta) \mid z\right] \bar{M}(z)\, dH(z). \tag{36}$$

Combining these two welfare effects, we arrive at the first term:

$$dW_1 = ds \int_z \bar{M}(z) \left(\mathrm{E}\left[\alpha(\theta)V'_a(a;\theta) \mid z\right] - \mathrm{E}_{\kappa \leq S(z)} \left[\alpha(\theta)V'_a(a;\theta) \mid z\right]\right) dH(z). \tag{37}$$

Then, by the law of total expectation, we have that $\mathrm{E}[\alpha(\theta)|z] = \bar{M}(z)\mathrm{E}_{\kappa \leq S(z)}[\alpha(\theta)|z] + (1 - \bar{M}(z))\mathrm{E}_{\kappa > S(z)}[\alpha(\theta)|z]$. Substituting this into Equation 36 and some algebra yields the following expression:

$$ds \int_z \bar{M}(z)(1 - \bar{M}(z)) \left(\mathrm{E}_{\kappa \leq S(z)} \left[\alpha(\theta)V'_a(a;\theta)|z\right] - \mathrm{E}_{\kappa > S(z)} \left[\alpha(\theta)V'_a(a;\theta)|z\right]\right) dH(z),$$

and then we can use the definition of the regression coefficient $\beta(z) = \mathrm{E}_{\kappa \leq S(z)}[\alpha(\theta)|z] - \mathrm{E}_{\kappa > S(z)}[\alpha(\theta)|z]$ to obtain

$$\int_z \frac{\bar{M}(z)(1 - \bar{M}(z))}{\int_z \bar{M}(z)(1 - \bar{M}(z))\, dH(z)} \beta(z)\, dH(z),$$

and then by the variance-weighting property of least squares, we reach the expression in the main text:

$$dW_1 = \beta \sigma_M^2\, ds, \tag{38}$$

where the within-income variance of take-up is $\int_z \bar{M}(z)(1 - \bar{M}(z))\, dH(z)$.

Second Term (Take-Up Costs). Due to the fiscal savings from marginal recipients, there is also a lump-sum transfer to all households. As the transfer benefit $S(z)$ declines by ds for all z, the total

[44] The derivative $\frac{d}{dz}\bar{M}(z)$ is a total derivative from the planner's perspective. That is, it shifts across people at different incomes z. It includes changes the $\bar{M}(z)$ in z due to both the the schedule $S(z)$ varying in z and the distribution of $\kappa \mid w$ varying in z.

fiscal savings across all incomes z is $\mathrm{E}\left[S(z)m(z;\theta)\right]$, where $m(z;\theta)$ is the density of $M(z;\theta)$ near indifference to take-up.

The resultant welfare gain from the lump-sum transfer is $\mathrm{E}\left[S(z)m(z;\theta)\right] ds$, recalling the normalization of the population-average welfare weight to one. We now move from the density $m(z;\theta)$ to a take-up elasticity by

$$\varepsilon_b(z;\theta) = S(z)\frac{m(z;\theta)}{M(z;\theta)}, \tag{39}$$

since $d\bar{M}(z)/dS(z) = \bar{m}(z)$. We can manipulate this expression to obtain $M(z;\theta)\varepsilon_b(z;\theta) = S(z)m(z;\theta)$. Using this as a substitution, the welfare gain is

$$dW_2 = ds\mathrm{E}[M(z;\theta)\varepsilon_b(z;\theta)] = ds\bar{M}(z)\bar{\varepsilon}_b(z), \tag{40}$$

where this weighted-average take-up elasticity is

$$\bar{\varepsilon}_b = \int_{z\times\Theta} \frac{M(z;\theta)\varepsilon_b(z;\theta)}{\int_{z\times\Theta} M(z;\theta)\, d\mu(z;\theta)}\, d\mu(z;\theta) \tag{41}$$

and $\bar{M}(z)$ reflects the overall take-up rate at income z.

Third Term (Labor Supply). The government has reduced tax burdens by $\bar{M}(z)ds$ at each z, which results in changes in marginal tax rates of $\bar{M}'(z)ds$ at each z. By assuming $\bar{M}'(z) < 0$, the implication is that the marginal rates will increase in this reform. Thus, labor supply contracts by $ds\bar{M}'(z)\frac{\partial z(\theta)}{\partial \tau}$ at each income z. The fiscal cost per $d\tau$ is $\frac{d}{dz}\left(T(z) - S(z)\bar{M}(z)\right)$. We assume this fiscal externality is offset via lump-sum taxes on all households, and thus a welfare weight of unity is applied. We thus obtain:

$$dW_3 = ds\int_z \bar{M}'(z)\frac{\partial z(\theta)}{\partial \tau}\cdot\frac{d}{dz}\left(T(z) - S(z)\bar{M}(z)\right)\, dH(z), \tag{42}$$

and applying in the definition of the elasticity $\varepsilon_\tau(z;\theta) = -(1-T'(z))/z\cdot\partial z(\theta)/\partial\tau$ yields the third term in the proposition:

$$dW_3 = ds\int_z \frac{\bar{M}'(z)z\bar{\varepsilon}_\tau(z)}{1-T'(z)}\frac{d}{dz}\left(T(z) - S(z)\bar{M}(z)\right) dH(z), \tag{43}$$

This completes the proof.

□

D.2 Proof of Proposition 2

In the main text, we treat a dollar of automatic transfer as equivalent to a cash dollar, allowing us to model changes in the automatic transfer as occurring through the tax system. Here, we explicitly distinguish between the tax system, an automatic transfer, and a voluntary transfer. We then derive an analog of Proposition 1 in this enriched setting.

The government must now choose a tax schedule $T(z)$, a voluntary transfer $S_V(z)$ and an automatic transfer $S_A(z)$. The voluntary transfer is what we labeled as $S(z)$ in the main paper: households must pay a cost κ to take up. Taxes and the automatic transfer are received automatically by households at income z (without a cost being paid), except a dollar of the latter is not valued equally to the a dollar of the former. Let λ represent the marginal utility for a dollar of S_A or S_V relative to a dollar of cash.

In the household's problem, the period budget constraint (Equation 8) becomes

$$c_t + a_{t+1} = z_t - T(z_t) + R_t a_t + \lambda S_A(z_t) + \lambda \int_0^{S(z_t)} [S(z_t) - \kappa]\, \mu(\kappa|w_t)\, d\kappa, \tag{44}$$

with all other equations left unchanged. We emphasize that the government's period budget constraint remains

$$\int_\Theta [T\,(z(\theta)) - S_A(z(\theta)) - \mathbb{1}_S S_V\,(z(\theta))]\, d\mu(a,\theta) = 0. \tag{45}$$

Even though a dollar of S_A or S_V is only valued at λ dollars by the household, that is, it costs the government a dollar to provide.

Reform. We define the reform analogously to the primary reform in Section 5. The voluntary transfer amount is cut by ds_V at all incomes. With the savings, at each income z, automatic transfers are increased by $s_A(z) = \bar{M}(z)ds_V$, so that people at each income level are compensated on average for the voluntary transfer cut. The slope of S_A rates (analogous to marginal tax rates) thus change by $S'_A(z) = \frac{d}{dz}\bar{M}(z)ds_V$ at z. Fiscal savings from marginal transfer recipients are redistributed as a lump sum automatic transfer. The revenue cost of any labor supply response is then paid for via lump-sum taxes (i.e. as cash, not in-kind).

We calculate the welfare effects of this reform analogously to Proposition 1.

Proposition 2. *The welfare effect of the reform is*

$$\frac{1}{\lambda}\frac{dW}{ds} = \underbrace{\beta\sigma_M^2}_{\textit{lost value of self-targeting}} + \underbrace{\bar{M}\bar{\varepsilon}_b}_{\textit{fiscal savings from marginals}} + \underbrace{\int \frac{\bar{M}'(z)z\bar{\varepsilon}_\tau(z)}{1-T'(z)}\left(\frac{d}{dz}(S_V(z)\bar{M}(z)) + S'_A(z) - T'(z)\right) dH(z).}_{\textit{labor-supply effect}} \tag{46}$$

where all other terms are as in Proposition 1.

While Proposition 1 moves money from from S to T, Proposition 2 moves money from S_V to S_A. In both cases, the dollar being moved has the same ex-post marginal utility (1 in S and T, λ in S_V and S_A). Moving a dollar from the those who take up the voluntary transfer to everyone decreases marginal utility by λ times the lost value of self-targeting from Proposition 1. Similarly, all the dollars saved from the marginals not taking up S_V are redistributed in-kind through S_A and so the utility value is λ times the fiscal savings from marginals term in Proposition 1.

As for the labor-supply effects, the first term is the change in the government budget due to labor supply changes that is then redistributed as an automatic transfer to everyone. But unlike Proposition 1, where marginal tax rates changed, here the slope of the S_A schedule changes, and so the relevant elasticity is ε_{S_A} not ε_z. However, per the household's problem, $\frac{dz}{ds_A} = \lambda\frac{dz}{d\tau}$.

The intuition is that the costs and benefits of this reform are all scaled by λ relative to the reform in Proposition 1. Utility benefits/costs are λ lower since they are paid in in-kind dollars rather than cash dollars, and impacts on the budget constraint, although intrinsically denominated in cash, are convertible to in-kind dollars since labor supply response to a dollar change in S_A is λ times the labor supply response to a dollar change in T.

Proof. The proof is very similar to the proof of Proposition 1. The first term, the lost value of self-targeting, is λ multiplied by the term in Proposition 1. The same is true for the second term, since the fiscal savings are redistributed in-kind. The key differences is in the labor supply term.

Since the slope of the automatic transfer schedule has risen (as $\bar{M}'(z) < 0$), labor supply contracts by $ds\bar{M}'(z)\frac{\partial z}{\partial s_A}$ at each income z. The fiscal cost per $d\tau$ is $T'(z) - \frac{d}{dz}\left(S(z)\bar{M}(z)\right) - S'_A(z)$. However, examining the first-order condition of the household, we have that $\frac{\partial z}{\partial s_A} = \lambda\frac{\partial z}{\partial \tau}$. Thus, $ds\bar{M}'(z)\frac{\partial z}{\partial s_A} = ds\bar{M}'(z)\lambda\frac{\partial z}{\partial \tau}$ and plugging in the definition of the elasticity: $\varepsilon_z(\theta) = -\frac{\partial z(\theta)}{\partial \tau}\frac{1-T'(z)}{z}$ and by assumption all fiscal costs are paid lump sum, that is by the average welfare weight $\mathrm{E}[\alpha(\theta)] = 1$, yields the first labor supply term after rearrangements similar to Proposition 1.

□

D.3 Additional Proofs

We now establish conditions under which there is a social benefit from self-targeting. These proofs formally establish claims in Section 5 about the signs of terms in Proposition 1.

Proposition 3. *Assuming the transfer $S(z)$ is positive, the first term in Equation 9 is negative if there exists advantageous selection, implying lost value from self-targeting.*

Proof. Directly from the definition of advantageous selection we get that $\mathrm{E}_{\kappa \leq S(z)}\left[\alpha(\theta)V_a'(a;\theta)\right] > \mathrm{E}\left[\alpha(\theta))V_a'(a;\theta)\right]$. This implies the integrand is positive for all values of z, and hence positive overall. □

We now establish conditions under which the policy reform we consider in the main text would raise or lower labor supply.

Proposition 4. *Suppose (1) the tax system is optimal and (2) take-up decreases in income (i.e., $\bar{M}'(z) < 0$). Then the labor supply effect in Equation 9 is negative.*

Proof. We derive a necessary condition from the optimal tax schedule that ensures the labor supply effect is signed as proposed. Suppose the planner increases the tax rate at income z by $d\tau$, and that the net fiscal gain/loss from this change is redistributed as a lump sum transfer/tax.

Breaking down the effect into fiscal and behavioural responses, we have the following changes to welfare:

1. Direct effect (fiscal and welfare):

$$d\tau \int_{x \geq z} (\mathrm{E}[\alpha(\theta)V_a'(a;\theta)] - \mathrm{E}[\alpha(\theta)V_a'(a;\theta) \mid z = x])\, dH(x)$$
$$= E_{x \geq z}\left[\mathrm{E}[\alpha(\theta)] - \mathrm{E}[\alpha(\theta)V_a'(a;\theta) \mid z = x]\right](1 - H(z)). \quad (47)$$

2. Compensated price effect (effect on tax receipts):

$$d\tau \frac{\partial z}{\partial \tau} \cdot T'(z) \cdot h(z) \cdot \mathrm{E}[\alpha(\theta)V_a'(a;\theta)] = d\tau \left(-\epsilon^z \cdot z \cdot \frac{T'(z)}{1 - T'(z)} h(z)\mathrm{E}[\alpha(\theta)V_a'(a;\theta)]\right). \quad (48)$$

3. Compensated price effect (effect on social program payments):

$$d\tau \frac{\partial z}{\partial \tau} \frac{d}{dz}\left[-S(z)\bar{M}(z)\right] h(z)\mathrm{E}[\alpha(\theta)V_a'(a;\theta)] \quad (49)$$
$$= d\tau \frac{\epsilon^z \cdot z}{1 - T'(z)} \frac{d}{dz}\left[S(z)\bar{M}(z)\right] h(z)\mathrm{E}[\alpha(\theta)V_a'(a;\theta)]. \quad (50)$$

A necessary condition for the optimality of the tax system is that the sum of these welfare effects is weakly negative. In particular, to convert to utility units, suppose the net fiscal externality from the change in marginal tax rates was redistributed as a lump sum. This need not be the optimal way to redistribute, but for optimality it cannot deliver a positive welfare benefit. Consequently,

$$\mathrm{E}_{x\geq z}\left[\mathrm{E}[\alpha(\theta)V_a'(a;\theta)] - \mathrm{E}[\alpha(\theta)V_a'(a;\theta)|z=x](1-H(z))\right. \tag{51}$$

$$\leq d\tau \mathrm{E}[\alpha(\theta)]\left(T'(z) - \frac{d}{dz}\left[S(z)\bar{M}(z)\right]\right)\left(\frac{h(z)\epsilon^z z}{1-T'(z)}\right). \tag{52}$$

The left hand side is positive, and therefore so must the right hand side be positive. Immediately we have that the labor supply term is negative, as hypothesized. □

D.4 Empirical Calibration

We now provide additional details on the empirical calibration.

Utility Parameter ψ. We calibrate $\psi = w/h^{1/\eta}$. This internal calibration arises from the household's labor–leisure first-order condition:

$$u_l = -w \cdot u_c \tag{53}$$

for hours l, consumption c, and hourly wage w. Obtaining expressions for u_l and u_c from GHH preferences, we obtain $\psi = w/l^{1/\eta}$.

Other Adjustments. Analysis uses household-level variables. We form utility using equivalized consumption and average hours worked across adults. We winsorize consumption and lifetime income at the first percentile, as well as social marginal utility at the 99th percentile, to diminish outliers that arise from isoelastic utility. We compute the hourly wage faced by the household by taking equivalized household wages in the PSID and dividing by total hours worked across adults.

Estimation. Estimation requires aggregating to income quantiles to compute the integrals in the labor-supply effect at the income-group level. We form 100 income-rank quantiles z, based on household equivalized income. We linearly interpolate the tax schedule across 100 quantiles. We form the elasticity of taxable income at the quantile level. For households with zero income, we replace their ETI as the minimum of households with positive income. To estimate the labor-supply term, we convert the keep rate, expressed in the text in units of dollars, to income-rank units.

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